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A comprehensive worldwide summary of strong-motion attenuation relationships for peak ground acceleration and spectral ordinates (1969 to 2000)

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1. INTRODUCTION

A number of reviews of attenuation studies have been made in the past which provide a good summary of the methods used, the results obtained and the problems associated with such relations. Trifunac & Brady (1975, 1976) provide a brief summary and comparison of published relations. Idriss (1978) presents a comprehensive review of published attenuation relations up until 1978, including a number which are not easily available. Boore & Joyner (1982) provide a review of attenuation studies published in 1981 and comment on empirical prediction of strong ground motion in general. Campbell (1985) contains a full survey of attenuation equations up until 1985. Joyner & Boore (1988) give an excellent analysis of ground motion prediction methodology in general, and attenuation relations in particular; Joyner & Boore (1996) update this by including more recent studies. Ambraseys & Bommer (1995) provide an overview of relations which are used for seismic design in Europe although they do not provide details about methods used.

After these studies were completed many more equations were derived. However, the data selection, processing and regression methods, and forms of equation used, are not fully covered by the authors of the studies. A summary of the methods used to derive the equations is therefore presented here. This report contains details of all studies for peak ground acceleration and response spectra which could be found in the literature (journals, conference proceedings and technical reports) although some may have been inadvertently missed.

Some of the studies included here have not been seen but are reported in other publications and hence the information given here may not be complete or correct.

Equations for single earthquakes (e.g. Bozorgnia *et al.*, 1995) or for earthquakes of approximately the same size (e.g. Sadigh *et al.*, 1978) are excluded due to their limited usefulness. Also excluded are those relations based on intensity measurements (e.g. Battis, 1981), those based on theoretical ground motions (stochastic source models etc.) (e.g Atkinson & Boore, 1990) or those originally developed to yield the magnitude of an earthquake (e.g. Espinosa, 1980), i.e. the regression is performed the wrong way round, which should not be used for the prediction of ground motion at a site. Studies which derive graphs to give predictions (e.g. Schnabel & Seed, 1973) are not considered in this report nor are those nonparametric formulations which provide predictions for different combinations of distance and magnitude (e.g. Anderson, 1997), both of which are more difficult to use for seismic hazard analysis than those which give a single formula.

All the studies which present the same attenuation relationship are mentioned at the top of the section and in the tables of general characteristics (Tables 3.1 & 5.1). The information contained within each section, and within the table, is the sum of information contained within each of the publications, i.e. not all the information may be in one study.

To make it easier to understand the functional form of attenuation equation adopted in each study the equations are given with variable names replacing actual coefficients and the derived coefficients and the standard deviation, σ, are given separately (for peak ground acceleration equations). These coefficients are given only for completeness and if an equation is to be used then the original reference should be consulted. If a coefficient is assumed before the analysis is performed then the number is given in the formula.

Obviously all the details from each publication cannot be included in this report because of lack of space but the most important details of the methods and data used are retained.

The number of records within each site and source mechanism category are given if this information was reported by the authors of the study. Sometimes these totals were found by counting the numbers in each category using the tables listing the data used.

In the equations unless otherwise stated, D, d, R, r, Δ or similar are distance and M or similar is magnitude and all other independent variables are stated. PGA is peak ground acceleration, PGV is peak ground

velocity and PSV is relative pseudo-velocity.

In Tables 3.1 & 5.1 the gross characteristics of the data used and equation obtained is only given for the main equation in the study. The reader should refer to the section on a particular publication for information on other equations derived in the study.

No discussion of the merits, ranges of applicability or limitations of any of the relationships is included herein except those mentioned by the authors or inherent in the data used. The ground motion models are reported in the form given in the original references except sometimes the equation is simplified if this can be easily done.

This report provides a comprehensive summary of strong motion attenuation studies which can be used for finding references to useful works and for using as a basis for reviews of previously published equations. Note however that the size of this report means that it may contain some errors or omissions.

2. SUMMARY OF PUBLISHED ATTENUATION RELATIONS FOR PEAK GROUND ACCELERATION

2.1 Milne & Davenport (1969)

- Ground motion model is:

$$A = \frac{a_1 \mathrm{e}^{a_2 M}}{a_3 \mathrm{e}^{a_4 M} + \Delta^2}$$

where A is in percentage of g, $a_1 = 0.69$, $a_2 = 1.64$, $a_3 = 1.1$ and $a_4 = 1.10$ (σ not given).

- Use data from Esteva & Rosenblueth (1964).

2.2 Esteva (1970)

- Ground motion model is:

$$a = c_1 \mathrm{e}^{c_2 M} (R + c_3)^{-c_4}$$

where a is in cms^{-2}, $c_1 = 1230$, $c_2 = 0.8$, $c_3 = 25$, $c_4 = 2$ and $\sigma = 1.02$ (in terms of natural logarithm).

- Records from soils comparable to stiff clay or compact conglomerate.
- Records from earthquakes of moderate duration.

2.3 Denham & Small (1971)

- Ground motion model is:

$$\log Y = b_1 + b_2 M + b_3 \log R$$

where Y is in g, $b_1 = -0.2$, $b_2 = 0.2$ and $b_3 = -1.1$ (σ not given).

- Records from near dam on recent unconsolidated lake sediments which are $\geq 50\,\mathrm{m}$ thick.
- Note need for more points and large uncertainty in b_1, b_2 and b_3.

2.4 Donovan (1973)

- Ground motion model is:

$$y = b_1 \mathrm{e}^{b_2 M} (R + 25)^{-b_3}$$

where y is in gal, $b_1 = 1080$, $b_2 = 0.5$, $b_3 = 1.32$ and $\sigma = 0.71$. 25 adopted from Esteva (1970).

- 214 (32%) records from San Fernando (9/2/1971) earthquake and 53% of records with PGA less than $0.5\,\mathrm{ms}^{-2}$.
- Considers portions of data and finds magnitude dependence increases with increasing distance from source and more small accelerations increase magnitude dependence. Thus magnitude and distance cannot be considered independent variables.

2.5 Esteva & Villaverde (1973)

- Ground motion model is:

$$Y_c = b_1 \mathrm{e}^{b_2 M} (R + b_4)^{-b_3}$$

where Y_c is in cms^{-2}, $b_1 = 5600$, $b_2 = 0.8$, $b_3 = 2$, $b_4 = 40$ and $\sigma = 0.64$ (in terms of natural logarithm).

2.6 Orphal & Lahoud (1974)

- Ground motion model is:

$$A = \lambda 10^{\alpha M} R^{\beta}$$

where A is in g, $\lambda = 6.6 \times 10^{-2}$, $\alpha = 0.40$, $\beta = -1.39$ and $\sigma = 1.99$ (this is multiplication factor).

- Use 113 records with distances between 15 to $350\,\mathrm{km}$ from San Fernando earthquake to find distance dependence, β.

- Use 27 records of Wiggins Jr. (1964) from El Centro and Ferndale areas, with magnitudes between 4.1 and 7.0 and distances between 17 and $94\,\mathrm{km}$ (assuming focal depth of $15\,\mathrm{km}$), to compute magnitude dependent terms assuming distance dependence is same as for San Fernando.

2.7 Ambraseys (1975)

- Ground motion model is:

$$\log Y = b_1 + b_2 M_L + b_3 \log R$$

where Y is in cms^{-2}, $b_1 = 0.46$, $b_2 = 0.63$, $b_3 = -1.10$ and $\sigma = 0.32$[1]

- Ambraseys & Bommer (1995) state that uses earthquakes with maximum focal depth of $15\,\mathrm{km}$.

2.8 Trifunac & Brady (1975),Trifunac (1976) & Trifunac & Brady (1976)

- Ground motion model is:

$$\begin{aligned}
\log_{10} a_{\max} &= M + \log_{10} A_0(R) - \log_{10} a_0(M,p,s,v) \\
\log_{10} a_0(M,p,s,v) &= \begin{cases} ap + bM + c + ds + ev + fM^2 - f(M - M_{\max})^2 & \\ \qquad \text{for} \quad M \geq M_{\max} & \\ ap + bM + c + ds + ev + fM^2 & \\ \qquad \text{for} \quad M_{\max} \geq M \geq M_{\min} & \\ ap + bM_{\min} + c + ds + ev + fM_{\min}^2 & \\ \qquad \text{for} \quad M \leq M_{\min} & \end{cases}
\end{aligned}$$

where $a_{\max}$ is in cms^{-2}, $\log_{10} A_0(R)$ is an empirically determined attenuation function from Richter (1958) used for calculation of M_L, p is confidence level and v is component direction ($v = 0$ for horizontal and 1 for vertical). Coefficients are: $a = -0.898$, $b = -1.789$, $c = 6.217$, $d = 0.060$, $e = 0.331$, $f = 0.186$, $M_{\min} = 4.80$ and $M_{\max} = 7.50$ ($\log_{10} A_0(R)$ not given here due to lack of space).

- Use three site categories:

$s = 0$ Alluvium or other low velocity 'soft' deposits: 63% of records.

$s = 1$ 'Intermediate' type rock: 23% of records.

[1] From Ambraseys & Bommer (1995).

$s = 2$ Solid 'hard' basement rock: 8% of records.

- Exclude records from tall buildings.
- Do not use data from other regions because attenuation varies with geological province and magnitude determination is different in other countries.
- Records baseline and instrument corrected. Accelerations thought to be accurate between 0.07 and $25\,\mathrm{Hz}$ or between 0.125 and $25\,\mathrm{Hz}$ for San Fernando records.
- Most records (71%) from earthquakes with magnitudes between 6.0–6.9, 22% are from 5.0–5.9, 3% are from 4.0–4.9 and 3% are from 7.0–7.7 (note barely adequate data from these two magnitude ranges). 63% of data from San Fernando earthquake.
- Note that for large earthquakes, i.e. long faults, $\log_{10} A_0(R)$ would have a tendency to flatten out for small epicentral distances and for low magnitude shocks curve would probably have a large negative slope. Due to lack of data $\lesssim 20\,\mathrm{km}$ this is impossible to check.
- Note difficulty in incorporating anelastic attenuation because representative frequency content of peak amplitudes change with distance and because relative contribution of digitization noise varies with frequency and distance.
- Note that $\log_{10} A_0(R)$ may be unreliable for epicentral distances less than $10\,\mathrm{km}$ because of lack of data.
- Change of slope in $\log_{10} A_0(R)$ at $R = 75\,\mathrm{km}$ because for greater distances main contribution to strong shaking from surface waves, which are attenuated less rapidly ($\sim 1/R^{1/2}$) than near-field and intermediate-field ($\sim 1/R^{2-4}$), or far-field body waves ($\sim 1/R$).
- Note lack of data to reliably characterise $\log_{10} a_0(M, p, s, v)$ over a sufficiently broad range of their arguments. Also note high proportion of San Fernando data may bias results.
- Firstly partition data into four magnitude dependent groups: 4.0–4.9, 5.0–5.9, 6.0–6.9 and 7.0–7.9. Subdivide each group into three site condition subgroups (for $s = 0$, 1 and 2). Divide each subgroup into two component categories (for $v = 0$ and 1). Calculate $\log_{10} a_0(M, p, s, v) = M + \log_{10} A_0(R) - \log_{10} a_{\max}$ within each of the 24 parts. Arrange each set of n $\log_{10} a_0$ values into decreasing order with increasing n. Then mth data point (where m equals integer part of pn) is estimate for upper bound of $\log_{10} a_0$ for $p\%$ confidence level. Then fit results using least squares to find $a, \ldots f$.
- Check number of PGA values less than confidence level for $p = 0.1, \ldots, 0.9$ to verify adequacy of bound. Find simplifying assumptions are acceptable for derivation of approximate bounds.

2.9 Blume (1977)

- Ground motion model is:
$$a = b_1 \mathrm{e}^{b_2 M_L} (R + 25)^{-b_3}$$
where a is in gal, for $M_L \leq 6\frac{1}{2}$ $b_1 = 0.318 \times 29^{1.14\bar{b}}$, $b_2 = 1.03$, $b_3 = 1.14\bar{b}$ and $\sigma = 0.930$ (in terms of natural logarithm) and for $M_L > 6\frac{1}{2}$ $b_1 = 26.0 \times 29^{1.22\bar{b}}$, $b_2 = 0.432$, $b_3 = 1.22\bar{b}$ and $\sigma = 0.592$ (in terms of natural logarithm).
- Assumes all earthquakes have focal depth of $8\,\mathrm{km}$.
- Makes no distinction for site conditions in first stage where uses only earthquake records.
- Studies effects of PGA cutoff (no cutoff, 0.01, 0.02 and $0.05\,\mathrm{ms}^{-2}$), distance cutoff (no cutoff and $< 150\,\mathrm{km}$) and magnitude cutoff (all, $\geq 5\frac{1}{2}$, ≥ 6, $\geq 6\frac{1}{2}$, $\geq 6\frac{3}{4}$ and $\leq 6\frac{1}{2}$).

- Selects $6\frac{1}{2}$ as optimum magnitude cutoff but uses all data to derive equation for $M_L \leq 6\frac{1}{2}$ because not much difference and dispersion is slightly lower (in terms of ± 1 standard deviation have 2.53 and 2.61).
- In second stage uses only records from underground nuclear explosions, consistent with natural earthquake records, to derive site factor.
- Uses 1911 alluvium and 802 rock records and derive PGA ratio of alluvium to rock assuming their PGAs equal at $4\,\mathrm{km}$.
- Finds site impedance ρV_s, where ρ is density and V_s is shear-wave velocity under site, is best measure of site condition. Use $2000\,\mathrm{fps}$ ($600\,\mathrm{ms}^{-1}$) as shear-wave velocity of alluvium stations.
- Multiplies equation (after taking logarithms) by $\bar{b} = \frac{1}{2}\log_{10}(\rho V_s)$ and normalise to $4\,\mathrm{km}$.
- Notes may not be a good model for other regions.

2.10 McGuire (1977)

- Ground motion model is:
$$E[v] = a10^{bM}(R+25)^{-c}$$
where E indicates expectation, v is in gal, $a = 472$, $b = 0.278$, $c = 1.301$.
- Excludes records for which significant soil amplification established but makes no distinction between rock and soil sites.
- Focal depths between 9 and $70\,\mathrm{km}$ with most about $10\,\mathrm{km}$. Most records from earthquakes with magnitudes about 6.5 and most distances less than $50\,\mathrm{km}$. Uses records from 21 different sites.
- Notes that physical laws governing ground motion near the source are different than those governing motion at greater distances therefore excludes records with epicentral distance or distance to fault rupture smaller than one-half of estimated length of rupture.
- Examines correlation among the records but find negligible effect.

2.11 Milne (1977)

- Ground motion model is:
$$\mathrm{ACC} = a_1 \mathrm{e}^{a_2 M} R^{a_3}$$
where ACC is in g, $a_1 = 0.04$, $a_2 = 1.00$ and $a_3 = -1.4$.

2.12 Donovan & Bornstein (1978)

- Ground motion model is:
$$\begin{aligned} y &= b_1 \mathrm{e}^{b_2 M}(R+25)^{-b_3} \\ \text{where} \quad b_1 &= c_1 R^{-c_2} \\ b_2 &= d_1 + d_2 \log R \\ b_3 &= e_1 + e_2 \log R \end{aligned}$$
where y is in gal, $c_1 = 2,154,000$, $c_2 = 2.10$, $d_1 = 0.046$, $d_2 = 0.445$, $e_1 = 2.515$, $e_2 = -0.486$, for $y = 0.01\,\mathrm{g}$ $\sigma = 0.5$, for $y = 0.05\,\mathrm{g}$ $\sigma = 0.48$, for $y = 0.10\,\mathrm{g}$ $\sigma = 0.46$ and for $y = 0.15\,\mathrm{g}$ $\sigma = 0.41$ (in terms of natural logarithm).

Use 25 because assume energy centre of Californian earthquakes to be at depth $5\,\mathrm{km}$.

- Consider two site conditions but do not model:
  1. Rock: (21 records)
  2. Stiff soil: (38 records)
- 32% of records from San Fernando (9/2/1971) but verifies that relationship is not significantly biased by this data.
- Most records within $50\,\mathrm{km}$ and most from earthquakes with magnitudes of about 6.5.
- Recognises that magnitude and distance are not independent variables.
- Find b_1, b_2 and b_3 by dividing data according to distance and computing b parameters for each set using least squares. Find a distinct trend with little scatter.

2.13 *Faccioli (1978)*

- Ground motion model is:
$$y = a10^{bM}(R+25)^{-c}$$
where y is in gal, $a = 108.60$, $b = 0.265$, $c = 0.808$ and $\sigma = 0.236$ (in terms of logarithm to base 10).
- Records from sites underlain by cohesive or cohesionless soils with shear-wave velocities less than about $100\,\mathrm{ms}^{-1}$ and/or standard penetration resistance $N \leq 10$ in uppermost $10\,\mathrm{m}$ with layers of considerably stiffer materials either immediately below or at depths not exceeding a few tens of metres.
- Focal depths between 9 and $100\,\mathrm{km}$.
- free-field accelerograms, to minimize soil-structure interaction.
- Excludes records with $\mathrm{PGA} < 0.4\,\mathrm{ms}^{-2}$.
- 21 Japanese records processed with frequency cutoffs of bandpass filter, for baseline correction, adjusted so as to account for length and mean sampling rate of records and response characteristics of SMAC-2. 4 of remaining 7 records processed in same way.

2.14 *McGuire (1978)*

- Ground motion model is:
$$\ln x = b_1 + b_2 M + b_3 \ln R + b_4 Y_s$$
where x is in cms^{-2}, $b_1 = 3.40$, $b_2 = 0.89$, $b_3 = -1.17$, $b_4 = -0.20$ and $\sigma = 0.62$.
- Uses two site categories:

$Y_s = 0$ Rock: sedimentary or basement rock or soil less than $10\,\mathrm{m}$ thick, 11 records.

$Y_s = 1$ Soil: alluvium or other soft material greater than $10\,\mathrm{m}$ thick, 59 records.

- Uses records from basement of buildings or from 'free-field'. Uses no more than seven records from same earthquake and no more than nine from a single site to minimize underestimation of calculated variance. Retains records which give a large distance and magnitude range.
- Notes that near-field ground motion governed by different physical laws than intermediate and far field so excludes near-field data, for example El Centro (19/5/1940) and Cholame-2, from Parkfield earthquake (28/6/1966)
- Considers a distance dependent site term but not statistically significant. Also uses a magnitude dependent site term and although it was statistically significant it did not reduce the scatter and also since largest magnitude for a rock site is 6.5, result may be biased.

2.15 A. Patwardhan, K. Sadigh, I.M. Idriss, R. Youngs (1978) reported in Idriss (1978)

- Ground motion model is:

$$\ln y = \ln A + BM_s + E \ln[R + d \exp(fM_s)]$$

where y is in cms^{-2}, $d = 0.864$ and $f = 0.463$ and for path A (rock): $A = 157$ (for median), $A = 186$ (for mean), $B = 1.04$ and $E = -1.90$, for path A (stiff soil): $A = 191$ (for median), $A = 224$ (for mean), $B = 0.823$ and $E = -1.56$ and for path B (stiff soil): $A = 284$ (for median), $A = 363$ (for mean), $B = 0.587$ and $E = -1.05$ (σ not given).

- Separate equations for two types of path:

  A Shallow focus earthquakes (California, Japan, Nicaragua and India), 63 records.

  B Subduction (Benioff) zone earthquakes (Japan and South America), 23 earthquakes, $5.3 \leq M_s \leq 7.8$, 32 records.

- Use two site categories for path A earthquakes for which derive separate equations:

  1. Rock: 21 records.
  2. Stiff soil: 42 records.

  Use only stiff soil records for deriving subduction zone equation.

- Most earthquakes for path A have $5 \leq M_s \leq 6.7$.

- All data corrected. PGA for corrected Japanese and South American records much higher than uncorrected PGA.

2.16 *Cornell* et al. *(1979)*

- Ground motion model is:

$$\ln A_p = a + bM_L + c \ln(R + 25)$$

where A_p is in cms^{-2}, $a = 6.74$, $b = 0.859$, $c = -1.80$ and $\sigma = 0.57$.

- No more than 7 records from one earthquake to avoid biasing results.

- Records from basements of buildings or free-field.

2.17 Aptikaev & Kopnichev (1980)

- Ground motion model is:

$$\log A_e = a_1 M + a_2 \log R + a_3$$

where A_e is in cms^{-2}, for $A_e \geq 160\,\mathrm{cms}^{-2}$ $a_1 = 0.28$, $a_2 = -0.8$ and $a_3 = 1.70$ and for $A_e < 160\,\mathrm{cms}^{-2}$ $a_1 = 0.80$, $a_2 = -2.3$ and $a_3 = 0.80$ (σ not given).

- As a rule, PGA corresponds to S-wave.

- Use five source mechanism categories (about 70 records, 59 earthquakes from W. N. America including Hawaii, Guatemala, Nicaragua, Chile, Peru, Argentina, Italy, Greece, Romania, central Asia, India and Japan):

  1. Contraction faulting (uplift and thrust), about 16 earthquakes.
  2. Contraction faulting with strike-slip component, about 6 earthquakes.

3. Strike-slip, about 17 earthquakes.
4. Strike-slip with dip-slip component, about 6 earthquakes.
5. Dip-slip, about 9 earthquakes.

- Use these approximately 70 records to derive ratios of mean measured, A_0, to predicted PGA, A_e, $\log(A_0/A_e)$, and for ratios of mean horizontal to vertical PGA, $\log A_h/A_v$, for each type of faulting. Use every earthquake with equal weight independent of number of records for each earthquake.

- Results are:

	Category 1	Category 2	Category 3	Category 4	Category 5
$\log A_0/A_e$	0.35 ± 0.13 (16)	0.11 ± 0.17 (5)	0.22 ± 0.08 (17)	0.06 ± 0.13 (6)	-0.06 ± 0.20 (9)
$\log A_h/A_v$	0.32 ± 0.13 (12)	0.32 ± 0.08 (5)	0.27 ± 0.07 (12)	0.18 ± 0.10 (5)	0.17 ± 0.11 (5)

  where $\pm$ gives 0.7 confidence intervals and number in brackets is number of earthquakes used.

- Also calculate mean envelope increasing speed for P-wave amplitudes, A, obtained at teleseismic distances: $n = \mathrm{d}\ln A/\mathrm{d}t$, where t is time for P-wave arrival and try to relate to ratios for each type of faulting.

2.18 *Blume (1980)*

- Ground motion model is:

$$a = b_1 \mathrm{e}^{b_2 M}(R+k)^{-b_3}$$

  where a is in gal, for method using distance partitioning $b_1 = 18.4$, $b_2 = 0.941$, $b_3 = 1.27$ and $k = 25$ and for ordinary one-stage method $b_1 = 102$, $b_2 = 0.970$, $b_3 = 1.68$ and $k = 25$ (σ not given).

- Does not use PGA cutoff because PGA is, by itself, a poor index of damage in most cases.

- Mean magnitude is 5.4 and mean distance is $84.4\,\mathrm{km}$.

- Notes problem of regression leverage for some attenuation studies. Lots of data in fairly narrow distance band, e.g. records from San Fernando earthquake, can dominate regression and lead to biased coefficients.

- Divides data into ten distance bands (A-J) which are $10\,\mathrm{km}$ wide up to $60\,\mathrm{km}$ and then 60-$99.9\,\mathrm{km}$, 100–$139.9\,\mathrm{km}$, 140–$199.9\,\mathrm{km}$ and $\geq 200\,\mathrm{km}$. Fits $\log_{10} a = bM - c$ to data in each band and fits ground motion model to selected point set in M, R and a.

- Also fits equation using all data using normal least squares.

- Adds 52 records ($3.2 \leq M \leq 6.5$, $5 \leq R \leq 15\,\mathrm{km}$) and repeats; finds little change.

2.19 *Iwasaki* et al. *(1980)*

- Ground motion model is:

$$\mathrm{PGA} = a_1 10^{a_2 M}(\Delta + 10)^{a_3}$$

  where PGA is in gal, for type I sites $a_1 = 46.0$, $a_2 = 0.208$ and $a_3 = -0.686$, for type II sites $a_1 = 24.5$, $a_2 = 0.333$ and $a_3 = -0.924$, for type III sites $a_1 = 59.0$, $a_2 = 0.261$ and $a_3 = -0.886$, for type IV sites $a_1 = 12.8$, $a_2 = 0.432$, $a_3 = -1.125$ and for all sites $a_1 = 34.1$, $a_2 = 0.308$ and $a_3 = -0.925$ (σ not given).

- Use four site categories:

Type I Tertiary or older rock (defined as bedrock) or diluvium with depth to bedrock, $H < 10\,\mathrm{m}$, 29 records.

Type II Diluvium with $H \geq 10\,\mathrm{m}$ or alluvium with $H < 10\,\mathrm{m}$, 74 records.

Type III Alluvium with $H < 25\,\mathrm{m}$ including soft layer (sand layer vulnerable to liquefaction or extremely soft cohesive soil layer) with thickness $< 5\,\mathrm{m}$, 130 records.

Type IV Other than above, usually soft alluvium or reclaimed land, 68 records.

- Select earthquakes with Richter magnitude ≥ 5.0, hypocentral depth $\leq 60\,\mathrm{km}$ and which include at least one record with PGA $\geq 50\,\mathrm{gals}$ ($0.5\,\mathrm{ms}^{-2}$). Exclude records with PGA $< 10\,\mathrm{gals}$ ($0.1\,\mathrm{ms}^{-2}$).
- All records for $M \geq 7.0$ are from distance $> 60\,\mathrm{km}$.
- Do regression separately for each soil category and also for combined data.

2.20 *Ohsaki* et al. *(1980a)*

- Ground motion model is:

$$A = 10^{a_1 M - a_2 \log x + a_3}$$

where A is in cms^{-2}, for horizontal PGA $a_1 = 0.440$, $a_2 = 1.381$ and $a_3 = 1.04$ and for vertical PGA $a_1 = 0.485$, $a_2 = 1.85$ and $a_3 = 1.38$ (σ not given).

- All records from free-field bedrock sites.

2.21 *Campbell (1981)*

- Ground motion model is:

$$\mathrm{PGA} = a \exp(bM)[R + c_1 \exp(c_2 M)]^{-d}$$

where PGA is in g, for unconstrained model $a = 0.0159$, $b = 0.868$, $c_1 = 0.0606$, $c_2 = 0.700$, $d = 1.09$ and $\sigma = 0.372$ (on natural logarithm) and for constrained model $a = 0.0185$, $b = 1.28$, $c_1 = 0.147$, $c_2 = 0.732$, $d = 1.75$ and $\sigma = 0.384$ (in terms of natural logarithm).

Uses this functional form because capable of modelling possible nonlinear distance scaling in near field and because distance at which transition from near field to far field occurs probably proportional to fault rupture zone size.

- Considers six site classifications but does not model:

A Recent alluvium: Holocene Age soil with rock $\geq 10\,\mathrm{m}$ deep, 71 records.

B Pleistocene deposits: Pleistocene Age soil with rock $\geq 10\,\mathrm{m}$ deep, 22 records.

C Soft rock: Sedimentary rock, soft volcanics, and soft metasedimentary rock, 14 records.

D Hard rock: Crystalline rock, hard volcanics, and hard metasedimentary rock, 9 records.

E Shallow soil deposits: Holocene or Pleistocene Age soil $< 10\,\mathrm{m}$ deep overlying soft or hard rock, 17 records. Not used in analysis.

F Soft soil deposits: extremely soft or loose Holocene Age soils, e.g. beach sand or recent floodplain, lake, swamp, estuarine, and delta deposits, 1 record. Not used in analysis.

- Notes that data from areas outside western USA may be substantially different than those from western USA due to tectonics and recording practices but far outweighed by important contribution these data can make to understanding of near-source ground motion.
- Notes use of only near-source data has made differences in anelastic attenuation negligible to inherent scatter from other factors.

- Selects data from shallow tectonic plate boundaries generally similar to western N. America, deep subduction events excluded because of differences in travel paths and stress conditions.
- Selects data from instruments with similar dynamic characteristics as those used in USA to avoid bias, therefore excludes data from SMAC accelerographs in Japan.
- Selects data which meet these criteria:
    1. Epicentres known with an accuracy of $5\,\mathrm{km}$ or less, or accurate estimate of closest distance to fault rupture surface known.
    2. Magnitudes accurate to within 0.3 units.
    3. Distances were within 20, 30, and $50\,\mathrm{km}$ for magnitudes less than 4.75 between 4.75 and 6.25 and greater than 6.25 respectively. Only uses data from earthquakes with magnitude ≥ 5.0 because of greatest concern for most design applications.
    4. Hypocentres or rupture zones within $25\,\mathrm{km}$ of ground surface.
    5. PGA$\geq\ 0.2\,\mathrm{ms}^{-2}$ for one component, accelerographs triggered early enough to capture strong phase of shaking.
    6. Accelerograms either free-field, on abutments of dams or bridges, in lowest basement of buildings, or on ground level of structures without basements. Excluded Pacoima Dam record, from San Fernando (9/2/1971) earthquake due to topographic, high-frequency resonance due to large gradation in wave propagation velocities and amplification due to E-W response of dam.
- Well distributed data, correlation between magnitude and distance only 6%.
- Uses PGA from digitised, unprocessed accelerograms or from original accelerograms because fully processed PGAs are generally smaller due to the $0.02\,\mathrm{s}$ decimation and frequency band-limited filtering of records.
- Uses mean of two horizontal components because more stable peak acceleration parameter than either single components taken separately or both components taken together.
- Magnitude scale chosen to be generally consistent with M_w. Division point between using M_L and M_s varied between 5.5 and 6.5; finds magnitudes quite insensitive to choice.
- Notes d_r is a statistically superior distance measure than epicentral or hypocentral and is physically consistent and meaningful definition of distance for earthquakes having extensive rupture zones.
- Does not use all data from San Fernando earthquake to minimize bias due to large number of records.
- Uses seven different weighting schemes, to control influence of well-recorded earthquakes (e.g. San Fernando and Imperial Valley earthquakes). Giving each record or each earthquake equal weight not reasonable representation of data. Uses nine distance dependent bins and weights each record by a relative weighting factor $1/n_{i,j}$, where $n_{i,j}$ is total number of recordings from ith earthquake in jth interval.
- Finds unconstrained coefficients and all coefficients statistically significant at 99%.
- Finds coefficients with d constrained to 1.75 (representative of far-field attenuation of PGA) and $c_2 = b/d$, which means PGA is independent of magnitude at the fault rupture surface. All coefficients statistically significant at 99%. Notes similarity between two models.
- Plots normalised weighted residuals against distance, magnitude[2] and predicted acceleration[2]. Finds that residuals uncorrelated, at 99%, with these variables.

[2] Not shown in paper.

- Normal probability plots, observed distribution of normalised weighted residuals and Kolmogorov-Smirnov test, at 90%, confirms that PGA can be accepted as being lognormally distributed.
- Finds effects of site geology, building size, instrument location and mechanism to be extensively inter-related so selects only records from free-field or small structures.
- Analyses all selected data, find sites of classes E and F significantly higher PGA , at 90% level, so removes records from E and F.
- Finds differences in PGA from other site categories to be negligible but notes that it cannot be extended to PGV, PGD, spectral ordinates or smaller magnitudes or further distances.
- Distribution with mechanism is: 69 from strike-slip, 40 from reverse, 5 from normal and 2 records from oblique. Finds that reverse fault PGAs are systematically higher, significant at 90%, than those from other fault types although size of bias is due to presence of data from outside N. America.
- Considers soil (A and B) records from small buildings (115 components) and in free-field and those obtained in lowest basement of large buildings (40 components). Finds PGA significantly lower, at 90% level, in large buildings.
- Finds topographic effects for 13 components used in final analysis (and for 11 components from shallow soil stations) to be significantly higher, at 90%, although states size of it may not be reliable due to small number of records.
- Removes Imperial Valley records and repeats analysis. Finds that saturation of PGA with distance is not strongly dependent on this single set of records. Also repeats analysis constraining $c_2 = 0$, i.e. magnitude independent saturation, and also constraining $c_1 = c_2 = 0$, i.e. no distance saturation, finds variance when no distance saturation is significantly higher, at 95%, than when there is saturation modelled.
- Finds that magnitude saturation effects in modelling near-source behaviour of PGA is important and c_2 is significantly greater than zero at levels of confidence exceeding 99%. Also variance is reduced when $c_2 \neq 0$ although not at 90% or above.
- Repeats analysis using distance to surface projection of fault, finds reduced magnitude saturation but similar magnitude scaling of PGA for larger events.

2.22 *Chiaruttini & Siro (1981)*

- Ground motion model is:

$$\log a = b_0 + b_{AN} X_{AN} + b_{AB} X_{AB} + b_M M_L + b_d \log d$$

  where a is in $\mathrm{g}/100$, $b_0 = 0.04$, $b_{AN} = 0.24$, $b_{AB} = 0.23$, $b_M = 0.41$ and $b_d = -0.99$ (σ not given).

- Use three site categories for Friuli records, although note that information is rather superficial:

  ThA Alluvium with depth $> 20\,\mathrm{m}$, 36 records.

  RI Rock-like: hard rock or stiff soil, 24[3] records.

  thA Alluvium-like with depth $\leq 20\,\mathrm{m}$: includes sites for which thickness of deposit is reported to be very small which accounts for a few metres of weathering of underlying bedrock, 60 records.

  Alpide belt records divided into two categories: rock-like (25 records) and alluvium-like (40 records).

[3] Typographic error in their Table 1 because only 14 records are listed for rock-like sites

- Use data from free-field instruments or from instruments in basements of small structures and divide data into three regions: those from 1976 Friuli shocks (120 records) $\Rightarrow X_{AN} = X_{AB} = 0$, those from 1972 Ancona swarm (40 records) $\Rightarrow X_{AN} = 1$ & $X_{AB} = 0$ and those from Alpide Belt (Azores to Pakistan excluding those from Friuli and Ancona) (64 records) $\Rightarrow X_{AN} = 0$ & $X_{AB} = 1$. Exclude records with PGA $< 0.15\,\mathrm{ms}^{-2}$ to avoid possible bias at low acceleration values.
- Assume average focal depth of $6\,\mathrm{km}$.
- Note some PGA values derived from velocity records which are retained because compatible with other data. No instrument corrections applied to Friuli records because correction does not substantially alter PGA.
- Use M_L because determined at short distances and allows homogenous determination from lowest values up to saturation at $M_L = 7.0$ and it is determined at frequencies of nearly $1\,\mathrm{Hz}$, close to accelerographic band.
- Perform regression on PGAs from each of the three regions and each soil types considered within that region.
- Group rock-like (R) and thick alluvium (ThA) records together for Friuli. Find b_d for Friuli equations derived for thin alluvium-like and rock and thick alluvium not significantly different but b_M is significantly different, at 95% level. Repeat analysis using only Tolmezzo records because of large scatter in residuals but decide it is in thA category.
- For Alpide belt equations find b_M is almost the same for Rl and Al records and the difference in b_d is less than standard error, thus repeat analysis using a dummy variable X_{Al} which equals 0 for Rl and 1 for Al records.

2.23 *Joyner & Boore (1981)*

- Ground motion model is:

$$\begin{aligned} \log y &= \alpha + \beta \mathbf{M} - \log r + br \\ \text{where } r &= (d^2 + h^2)^{1/2} \end{aligned}$$

  where y is in g, $\alpha = -1.02$, $\beta = 0.249$, $b = -0.00255$, $h = 7.3$ and $\sigma = 0.26$.

- Use two site categories (not all records have category):

$S = 0$ Rock: sites described as granite, diorite, gneiss, chert, greywacke, limestone, sandstone or siltstone and sites with soil material less than 4 to $5\,\mathrm{m}$ thick overlying rock, 29 records. Indicate caution in applying equations for $\mathbf{M} > 6.0$ due to limited records.

$S = 1$ Soil: sites described as alluvium, sand, gravel, clay, silt, mud, fill or glacial outwash except where soil less than 4 to $5\,\mathrm{m}$ thick, 96 records.

- Restrict data to western North American shallow earthquakes, depth less than $20\,\mathrm{km}$, with $\mathbf{M} > 5.0$. Most records from earthquakes with magnitudes less than 6.6.
- Exclude records from base of buildings three or more storeys high and from abutments of dams.
- Exclude records associated with distances which had an uncertainty greater than $5\,\mathrm{km}$.
- Exclude records from distances greater than or equal to the shortest distance to an instrument which did not trigger.
- Six earthquakes recorded at only one station so not included in second stage regression.

- Include quadratic dependence term, $\gamma \mathbf{M}^2$, but not significant at 90% level so omitted.
- Include site term, cS, but not significant so omitted.
- Examine residuals against distance for difference magnitude ranges, no obvious differences in trends are apparent among the different magnitude classes.
- Consider a magnitude dependent $h = h_1 \exp(h_2[\mathbf{M} - 6.0])$ but reduction in variance not significant. Also prefer magnitude independent h because requires fewer parameters.
- Examine effect of removing records from different earthquakes from data.
- Examine effect of different h on residuals and b. Note coupling between h and b.
- Note coincidence of anelastic coefficient, b, and measured Q values. Also note similarity between h and proportions of depth of seismogenic zone in California.

2.24 Bolt & Abrahamson (1982)

- Ground motion model is:
$$y = a\{(x+d)^2 + 1\}^c \mathrm{e}^{-b(x+d)}$$
where y is in g, for $5 \leq \mathbf{M} < 6$ $a = 1.2$, $b = 0.066$, $c = 0.033$, $d = 23$ and standard error for one observation of $0.06\,\mathrm{g}$, for $6 \leq \mathbf{M} < 7$ $a = 1.2$, $b = 0.044$, $c = 0.042$, $d = 25$ and standard error for one observation of $0.10\,\mathrm{g}$, for $7 \leq \mathbf{M} \leq 7.7$ $a = 0.24$ $b = 0.022$, $c = 0.10$, $d = 15$ and standard error for one observation of $0.05\,\mathrm{g}$ and for $6 \leq \mathbf{M} \leq 7.7$ $a = 1.6$, $b = 0.026$, $c = -0.19$, $d = 8.5$ and standard error for one observation of $0.09\,\mathrm{g}$.
- Use data of Joyner & Boore (1981).
- Form of equation chosen to satisfy plausible physical assumptions but near-field behaviour is not determined from overwhelming contributions of far-field data.
- Apply nonlinear regression on y not on $\log y$ to give more weight to near-field values.
- Split data into four magnitude dependent groups: $5 \leq \mathbf{M} < 6$, $6 \leq \mathbf{M} < 7$, $7 \leq \mathbf{M} \leq 7.7$ and $6 \leq \mathbf{M} \leq 7.7$.
- Use form of equation and regression technique of Joyner & Boore (1981), after removing 25 points from closer than $8\,\mathrm{km}$ and find very similar coefficients to Joyner & Boore (1981). Conclude from this experiment and their derived coefficients for the four magnitude groups that using their form of equation predicted near-field accelerations are not governed by far-field data.
- Find no evidence of systematic increase in PGA near the source as a function of magnitude and that the large scatter prevents attaching significance to differences in near-field PGA which are predicted using their attenuation relations for different magnitude ranges.

2.25 PML (1982)

- Ground motion model is:
$$\ln(a) = C_1 + C_2 M + C_3 \ln[R + C_4 \exp(C_5 M)]$$
where a is in g, $C_1 = -1.17$, $C_2 = 0.587$, $C_3 = -1.26$, $C_4 = 2.13$, $C_5 = 0.25$ and $\sigma = 0.543$.

- Use data from Italy (6 records, 6 earthquakes), USA (18 records, 8 earthquakes), Greece (13 records, 9 earthquakes), Iran (3 records, 3 earthquakes), Pakistan (3 records, 1 earthquake), Yugoslavia (3 records, 1 earthquake), USSR (1 record, 1 earthquake), Nicaragua (1 record, 1 earthquake), India (1 record, 1 earthquake) and Atlantic Ocean (1 record, 1 earthquake).
- Develop for use in UK.

2.26 Schenk (1982)

- Ground motion model is:

$$\log A_{\mathrm{mean}} = aM - b\log R + c$$

where A_{mean} is in cms^{-2}, $a = 1.1143$, $b = 1.576$ and $c = 2.371$ (σ not given).

- Fits equation by eye because least squares method is often strictly dependent on marginal observations, particularly for little pronounced dependence.

2.27 Joyner & Fumal (1984)

- Ground motion model is:

$$\begin{aligned} \log y &= c_0 + c_1(\mathbf{M}-6) + c_2(\mathbf{M}-6)^2 + c_3\log r + c_4 r + S \\ \text{where } r &= (d^2+h^2)^{\frac{1}{2}} \\ \text{and: } S &= \begin{cases} 0 & \text{for rock site} \\ c_6 \log \frac{V}{V_0} & \text{for soil site} \end{cases} \end{aligned}$$

where y is in g, coefficients c_0 to c_4, h and σ are from Joyner & Boore (1981) and c_6 and V_0 are not significant at 90% level so do not report them.

- Use data of Joyner & Boore (1981).
- Continuous site classification for soil sites in terms of shear-wave velocity, V, to depth of one quarter wavelength of waves of period of concern. V measured down to depths of at least $30\,\mathrm{m}$ and then extrapolated using geological data. V known for 33 stations.
- Soil amplification factor based on energy conservation along ray tubes, which is a body wave argument and may not hold for long periods for which surface waves could be important. Does not predict resonance effects.
- Regress residuals, R_{ij}, w.r.t. motion predicted for rock sites on $\log R_{ij} = P_i + c_6 V_j$, where j corresponds to jth station and i to ith earthquake. Decouples site effects variation from earthquake-to-earthquake variation. Find unique intercept by requiring average site effect term calculated using shear-wave velocity to be same as that calculated using rock/soil classification.
- No significant, at 90%, correlation between residuals and V for PGA.
- Repeat regression on residuals using V and depth to underlying rock (defined as either shear-wave velocity $> 750\,\mathrm{ms}^{-1}$ or $> 1500\,\mathrm{ms}^{-1}$). Find no correlation.

2.28 *Kawashima* et al. *(1984) & Kawashima* et al. *(1986)*

- Ground motion model is:

$$X(M, \Delta, \mathrm{GC}_i) = a(\mathrm{GC}_i) 10^{b(\mathrm{GC}_i)M} (\Delta + 30)^c$$

where $X(M, \Delta, \mathrm{GC}_i)$ is in gal, $c = -1.218$, for group 1 sites $a(\mathrm{GC}_1) = 987.4$, $b(\mathrm{GC}_1) = 0.216$ and $\sigma = 0.216$, for group 2 sites $a(\mathrm{GC}_2) = 232.5$, $b(\mathrm{GC}_2) = 0.313$ and $\sigma = 0.224$ and for group 3 sites $a(\mathrm{GC}_3) = 403.8$, $b(\mathrm{GC}_3) = 0.265$ and $\sigma = 0.197$.

- Use three site categories:

Group 1 Tertiary or older rock (defined as bedrock) or diluvium with $H < 10\,\mathrm{m}$ or fundamental period $T_G < 0.2\,\mathrm{s}$.

Group 2 Diluvium with $H \geq 10\,\mathrm{m}$, alluvium with $H < 10\,\mathrm{m}$ or alluvium with $H < 25\,\mathrm{m}$ including soft layer with thickness $< 5\,\mathrm{m}$ or fundamental period $0.2 < T_G < 0.6\,\mathrm{s}$.

Group 3 Other than above, normally soft alluvium or reclaimed land.

- Only includes free-field records with $M_{\mathrm{JMA}} \geq 5.0$ and focal depths $D_p < 60\,\mathrm{km}$. Excludes records from structures with first floor or basement.

- Records instrument corrected, because Japanese instruments substantially suppress high frequencies, considering accuracy of digitization for frequencies between $\frac{1}{3}$ and $12\,\mathrm{Hz}$.

- Note that M_{JMA} and Δ not necessarily most suitable parameters to represent magnitude and distance but only ones for all records in set.

- Note lack of near-field data for large magnitude earthquakes, approximately $\frac{3}{4}$ of records from $M_{\mathrm{JMA}} < 7.0$.

- Use $30\,\mathrm{km}$ in distance dependence term because focal depth of earthquakes with magnitudes between 7.5 and 8.0 are between 30 and $100\,\mathrm{km}$ so 30 is approximately half the fault length.

- Try equation: $\log X = f_1 + f_2 M + f_3 \log(\Delta + 30) + f_4 D_p + f_5 M \log(\Delta + 30) + f_6 M D_p + f_7 D_p \log(\Delta + 30) + f_8 M^2 + f_9 \{\log(\Delta + 30)\}^2 + f_{10} D_p^2$ where f_i are coefficients to be found considering each soil category separately. Apply multiple regression analysis to 36 combinations of retained coefficients, f_i, and compute multiple correlation coefficient, R, and adjusted multiple correlation coefficient, R^*. Find that inclusion of more than three coefficients does not give significant increase in R^*, and can lead to unrealistic results. Conclude due to insufficient data.

- Consider a, b and c dependent and independent of soil type and examine correlation coefficient, R, and adjusted correlation coefficient, R^*. Find that c is not strongly dependent on soil type.

- Find match between normal distribution and histograms of residuals.

2.29 *McCann Jr. & Echezwia (1984)*

- Four ground motion models:

$$\begin{aligned}
\log_{10} Y &= a + bM + d \log_{10}[(R^2 + h^2)^{1/2}] \qquad \text{Model I} \\
\log_{10} Y &= a + bM + d \log_{10}[R + c_1 \exp(c_2 M)] \qquad \text{Model II} \\
\log_{10} Y &= a + bM + d \log_{10}\left[\frac{c_1}{R^2} + \frac{c_2}{R}\right] + eR \qquad \text{Model III} \\
\log_{10} Y &= a + bM + d \log_{10}[R + 25] \qquad \text{Model IV}
\end{aligned}$$

where Y is in g, for model I $a = -1.320$, $b = 0.262$, $d = -0.913$, $h = 3.852$ and $\sigma = 0.158$, for model II $a = -1.115$, $b = 0.341$, $c_1 = 1.000$, $c_2 = 0.333$, $d = -1.270$ and $\sigma = 0.154$, for model III $a = -2.000$, $b = 0.270$, $c_1 = 0.968$, $c_2 = 0.312$, $d = 0.160$, $e = -0.0105$ and $\sigma = 0.175$ and for model IV $a = 1.009$, $b = 0.222$, $d = -1.915$ and $\sigma = 0.174$.

- Note 25 in Model IV should not be assumed but should be found by regression.
- Note tectonics and travel paths may be different between N. American and foreign records but consider additional information in near field more relevant.
- Selection procedure composite of Campbell (1981) and Joyner & Boore (1981). Exclude data from buildings with more than two storeys.
- Weighted least squares, based on distance, applied to control influence of well recorded events (such as San Fernando and Imperial Valley). Similar to Campbell (1981)
- Test assumption that logarithm of residuals are normally distributed. Cannot disprove assumption.
- Variability between models not more than $\pm 20\%$ at distances $> 10\,\mathrm{km}$ but for distances $< 1\,\mathrm{km}$ up to $\pm 50\%$.

2.30 *Schenk (1984)*

- Ground motion model is:
$$\log A_{\mathrm{mean}} = aM - b \log R + c$$
where A_{mean} is in cms^{-2}, $a = 0.37$, $b = 1.58$ and $c = 2.35$ (σ not given).
- Considers two site conditions but does not model:
  1. Solid
  2. Soft
- Fits equation by eye.
- States applicable approximately for: $R_{\mathrm{lower}} \leq R \leq R_{\mathrm{upper}}$ where $\log R_{\mathrm{lower}} \doteq 0.1M + 0.5$ and $\log R_{\mathrm{upper}} \doteq 0.35M + 0.4$, due to distribution of data.
- Notes great variability in recorded ground motions up to $R = 30\,\mathrm{km}$ due to great influence of different site conditions.
- Notes for $M \leq 4$ source can be assumed spherical but for $M > 4$ elongated (extended) shape of focus should be taken into account.

2.31 *Xu* et al. *(1984)*

- Ground motion model is:
$$\mathrm{PGA} = a_1 \exp(a_2 M)(R + a_3)^{-a_4}$$
where PGA is in g, $a_1 = 0.1548$, $a_2 = 0.5442$, $a_3 = 8$ and $a_4 = 1.002$ (σ not given).
- All records from aftershocks of 1975 Haicheng earthquake and from 1976 Tangshan earthquake and aftershocks.
- Most records from earthquakes with magnitude less than 5.8 and from distances $< 30\,\mathrm{km}$.
- Exclude records with PGA $< 0.5\,\mathrm{ms}^{-2}$ to avoid too much contribution from far field.

- Due to small number of records simple regression technique justified.
- States valid for $4 \leq M \leq 6.5$ and $R \leq 100\,\mathrm{km}$.
- Also use 158 records from western N. America to see whether significantly different than N. Chinese data. Derive equations using both western N. American and N. Chinese data and just western N. American data and find that predicted PGAs are similar, within uncertainty.
- Insufficient data to find physically realistic anelastic term.

2.32 *Kawashima* et al. *(1985)*

- Use very similar data to Kawashima *et al.* (1984); do not use some records because missing due to recording and digitising processes. Use equation and method (although do not check all 36 combinations of forms of equation) used by Kawashima *et al.* (1984), see section 2.28.
- $X(M, \Delta, \mathrm{GC}_i)$ is in gal. Coefficients are: $c = -1.190$ and for ground group 1 $a = 117.0$ and $b = 0.268$ and for ground group 2 $a = 88.19$ and $b = 0.297$ and for group ground 3 $a = 13.49$ and $b = 0.402$ with $\sigma = 0.253$.

2.33 *Peng* et al. *(1985)*

- Ground motion model is:
$$\log_{10} a = A + BM + C \log_{10} R + DR$$
where a is in cms^{-2}, for N.E. China $A = -0.474$, $B = 0.613$, $C = -0.873$ and $D = -0.00206$ (σ not given) and for S.W. China $A = 0.437$, $B = 0.454$, $C = -0.739$ and $D = -0.00279$ (σ not given).
- Consider two site conditions for NE records but do not model:
  1. Rock: 28 records.
  2. Soil: 45 records.
- Consider all records to be free-field.
- Note that Chinese surface-wave magnitude, M, is different than M_s and may differ by 0.5 or more. Use m_b or M_s and find larger residuals.
- Most records from $M \leq 5.8$.
- Note isoseismals are not elongated for these earthquakes so use of another distance measure will not change results by much.
- Also derives equation for SW China ($3.7 \leq M \leq 7.2$, $6.0 \leq R \leq 428.0\,\mathrm{km}$ all but one record $\leq 106.0\,\mathrm{km}$, 36 records from 23 earthquakes) and note difference between results from NE China although use less data.
- Note that some scatter may be due to radiation pattern.
- Note that data is from limited distance range so need more data to confirm results.

2.34 PML (1985)

- Ground motion model is:

$$\ln(a) = C_1 + C_2 M + C_3 \ln[R + C_4 \exp(C_5 M)] + C_6 F$$

where a is in g, $C_1 = -0.855$, $C_2 = 0.46$, $C_3 = -1.27$, $C_4 = 0.73$, $C_5 = 0.35$, $C_6 = 0.22$ and $\sigma = 0.49$.

- Use data from Italy (47 records, 9 earthquakes), USA (128 records, 18 earthquakes), Greece (11 records, 8 earthquakes), Iran (2 records, 2 earthquakes), Yugoslavia (7 records, 2 earthquake), Nicaragua (1 record, 1 earthquake), New Zealand (3 records, 3 earthquakes), China (2 records, 2 earthquakes) and Canada (2 records, 1 earthquake).
- Develop for use in UK.
- Select earthquakes with $M_s < 7$ and $R \leq 40\,\mathrm{km}$.
- Focal depths $< 40\,\mathrm{km}$.
- Use two source mechanism categories (40 records have no source mechanism given):

$F = 0$ Strike-slip and normal, 85 records.

$F = 1$ Thrust, 78 records.

- Also derive equation not considering source mechanism, i.e. $C_6 = 0$.

2.35 McCue (1986)

- Ground motion model is:

$$A = a_1(\mathrm{e}^{a_2 M_L})(d_h)^{a_3}$$

where A is in g, $a_1 = 0.00205$, $a_2 = 1.72$ and $a_3 = -1.58$ (σ not given).

2.36 C.B. Crouse (1987) reported in Joyner & Boore (1988)

- Ground motion model is:

$$\ln y = a + bM_s + cM_s^2 + d\ln(r+1) + kr$$

where y is in gal, $a = 2.48456$, $b = 0.73377$, $c = -0.01509$, $d = -0.50558$, $k = -0.00935$ and $\sigma = 0.58082$.

- Records from deep soil sites (generally greater than $60\,\mathrm{m}$ in thickness).
- Data from shallow crustal earthquakes.

2.37 Sabetta & Pugliese (1987)

- Ground motion model is:

$$\log y = \boldsymbol{a} + \boldsymbol{b}M - \log(R^2 + \boldsymbol{h}^2)^{1/2} + \boldsymbol{e}S$$

where y is in g and for distance to surface projection of fault $\boldsymbol{a} = -1.562$, $\boldsymbol{b} = 0.306$, $\boldsymbol{e} = 0.169$, $\boldsymbol{h} = 5.8$ and $\sigma = 0.173$.

- Use two site categories:

$S = 0$ Stiff and deep soil: limestone, sandstone, siltstone, marl, shale and conglomerates ($V_s > 800\,\mathrm{ms}^{-1}$) or depth of soil, H, $> 20\,\mathrm{m}$, 74 records.

$S = 1$ Shallow soil: depth of soil, H, $5 \leq H \leq 20\,\mathrm{m}$, 21 records.

- Select records which satisfy these criteria:

  1. Reliable identification of the triggering earthquake.
  2. Magnitude greater than 4.5 recorded by at least two stations.
  3. Epicentres determined with accuracy of $5\,\mathrm{km}$ or less.
  4. Magnitudes accurate to within 0.3 units.
  5. Accelerograms from free-field. Most are from small electric transformer cabins, 4 from one- or two-storey buildings with basements and 5 from near abutments of dams.

- Depths between 5.0 and $16.0\,\mathrm{km}$ with mean $8.5\,\mathrm{km}$.

- Focal mechanisms are: normal and oblique (7 earthquakes, 48 records), thrust (9 earthquakes, 43 records) and strike-slip (1 earthquake, 4 records).

- Notes lack of records at short distances from large earthquakes.

- Records baseline-, instrument-corrected and filtered with cutoff frequencies determined by visual inspection in order to maximise signal to noise ratio within band. Cutoff frequencies ranged from 0.2 to $0.4\,\mathrm{Hz}$ and from 25 to $35\,\mathrm{Hz}$. This correction routine thought to provide reliable estimates of PGA so uncorrected PGA do not need to be used.

- For well separated multiple shocks, to which magnitude and focal parameters refer, use only first shock.

- Magnitude scale assures a linear relationship between logarithm of PGA and magnitude and avoids saturation effects of M_L.

- Distance to surface projection of fault rupture thought to be a more physically consistent definition of distance for earthquakes having extensive rupture zones and is easier to predict for future earthquakes. Also reduces correlation between magnitude and distance.

- Use Exploratory Data Analysis using the ACE procedure to find transformation functions of distance, magnitude and PGA.

- Include anelastic attenuation term but it is positive and not significant.

- Include magnitude dependent $\boldsymbol{h}$ equal to $\boldsymbol{h_1}\exp(\boldsymbol{h_2}M)$ but find $\boldsymbol{h_2}$ not significantly different than zero. Note distribution of data makes test not definitive.

- Find geometric attenuation coefficient, $\boldsymbol{c}$, is close to -1 and highly correlated with $\boldsymbol{h}$ so constrain to -1 so less coefficients to estimate.

- Consider deep soil sites as separate category but find difference between them and stiff sites is not significant.

- Also use two-stage method but coefficients and variance did not change significantly with respect to those obtained using one-stage method, due to uniform distribution of recordings among earthquakes.

- Find no significant trends in residuals, at 99% level and also no support for magnitude dependent shape for attenuation curves.

- Exclude records from different seismotectonic and geological regions and repeat analysis. Find that predicted PGA are similar.
- Plot residuals from records at distances $15\,\mathrm{km}$ or less against magnitude; find no support for magnitude dependence of residuals.
- Note some records are affected by strong azimuthal effects, but do not model them because they require more coefficients to be estimated, direction of azimuthal effect different from region to region and azimuthal effects have not been used in other relationships.

2.38 *K. Sadigh (1987) reported in Joyner & Boore (1988)*

- Ground motion model is:

$$\ln y = a + b\mathbf{M} + c_1(8.5 - \mathbf{M})^{c_2} + d\ln[r + h_1\exp(h_2\mathbf{M})]$$

  where y is in g. For strike-slip earthquakes: $b = 1.1$, $c_1 = 0$, $c_2 = 2.5$, for PGA at soil sites $a = -2.611$ and $d = -1.75$, for $\mathbf{M} < 6.5$ $h_1 = 0.8217$, $h_2 = 0.4814$ and for $\mathbf{M} \geq 6.5$ $h_1 = 0.3157$ and $h_2 = 0.6286$, for PGA at rock sites $a = -1.406$ and $d = -2.05$, for $\mathbf{M} < 6.5$ $h_1 = 1.353$ and $h_2 = 0.406$ and for $\mathbf{M} \geq 6.5$ $h_1 = 0.579$ and $h_2 = 0.537$. For reverse-slip increase predicted values by 20%. For $\mathbf{M} < 6.5$ $\sigma = 1.26 - 0.14\mathbf{M}$ and for $\mathbf{M} \geq 6.5$ $\sigma = 0.35$.
- Uses two site categories:
  1. Soil
  2. Rock
- Use two source mechanism categories:
  1. Strike-slip
  2. Reverse-slip
- Supplement data with significant recordings of earthquakes with focal depths $< 20\,\mathrm{km}$ from other parts of world.
- Different equations for $\mathbf{M} < 6.5$ and $\mathbf{M} \geq 6.5$.

2.39 *Singh* et al. *(1987)*

- Ground motion model is:

$$\log y_{\max} = \alpha M_s - c\log R + \beta$$

  where $y_{\max}$ is in cms^{-2}, $\alpha = 0.429$, $c = 2.976$, $\beta = 5.396$ and $\sigma = 0.15$.

  More complicated functional form unwarranted due to limited distance range.
- Depths between 15 and $20\,\mathrm{km}$.
- Only use data from a single firm site (Ciudad Universitaria), on a surface layer of lava flow or volcanic tuff.
- Only records from coastal earthquakes.
- Residuals plotted against distance, no trends seen.
- Give amplification factor for lake bed sites (25 to $80\,\mathrm{m}$ deposit of highly compressible, high water content clay underlain by resistant sands), but note based on only a few sites so not likely to be representative of entire lake bed.

2.40 *Algermissen* et al. *(1988)*

- Ground motion model is:
$$\ln(A) = a_1 + a_2 M_s + a_3 \ln(R) + a_4 R$$
where A is in g, $a_1 = -1.987$, $a_2 = 0.604$, $a_3 = -0.9082$, $a_4 = -0.00385$ and $\sigma = 0.68$.

2.41 *Annaka & Nozawa (1988)*

- Ground motion model is:
$$\log A = C_m M + C_h H - C_d \log(R + A \exp BM) + C_o$$
where A is in cms^{-2}, A and B so PGA becomes independent of magnitude at fault rupture, H is depth of point on fault plane when R becomes closest distance to fault plane, $C_m = 0.627$, $C_h = 0.00671$, $C_d = 2.212$, $C_o = 1.711$ and $\sigma = 0.211$.
- Focal depths $< 100\,\mathrm{km}$.
- Convert records from sites with $V_s < 300\,\mathrm{ms}^{-1}$ into records from sites with $V_s > 300\,\mathrm{ms}^{-1}$ using 1-D wave propagation theory.
- Introduce term $C_h H$ because it raises multiple correlation coefficient for PGA.
- Note equations apply for site where $300 \leq V_s \leq 600\,\mathrm{ms}^{-1}$.

2.42 *K.W. Campbell (1988) reported in Joyner & Boore (1988)*

- Ground motion model is:
$$\begin{aligned} \ln y &= a + bM + d \ln[r + h_1 \exp(h_2 M)] + s \\ \text{where } s &= e_1 K_1 + e_2 K_2 + e_3 K_3 + e_4 K_4 + e_5 K_5 + e_6 (K_4 + K_5) \tanh(e_7 r) \end{aligned}$$
where y is in g, $a = -2.817$, $b = 0.702$, $d = -1.20$, $h_1 = 0.0921$, $h_2 = 0.584$, $e_1 = 0.32$, $e_2 = 0.52$, $e_3 = 0.41$, $e_4 = -0.85$, $e_5 = -1.14$, $e_6 = 0.87$, $e_7 = 0.068$ and $\sigma = 0.30$.
- Uses two site categories:

$K_3 = 1$ Soils $\leq 10\,\mathrm{m}$ deep.
$K_3 = 0$ Other.

- Uses three embedment categories:

$K_4 = 1$, $K_5 = 0$ Basements of buildings 3–9 storeys.
$K_5 = 1$, $K_4 = 0$ Basements of buildings ≥ 10 storeys.
$K_4 = 0$, $K_5 = 0$ Other.

- Selects data using these criteria:
  1. Largest horizontal component of peak acceleration was $\geq 0.02\,\mathrm{g}$ [$\geq 0.2\,\mathrm{ms}^{-2}$].
  2. Accelerograph triggered early enough to record strongest phase of shaking.
  3. Magnitude of earthquake was ≥ 5.0.
  4. Closest distance to seismogenic rupture was < 30 or $< 50\,\mathrm{km}$, depending on whether magnitude of earthquake was < 6.25 or > 6.25.

5. Shallowest extent of seismogenic rupture was $\leq 25\,\mathrm{km}$.
6. Recording site located on unconsolidated deposits.

- Excludes records from abutments or toes of dams.
- Derives two equations: unconstrained (coefficients given above) and constrained which includes a anelastic decay term kr which allows equation to be used for predictions outside near-source zone (assumes $k = -0.0059$ for regression, a value appropriate for region of interest should be chosen).
- Uses two source mechanism categories:

$K_1 = 0$ Strike-slip.
$K_1 = 1$ Reverse.

- Uses two directivity categories:

$K_2 = 1$ Rupture toward site.
$K_2 = 0$ Other.

2.43 *Fukushima* et al. *(1988) & Fukushima & Tanaka (1990)*

- Ground motion model is:

$$\log A = aM - \log(R + c10^{aM}) - bR + d$$

where A is in cms^{-2}, $a = 0.41$, $b = 0.0034$, $c = 0.032$, $d = 1.30$ and $\sigma = 0.21$.

- Use four site categories for some Japanese stations (302 Japanese records not classified):

1. Rock: 41 records
2. Hard: ground above Tertiary period or thickness of diluvial deposit above bedrock $< 10\,\mathrm{m}$, 44 records.
3. Medium: thickness of diluvial deposit above bedrock $> 10\,\mathrm{m}$, or thickness of alluvial deposit above bedrock $< 10\,\mathrm{m}$, or thickness of alluvial deposit $< 25\,\mathrm{m}$ and thickness of soft deposit is $< 5\,\mathrm{m}$, 66 records.
4. Soft soil: other soft ground such as reclaimed land, 33 records.

- Use 1100 mean PGA values from 43 Japanese earthquakes ($6.0 \leq M_{\mathrm{JMA}} \leq 7.9$, focal depths $\leq 30\,\mathrm{km}$) recorded at many stations to investigate one and two-stage methods. Fits $\log A = c - b \log X$ (where X is hypocentral distance) for each earthquake and computes mean of b, $\bar{b}$. Also fits $\log A = aM - b^* \log X + c$ using one-stage method. Find that $\bar{b} > b^*$ and shows that this is because magnitude and distance are strongly correlated (0.53) in data set. Find two-stage method of Joyner & Boore (1981) very effective to overcome this correlation and use it to find similar distance coefficient to $\bar{b}$. Find similar effect of correlation on distance coefficient for two other models: $\log A = aM - b \log(\Delta + 30) + c$ and $\log A = aM - \log X - bX + c$, where Δ is epicentral distance.
- Japanese data selection criteria: focal depth $< 30\,\mathrm{km}$, $M_{\mathrm{JMA}} > 5.0$ and predicted PGA $\geq 0.1\,\mathrm{ms}^{-2}$. US data selection criteria: $d_r \leq 50\,\mathrm{km}$, use data from Campbell (1981).
- Because a affects distance and magnitude dependence, which are calculated during first and second steps respectively use an iterative technique to find coefficients. Allow different magnitude scaling for US and Japanese data.
- For Japanese data apply station corrections before last step in iteration to convert PGAs from different soil conditions to standard soil condition using residuals from analysis.

- Two simple numerical experiments performed. Firstly a two sets of artificial acceleration data was generated using random numbers based on attenuation relations, one with high distance decay and which contains data for short distance and one with lower distance decay, higher constant and no short distance data. Find that the overall equation from regression analysis has a smaller distance decay coefficient than individual coefficients for each line. Secondly find the same result for the magnitude dependent coefficient based on similar artificial data.
- Exclude Japanese data observed at long distances where average acceleration level was predicted (by using an attenuation relation derived for the Japanese data) to be less than the trigger level (assume to be about $0.05\,\mathrm{ms}^{-2}$) plus one standard deviation (assume to be 0.3), i.e. $0.1\,\mathrm{ms}^{-2}$, to avoid biasing results and giving a lower attenuation rate.
- Use the Japanese data and same functional form and method of Joyner & Boore (1981) to find an attenuation relation; find the anelastic coefficient is similar so conclude attenuation rate for Japan is almost equal to W. USA.
- Find difference in constant, d, between Japanese and W. USA PGA values.
- Plot residuals against distance and magnitude and find no bias or singularity.

2.44 Gaull (1988)

- Ground motion model is:

$$\log \mathrm{PGA} = [(a_1 \log R + a_2)/a_3](M_L - a_4) - a_5 \log R - a_6 R + a_7$$

where PGA is in ms^{-2}, $a_1 = 5$, $a_2 = 3$, $a_3 = 20$, $a_4 = 6$, $a_5 = 0.77$, $a_6 = 0.0045$ and $a_7 = 1.2$ (σ not given).

- Considers three site categories but does not model:
  1. Rock: 6 records
  2. Alluvium: 5 records
  3. Average site: 10 records
- Most records from earthquakes with magnitudes about 3 and most from distances below about $20\,\mathrm{km}$.
- Band pass filter records to get PGA associated with waves with periods between 0.1 and $0.5\,\mathrm{s}$ because high frequency PGA from uncorrected records not of engineering significance.
- Adds 4 near source ($5 \leq R \leq 10\,\mathrm{km}$) records from US, Indian and New Zealand earthquakes with magnitudes between 6.3 and 6.7 to supplement high magnitude range.
- Add some PGA points estimated from intensities associated with 14/10/1968 $M_L = 6.9$ Meckering earthquake in Western Australia.
- Plot 6 records from one well recorded event with $M_L = 4.5$ and fit an attenuation curve of form $\log \mathrm{PGA} = b_1 - b_2 \log R - b_3 R$ by eye. Plot PGA of all records with $2 \leq R \leq 20\,\mathrm{km}$ against magnitude, fit an equation by eye. Use these two curves to normalise all PGA values to $M_L = 4.5$ and $R = 5\,\mathrm{km}$ from which estimates attenuation relation.

2.45 Joyner & Boore (1988)

- Use same data and very similar method to Joyner & Boore (1981), see Section 2.23, and find: $\beta = 0.23$, $b = -0.0027$, $h = 8.0$ and $\sigma = 0.28$, for randomly oriented component $\alpha = 0.43$ and for larger component $\alpha = 0.49$.

2.46 *McCue* et al. *(1988)*

- Ground motion model is:
$$A = a(\exp(bM)) \left(\frac{R}{R_0} + c \right)^{-d}$$
where A is in g, $\ln a = -5.75$, $b = 1.72$, $c = 0$, $d = 1.69$ and $R_0 = 1$ (σ not given).
- Few records from free-field, most are in dams or special structures.,
- Because only 62 records, set $R_0 = 1$ and $c = 0$.
- Most records from earthquakes with M_L between 1.5 and 2.0.
- Maximum PGA in set $3.05\,\mathrm{ms}^{-2}$.
- Nonuniform distribution of focal distances. One quarter of records from same hypocentral distance. Therefore plot PGA of these records against magnitude ($1.2 \lesssim M_L \lesssim 4.3$ most less than 2.1) to find b. Then plot $bM - \ln A$ against $\ln(R/R_0)$ for all records to find a and d.
- Notes limited data.

2.47 *Petrovski & Marcellini (1988)*

- Ground motion model is:
$$\ln(a) = a_1 + a_2 M + a_3 \ln(R + 20)$$
where a is in cms^{-2}, $a_1 = 6.48$, $a_2 = 0.544$, $a_3 = -1.33$ and $\sigma = 0.67$.
- Data from Greece, Italy and former Yugoslavia.
- Maximum focal depth is $40\,\mathrm{km}$.

2.48 *Tong & Katayama (1988)*

- Ground motion model is:
$$\log \bar{A} = \alpha M - \beta \log(\Delta + 10) + \gamma T + \delta$$
where $\bar{A}$ is in gal, T is predominant period of site, $\alpha = 0.509$, $\beta = 2.32$, $\gamma = 0.039$ and $\delta = 2.33$ (σ not given).
- Correlation coefficient between magnitude and distance is 0.84, so magnitude and distance cannot be considered independent, so attenuation rate, β, is difficult to find.
- First step fit $\log \bar{A} = -\beta_i \log(\Delta + 10) + \delta_i$ to each earthquake. Define reliability parameter, $\psi_i = N_i R_i^2$, where N_i is degrees of freedom for i earthquake and R_i is correlation coefficient. Plot ψ_i against β_i and find attenuation rate scattered, between -6 and 9, for $\psi_i < 1$ (Group B) and for $\psi_1 > 1$ attenuation rate converges (Group U).
- Group B includes earthquakes with focal depths $> 388\,\mathrm{km}$, earthquakes with small magnitudes and records from distances $\approx 100\,\mathrm{km}$, earthquakes with records from great distances where spread of distances is small, earthquakes recorded by only 3 stations and earthquakes with abnormal records. Exclude these records.
- Apply multiple regression on Group U to find α, β, γ and δ simultaneously. Also fix $\beta = \sum \psi_i \beta_i / \sum \psi_i$ and find α, γ and δ. Find different coefficients but similar correlation coefficient. Conclude due to strong correlation between M and Δ so many regression planes exist with same correlation coefficient.

- Perform Principal Component Analysis (PCA) on $\log A$, M, $\log(\Delta + 10)$, T and $\log \bar{A}/A$ and find that equation found by fixing β is not affected by ill-effect of correlation between M and Δ.
- Omit T from regression and find little effect in estimation.

2.49 *Yamabe & Kanai (1988)*

- Ground motion model is:

$$\begin{aligned} \log_{10} a &= \beta - \nu \log_{10} x \\ \text{where } \beta &= b_1 + b_2 M \\ \text{and: } \nu &= c_1 + c_2 M \end{aligned}$$

  where a is in gal, $b_1 = -3.64$, $b_2 = 1.29$, $c_1 = -0.99$ and $c_2 = 0.38$ (σ not given).

- Focal depths between 0 and $130\,\mathrm{km}$.
- Regress recorded PGA of each earthquake, i, on $\log_{10} a = \beta_i - \nu_i \log_{10} x$, to find β_i and ν_i. Then find b_1 and b_2 from $\beta = b_1 + b_2 M$ and c_1 and c_2 from $\nu = c_1 + c_2 M$.
- Also consider $\nu = d_1 \beta$.
- Find β and ν from 6 earthquakes (magnitudes between 5.4 and 6.1) from Tokyo-Yokohama area are much higher than for other earthquakes, so ignore them. Conclude that this is due to effect of buildings on ground motion.

2.50 *Youngs* et al. *(1988)*

- Ground motion model is:

$$\ln(a_{\max}) = C_1 + C_2 M_w - C_3 \ln[R + C_4 \exp(C_5 M_w)] + B Z_t$$

  where $a_{\max}$ is in g, $C_1 = 19.16$, $C_2 = 1.045$, $C_3 = -4.738$, $C_4 = 205.5$, $C_5 = 0.0968$, $B = 0.54$ and $\sigma = 1.55 - 0.125 M_w$.

- Use only rock records to derive equation but use some (389 records) for other parts of study. Classification using published shear-wave velocities for some sites.
- Exclude data from very soft lake deposits such as those in Mexico City because may represent site with special amplification characteristics.
- Data from subduction zones of Alaska, Chile, Peru, Japan, Mexico and Solomon Islands.
- Use two basic types of earthquake:

$Z_t = 0$ Interface earthquakes: low angle, thrust faulting shocks occurring on plate interfaces.

$Z_t = 1$ Intraslab earthquakes: high angle, predominately normal faulting shocks occurring within down going plate.

  Classification by focal mechanisms or focal depths (consider earthquakes with depths $> 50\,\mathrm{km}$ to be intraslab). Note that possible misclassification of some intraslab shocks as interface events because intraslab earthquakes do occur at depths $< 50\,\mathrm{km}$.

- Plots PGA from different magnitude earthquakes against distance; find near-field distance saturation.
- Originally include anelastic decay term $-C_6 R$ but C_6 was negative (and hence nonphysical) so remove.

- Plot residuals from original PGA equation (using rock and soil data) against M_w and R; find no trend with distance but reduction in variance with increasing M_w. Assume standard deviation is a linear function of M_w and find coefficients using combined rock and soil data (because differences in variance estimation from rock and soil are not significant).

- Use derived equation connecting standard deviation and M_w for weighted (weights inversely proportional to variance defined by equation) nonlinear regression in all analyses.

- Plot residuals from original PGA equation; find that hypothesis that coefficients of equations for interface and intraslab earthquakes are the same can be rejected (using likelihood ratio test for nonlinear regression models) at 0.05 percentile level for both soil and rock. Try including a term proportional to depth of rupture into equation (because intraslab deeper than interface events) but find no significant reduction in standard error. Introduce BZ_t term into equation; find B is significant at 0.05 percentile level. Try including rupture type dependence into other coefficients but produces no further decrease in variance so reject.

- Use only data from sites with multiple recordings of both interface and intraslab earthquakes and include dummy variables, one for each site, to remove differences due to systematic site effects. Fix C_1 to C_5 to values from entire set and find individual site terms and B; find B is very similar to that from unconstrained regression.

- Examine residuals for evidence of systematic differences between ground motion from different subduction zones; find no statistically significant differences in PGA among different subduction zones.

- Use geometric mean of two horizontal components to remove effect of component-to-component correlations that affect validity of statistical tests assuming individual components of motion represent independent measurements of ground motion. Results indicate no significant difference between estimates of variance about median relationships obtained using geometric mean and using both components as independent data points.

- Extend to $M_w > 8$ using finite difference simulations of faulting and wave propagation modelled using ray theory. Method and results not reported here.

2.51 Abrahamson & Litehiser (1989)

- Ground motion model is:

$$\log_{10} a = \alpha + \beta M - \bar{c} \log_{10}[r + \exp(h_2 M)] + F\phi + Ebr$$

  where $F = 1$ for reverse or reverse oblique events and 0 otherwise and $E = 1$ for interplate events and 0 otherwise, a is in g, for horizontal PGA $\alpha = -0.62$, $\beta = 0.177$, $\bar{c} = 0.982$, $h_2 = 0.284$, $\phi = 0.132$, $b = -0.0008$ and $\sigma = 0.277$ and for vertical PGA $\alpha = -1.15$, $\beta = 0.245$, $\bar{c} = 1.096$, $h_2 = 0.256$, $\phi = 0.096$, $b = -0.0011$ and $\sigma = 0.296$.

- Consider three site classifications, based on Joyner & Boore (1981):

  1. Rock: corresponds to C, D & E categories of Campbell (1981), 159 records.
  2. Soil: corresponds to A,B & F categories of Campbell (1981), 324 records.
  3. Unclassified: 102 records.

  Use to examine possible dependence in residuals not in regression because of many unclassified stations.

- Data based on Campbell (1981).

- Fault mechanisms are: strike-slip (256 records from 28 earthquakes), normal (14 records from 7 earthquakes), normal oblique (42 records from 12 earthquakes), reverse (224 records from 21 earthquakes) and reverse oblique (49 records from 8 earthquakes). Grouped into normal-strike-slip and reverse events. Weakly correlated with magnitude (0.23), distance (0.18) and tectonic environment (0.03).
- Tectonic environments are: interplate (555 records from 66 earthquakes) and intraplate (30 records from 10 earthquakes) measurements. Weakly correlated with magnitude (-0.26), distance (-0.17) and fault mechanism (0.03).
- Depths less than $25\,\mathrm{km}$.
- Use array average (37 instruments are in array) from 10 earthquakes recorded at SMART 1 array in Taiwan.
- Most records from distances less than $100\,\mathrm{km}$ and magnitude distribution is reasonably uniform but correlation between magnitude and distance of 0.52.
- Try two-stage technique and model (modified to include fault mechanism and tectonic environment parameters) of Joyner & Boore (1981), find inadmissable positive anelastic coefficient, so do not use it.
- Use a hybrid regression technique based on Joyner & Boore (1981) and Campbell (1981). A method to cope with highly correlated magnitude and distance is required. First step: fit data to $f_2(r) = \bar{c}\log_{10}(r+h)$ and have separate constants for each earthquake (like in two-stage method of Joyner & Boore (1981)). Next holding $\bar{c}$ constant find α, β, b and h_2 from fitting $h = \exp(h_2 M)$. Weighting based on Campbell (1981) is used.
- Form of h chosen using nonparametric function, $H(M)$, which partitions earthquakes into 0.5 unit bins. Plot $H(M)$ against magnitude. Find that $H(M) = h_1\exp(h_2 M)$ is controlled by Mexico (19/9/1985) earthquake and h_1 and h_2 are highly correlated, 0.99, although does given lower total variance. Choose $H(M) = \exp(h_2 M)$ because Mexico earthquake does not control fit and all parameters are well-determined, magnitude dependent h significant at 90%.
- Try removing records from single-recorded earthquakes and from shallow or soft soil but effect on predictions and variance small ($< 10\%$).
- Plot weighted residuals within $10\,\mathrm{km}$ no significant, at 90%, trends are present.
- Find no significant effects on vertical PGA due to site classification.

2.52 Campbell (1989)

- Ground motion model is:
$$\ln \mathrm{PHA} = a + bM_L - 1.0\ln[R + c_1]$$
where PHA is in g, $a = -2.501$, $b = 0.623$, $c_1 = 7.28$ and $\sigma = 0.506$.
- Selects records from deep soil ($> 10\,\mathrm{m}$). Excludes data from shallow soil ($\leq 10\,\mathrm{m}$) and rock sites and those in basements of buildings or associated with large structures, such as dams and buildings taller than two storeys. Selects records with epicentral distances $\leq 20\,\mathrm{km}$ for $M_L < 4.75$ and distances $\leq 30\,\mathrm{km}$ for $M_L \geq 4.75$ to minimize regional differences in anelastic attenuation and potential biases associated with nontriggering instruments and unreported PGAs.
- Focal depths, H, between 1.8 and $24.3\,\mathrm{km}$ with mean of $8.5\,\mathrm{km}$.
- PGAs scaled from either actual or uncorrected accelerograms in order to avoid potential bias due to correction.

- Uses weighted nonlinear least squares technique of Campbell (1981).
- Tries two other forms of equation: $\ln \mathrm{PHA} = a + bM_L - 1.0 \ln[R + c_1] + e_1 H$ and $\ln \mathrm{PHA} = a + bM_L - 1.0 \ln[R + c_1] + e_2 \ln H$ for epicentral and hypocentral distance. Allows saturation of PGA for short distances but finds nonsignificant coefficients, at 90%. Also tries distance decay coefficient other than -1.0 but finds instability in analysis.
- Examines normalised weighted residuals against focal depth, M_L and distance. Finds that although residuals seem to be dependent on focal depth there are probably errors in focal depth estimation for deep earthquakes in the study so the dependence may not be real. Finds residuals not dependent on magnitude or distance.
- Uses 171 records ($0.9 \leq R \leq 28.1\,\mathrm{km}$) from 75 earthquakes ($2.5 \leq M_L \leq 5.0$, $0.7 \leq H \leq 24.3\,\mathrm{km}$) excluded from original analysis because they were on shallow soil, rock and/or not free-field, to examine importance of site geology and building size. Considers difference between PGA from records grouped according to instrument location, building size, embedment, and site geology and the predicted PGA using the attenuation equation to find site factors, S. Groups with nonsignificant, at 90%, values of S are grouped together. Finds two categories: embedded alluvial sites from all building sizes (38 records) and shallow-soil (depth of soil $\leq 10\,\mathrm{m}$) sites (35 records) to have statistically significant site factors.
- Performs regression analysis on all records (irrespective of site geology or building size) from Oroville (172 records from 32 earthquakes) and Imperial Valley (71 records from 42 earthquakes) to find individual sites that have significant influence on prediction of PGA (by using individual site coefficients for each station). Finds equations predict similar PGA to those predicted by original equation. Finds significant differences between PGA recorded at different stations in the two regions some related to surface geology but for some finds no reason.
- Uses 27 records ($0.2 \leq R \leq 25.0\,\mathrm{km}$) from 19 earthquakes ($2.5 \leq M_{bLG} \leq 4.8$, $0.1 \leq H \leq 9\,\mathrm{km}$) from E. N. America to examine whether they are significantly different than those from W. N. America. Finds residuals significantly, at 99% level, higher than zero and concludes that it is mainly due to site effects because most are on shallow soils or other site factors influence ground motion. Correcting the recorded PGAs using site factors the difference in PGA between E. N. America and W. N. America is no longer significant although notes may not hold for all of E. N. America.

2.53 *Alfaro* et al. *(1990)*

- Ground motion model for near field is:

$$\log(A) = a_1 + a_2 M_s - \log(r^2 + a_3^2)^{\frac{1}{2}}$$

where A is in g, $a_1 = -1.116$, $a_2 = 0.312$, $a_3 = 7.9$ and $\sigma = 0.21$.

Ground motion model for far field is:

$$\log(A) = b_1 + b_2 M_s + b_3 \log(r^2 + b_4^2)^{\frac{1}{2}}$$

where A is in g, $b_1 = -1.638$, $b_2 = 0.438$, $b_3 = -1.181$, $b_4 = 70.0$ and $\sigma = 0.21$.

- Separate crustal and subduction data because of differences in travel path and stress conditions:
  1. Near field
  2. Far field, 20 records from San Salvador, 20 earthquakes, $4.2 \leq M_s \leq 7.2$, depths between 36 and $94\,\mathrm{km}$, $31 \leq r \leq 298\,\mathrm{km}$.

2.54 Ambraseys (1990)

- Ground motion model is:

$$\begin{aligned} \log y &= \alpha + \beta M_w - \log r + br \\ \text{where } r &= (d^2 + h^2)^{1/2} \end{aligned}$$

where y is in g, $\alpha = -1.101$, $\beta = 0.2615$, $b = -0.00255$, $h = 7.2$ and $\sigma = 0.25$.

- Uses data and method of Joyner & Boore (1981) but re-evaluates M_w for all earthquakes. Finds some large changes, e.g. Santa Barbara changes from $M_w = 5.1$ to $M_w = 5.85$. Uses M_L for 2 earthquakes ($M_L = 5.2, 6.2$).
- Find effect of uncertainty in M_w causes less than 10% change in σ.
- Also calculates equation using M_s instead of M_w.
- Finds assumption $M_s = M_w$ introduces bias, particularly for small magnitude shocks, on unsafe side, and this can be significant in cases where there is a preponderance of small earthquakes in set.

2.55 Campbell (1990)

- Ground motion model is:

$$\begin{aligned} \ln(Y) &= a + bM + d\ln[R + c_1\exp(c_2 M)] + eF + f_1\tanh[f_2(M + f_3)] + g_1\tanh(g_2 D) + h_1 K_1 \\ &\quad + h_2 K_2 + h_3 K_3 \end{aligned}$$

where Y is in g, $a = -2.245$, $b = 1.09$, $c_1 = 0.361$, $c_2 = 0.576$, $d = -1.89$, $e = 0.218$, $f_1 = 0$, $f_2 = 0$, $f_3 = 0$, $g_1 = 0$, $g_2 = 0$, $h_1 = -0.137$, $h_2 = -0.403$ and $h_3 = 0$. $\sigma = 0.517$ for $M \leq 6.1$ and $\sigma = 0.387$ for $M \geq 6.2$. Also given is $\sigma = 0.450$ for $M \geq 4.7$.

- Records from firm soil and soft rock sites. Characterises site conditions by depth to basement rock (sediment depth) in km, D.
- Records from different size buildings. $K_1 = 1$ for embedded buildings 3–11 storeys, $K_2 = 1$ for embedded buildings with >11 storeys and $K_3 = 1$ for non-embedded buildings >2 storeys in height. $K_1 = K_2 = K_3 = 0$ otherwise.
- Uses two fault mechanisms:

$F = 0$ Strike-slip

$F = 1$ Reverse

2.56 Dahle et al. (1990b) & Dahle et al. (1990a)

- Ground motion model is:

$$\begin{aligned} \ln A &= c_1 + c_2 M + c_4 R + \ln G(R, R_0) \\ \text{where } G(R, R_0) &= R^{-1} \quad \text{for} \quad R \leq R_0 \\ \text{and: } G(R, R_0) &= R_0^{-1}\left(\frac{R_0}{R}\right)^{5/6} \quad \text{for} \quad R > R_0 \end{aligned}$$

where A is in ms^{-2}, $c_1 = -1.471$, $c_2 = 0.849$, $c_4 = -0.00418$ and $\sigma = 0.83$.

- Use records from rock sites (presumably with hard rock or firm ground conditions).
- Assume intraplate refers to area that are tectonically stable and geologically more uniform than plate boundary areas. Select records from several 'reasonably' intraplate areas (eastern N. America, China, Australia, and some parts of Europe), due to lack of data.
- Select records which are available unprocessed and with sufficient information on natural frequency and damping of instrument.
- Use M_s, when available, because reasonably unbiased with respect to source dimensions and there is globally consistent calculation method.
- Most (72%) records from earthquakes with $M \leq 5.5$. Tangshan and Friuli sequence comprise a large subset. Correlation coefficient between magnitude and distance is 0.31.
- Instrument correct records and elliptical filter with pass band 0.25 to $25.0\,\mathrm{Hz}$.
- If depth unknown assume $15\,\mathrm{km}$.
- Choose $R_0 = 100\,\mathrm{km}$ although depends on crustal structure and focal depth. It is distance at which spherical spreading for S waves overtaken by cylindrical spreading for Lg waves.
- PGA attenuation relation is pseudo-acceleration equation for $0.025\,\mathrm{s}$ period and 5% damping.
- Plot residuals against magnitude and distance.
- Note 'first order' results, because data from several geological regions and use limited data base.

2.57 *Jacob* et al. *(1990)*

- Ground motion model is:
$$A = 10^{(a_1 + a_2 M + a_3 \log d + a_4 d)}$$
where A is in g, $a_1 = -1.43$, $a_2 = 0.31$, $a_3 = -0.62$ and $a_4 = -0.0026$ (σ not given).
- Note equation only for hard rock sites.
- Equation from a composite of two separate regressions: one using data from 6 earthquakes, $4.7 \leq M \leq 6.4$ and d primarily between 40 and $820\,\mathrm{km}$ and one using the same data supplemented with data from 2 earthquakes with $M = 1.8$ and $M = 3.2$ and $d \leq 20\,\mathrm{km}$ to extend results to smaller M and d. Give no details of this composite regression.
- Note regressions are preliminary and should be tested against more data.
- Note careful assessment of uncertainties is required.

2.58 *Sen (1990)*

- Ground motion model is:
$$\ln \mathrm{PGA} = a + bM + c \ln(r + h) + \phi F$$
where PGA is in cms^{-2}, $a = 1.375$, $b = 1.672$, $c = -1.928$ and $\phi = 0.213$ (h not given). Standard deviation is composed of two parts, inter-site $\tau = 0.261$ and intra-site $\sigma = 0.653$. $F = 1$ for thrust mechanism and 0 otherwise.
- Computes theoretical radiation pattern and finds a linear trend between residuals and radiation pattern but does not model.

2.59 *Tsai* et al. *(1990)*

- Ground motion model is:

$$\ln y = C_0 + C_1 M + C_2 (8.5 - M)^{2.5} + C_3 \ln[D + C_4 \exp(C_5 M)]$$

  where $C_3 = -2.1$, $C_4 = 0.616$, $C_5 = 0.524$ and for $M \geq 6.5$ $C_0 = -1.092$, $C_1 = 1.10$, $C_2 = 0$ and $\sigma = 0.36$ and for $M < 6.5$ $C_0 = -0.442$, $C_1 = 1.0$, $C_2 = 0$ and $\sigma = 1.27 - 0.14M$.

- All records from rock or rock-like sites.

- Separate equation for $M < 6.5$ and $M \geq 6.5$.

- Use only shallow crustal thrust earthquakes.

- Use another database of rock and soil site records and simulated acceleration time histories to find conversion factors to predict strike-slip and oblique ground motions from the thrust equation given above. For strike-slip conversion factor is 0.83 and for oblique conversion factor is 0.91.

- Standard deviation, σ, for $M \geq 6.5$ from regression whereas σ for $M < 6.5$ from previous results. Confirm magnitude dependence of standard deviation using 803 recordings from 124 earthquakes, $3.8 \leq M_w \leq 7.4$, $D < 100\,\mathrm{km}$.

2.60 *Ambraseys & Bommer (1991) & Ambraseys & Bommer (1992)*

- Ground motion model is:

$$\begin{aligned} \log a &= \alpha + \beta M - \log r + br \\ \text{where } r &= (d^2 + h_0^2)^{1/2} \\ \text{or: } r &= (d^2 + h^2)^{1/2} \end{aligned}$$

  where a is in g, for horizontal PGA $\alpha = -1.09$, $\beta = 0.238$, $b = -0.00050$, $h = 6.0$ and $\sigma = 0.28$ and for vertical PGA $\alpha = -1.34$, $\beta = 0.230$, $b = 0$, $h = 6.0$ and $\sigma = 0.27$. When use focal depth explicitly: for horizontal PGA $\alpha = -0.87$, $\beta = 0.217$, $b = -0.00117$ and $\sigma = 0.26$ and for vertical PGA $\alpha = -1.10$, $\beta = 0.200$, $b = -0.00015$ and $\sigma = 0.26$.

- Consider two site classifications (without regard to depths of deposits) but do not model:

  1. Rock
  2. Alluvium

- Select records which have: $M_s \geq 4.0$ and standard deviation of M_s known and reliable estimates of source-site distance and focal depth, $h \leq 25\,\mathrm{km}$, regardless of local soil conditions from free-field and bases of small buildings. No reliable data or outliers excluded. Records from instruments at further distances from the source than the closest non-triggered instrument were non-excluded because of non-homogeneous and irregularly spaced networks and different and unknown trigger levels.

- Most data, about 70%, with distances less than $40\,\mathrm{km}$. Note strong bias towards smaller values of magnitude and PGA.

- PGA read from analogue and digitised data, with different levels of processing. Differences due to different processing usually below 5%, but some may be larger.

- Errors in distances for small shocks may be large.

- Prefer one-stage technique because second step of two-stage method would ignore records from singly-recorded earthquakes which compose over half the events, also find more realistic, b, and h_0 using one-stage method. Do not use weighting because involves assumptions which are difficult to verify.
- Find inadmissable and positive b for vertical PGA so remove and repeat.
- Remove records from distances less than or equal to half their focal depth and also less than or equal to their focal depth, find that h_0 is governed by near-field data.
- Use focal depth explicitly, by replacing $r = (d^2 + h_0^2)^{1/2}$ by $r = (d^2 + h^2)^{1/2}$. Find lower standard deviation and that it is very significant.
- Repeat analysis on subsets of records grouped by focal depth. Find no correlation between h_0 and focal depth of subset. Use h_0 equal to mean focal depth in each subset and find similar results to when focal depth used explicitly.
- Repeat analysis with geometric attenuation coefficient equal to -0.83, corresponding to the Airy phase, as opposed to -1.0.
- Find small dependence of horizontal PGA on site classification, note due to level of information available.

2.61 Crouse (1991)

- Ground motion model is:

$$\ln \mathrm{PGA} = p_1 + p_2 M + p_4 \ln[R + p_5 \exp(p_6 M)] + p_7 h$$

  where PGA is in gal, using all PGA values $p_1 = 6.36$, $p_2 = 1.76$, $p_4 = -2.73$, $p_5 = 1.58$, $p_6 = 0.608$, $p_7 = 0.00916$ and $\sigma = 0.773$.
- Use data from stiff soil sites (depth of soil $< 25\,\mathrm{m}$).
- Include data from any zones with strong seismic coupling, such as the younger subduction zones (S.W. Japan, Alaska, C. America (Mexico), C. Chile, Peru and northern Honshu and Kuril subduction zones in Japan) unless compelling reasons to exclude data. Do this because lack of data from Cascadia. Most ($> 70\%$) are from Japan.
- Focal depths, h, between 0 and $238\,\mathrm{km}$.
- Compare Japanese and Cascadia PGA values for earthquakes with similar magnitude and depths and find similar.
- Do not exclude data from buildings or which triggered on S-wave. Note could mean some PGAs are underestimated.
- Plot ground motion amplitude (PGA and also some maximum displacements from seismograms) against distance for a number of large magnitude shocks (including some data from rock sites which not included in set for regression). Find that rate of attenuation becomes smaller for shorter distances and process is magnitude dependent. Also plot Japanese PGA data, from earthquakes with $h \leq 50\,\mathrm{km}$, split into three distance groups (between 50 and $75\,\mathrm{km}$, between 100 and $150\,\mathrm{km}$ and between 250 and $300\,\mathrm{km}$) find as distance increases magnitude scaling becomes larger and possible saturation in PGA for large magnitudes. Fit $\ln \mathrm{PGA} = p_1 + p_2 \ln(R + C)$ to some PGA values from large magnitude shocks for $C = 0$ and $C > 0$, find lower standard deviation for $C > 0$.

- Fit $\ln \mathrm{PGA} = a + bM$ and $\ln \mathrm{PGA} = a + bM + cM^2$ to Japanese data split into the three distance groups (mentioned above); find b increases with increasing distance range but both equations fit data equally well.
- Constrain p_4 to negative value and p_5 and p_6 to positive values.
- Include quadratic magnitude term, $p_3 M^2$, but find equal to zero.
- Plot residuals against M; find uniformly distributed and evidence for smaller residuals for larger M.
- Plot residuals against R[4] and find decreasing residuals for increasing R.
- Give equation using only those records available in digital form (235 records).

2.62 *Huo & Hu (1991)*

- Ground motion model is (case II):

$$\log y = C_1 + C_2 M - C_4 \log[R + C_5 \exp(C_6 M)]$$

  where y is in gal, $C_5 = 0.231$ and $C_6 = 0.626$, for rock $C_1 = 0.894$, $C_2 = 0.563$, $C_4 = 1.523$ and $\sigma = 0.220$ and for soil $C_1 = 1.135$, $C_2 = 0.462$, $C_4 = 1.322$ and $\sigma = 0.243$ (these coefficients are from regression assuming M and R are without error).
- Use two site categories:
  1. Rock
  2. Soil
- Supplement western USA data in large magnitude range with 25 records from 2 foreign earthquakes with magnitudes 7.2 and 7.3.
- Note that there are uncertainties associated with magnitude and distance and these should be considered in derivation of attenuation relations.
- Develop method, based on weighted consistent least-square regression, which minimizes residual error of all random variables not just residuals between predicted and measured ground motion. Method considers ground motion, magnitude and distance to be random variables and also enables inverse of attenuation equation to be used directly.
- Note prediction for $R > 100\,\mathrm{km}$ may be incorrect due to lack of anelastic attenuation term.
- Use both horizontal components to maintain their actual randomness.
- Note most data from moderate magnitude earthquakes and from intermediate distances therefore result possibly unreliable outside this range.
- Use weighted analysis so region of data space with many records are not overemphasized. Use M-R subdivisions of data space: for magnitude $M < 5.5$, $5.5 \le M \le 5.9$, $6.0 \le M \le 6.4$, $6.5 \le M \le 6.9$, $7.0 \le M \le 7.5$ and $M > 7.5$ and for distance $R < 3$, $3 \le R \le 9.9$, $10 \le R \le 29.9$, $30 \le R \le 59.9$, $60 \le R \le 99.9$, $100 \le R \le 300$ and $R > 300\,\mathrm{km}$. Assign equal weight to each subdivision, and any data point in subdivision i containing n_i data has weight $1/n_i$ and then normalise.
- To find C_5 and C_6 use 316 records from 7 earthquakes ($5.6 \le M \le 7.2$) to fit $\log Y = \sum_{i=1}^{m} C_{2,i} E_i - C_4 \log[r + \sum_{i=1}^{m} R_{0,i} E_i]$, where $E_i = 1$ for ith earthquake and 0 otherwise. Then fit $R_0 = C_5 \exp(C_6 M)$ to results.

[4] Not shown in paper.

- Also try equations: $\log y = C_1 + C_2 M - C_4 \log[R + C_5]$ (case I) and $\log y = C_1 + C_2 M - C_3 M^2 - C_4 \log[R + C_5 \exp(C_6 M)]$ (case III) for $M \leq M_c$, where impose condition $C_3 = (C_2 - C_4 C_6 / \ln 10)/(2M_c)$ so ground motion is completely saturated at $M = M_c$ (assume $M_c = 8.0$).
- Find equations for rock and soil separately and for both combined.

2.63 I.M. Idriss (1991) reported in Idriss (1993)

- Ground motion model is:

$$\ln(Y) = [\alpha_0 + \exp(\alpha_1 + \alpha_2 M)] + [\beta_0 - \exp(\beta_1 + \beta_2 M)] \ln(R + 20) + aF$$

  where Y is in g, $a = 0.2$, for $M \leq 6$ $\alpha_0 = -0.150$, $\alpha_1 = 2.261$, $\alpha_2 = -0.083$, $\beta_0 = 0$, $\beta_1 = 1.602$, $\beta_2 = -0.142$ and $\sigma = 1.39 - 0.14M$ and for $M > 6$ $\alpha_0 = -0.050$, $\alpha_1 = 3.477$, $\alpha_2 = -0.284$, $\beta_0 = 0$, $\beta_1 = 2.475$, $\beta_2 = -0.286$ and for $M < 7\frac{1}{4}$ $\sigma = 1.39 - 0.14M$ and for $M \geq 7\frac{1}{4}$ $\sigma = 0.38$.
- Records from rock sites.
- Uses three fault mechanisms:

  F=0 Strike slip

  F=0.5 Oblique

  F=1 Reverse
- Separate equations for $M \leq 6$ and $M > 6$.
- Examines residuals for PGA. Finds average residual almost zero over entire distance range; trend reasonable up to about $60\,\mathrm{km}$ but beyond $60\,\mathrm{km}$ relationship would underestimate recorded PGA.
- Finds standard deviation to be linear function of magnitude.

2.64 Niazi & Bozorgnia (1991)

- Ground motion model is:

$$\ln Y = a + bM + d \ln[R + c_1 \mathrm{e}^{c_2 M}]$$

  where Y is in g, for horizontal PGA $a = -5.503$, $b = 0.936$, $c_1 = 0.407$, $c_2 = 0.455$, $d = -0.816$ and $\sigma = 0.461$ and for vertical PGA $a = -5.960$, $b = 0.989$, $c_1 = 0.013$, $c_2 = 0.741$, $d = -1.005$ and $\sigma = 0.551$.
- All records from SMART-1 array so essentially identical site conditions and travel paths.
- All records from free-field instruments mounted on $4\mathrm{inch}$ ($10\,\mathrm{cm}$) thick concrete base mats, approximately 2 by $3\,\mathrm{feet}$ (60 by $90\,\mathrm{cm}$) across.
- Select earthquakes to cover a broad range of magnitude, distance and azimuth and ensuring thorough coverage of the array. Criteria for selection is: at least 25 stations recorded shock, focal depth $< 30\,\mathrm{km}$, hypocentral distance $< 50\,\mathrm{km}$ except for two large earthquakes from beyond $50\,\mathrm{km}$ to constrain distance dependence.
- Focal depths between 0.2 and $27.2\,\mathrm{km}$ with all but one $\leq 13.9\,\mathrm{km}$.
- Azimuths between $60°$ and $230°$.
- Most records (78%) have magnitudes between 5.9 and 6.5. Note magnitude and distance are not independent (correlation coefficient is 0.6).

- Records have sampling interval of $0.01\,\mathrm{s}$. Processed using trapezoidal band passed filter with corner frequencies 0.07, 0.10, 25.0 and $30.6\,\mathrm{Hz}$.
- Not enough information to use distance to rupture zone.
- Source mechanisms of earthquakes are: 4 normal, 2 reverse, 1 reverse oblique and 1 normal oblique with 4 unknown. Do not model source mechanism dependence because of 4 unknown mechanisms.
- Use weighted regression, give equal weight to recordings from each earthquake within each of 10 distance bins (< 2.5, 2.5–5.0, 5.0–7.5, 7.5–10.0, 10.0–14.1, 14.1–20.0, 20–28.3, 28.3–40.0, 40.0–56.6 and 56.6–$130\,\mathrm{km}$). Do this so earthquakes with smaller number of recordings are not overwhelmed by those with a larger coverage and also to give additional weight to shocks recorded over multiple distance bins. Apply two-stage regression, because of high correlation between magnitude and distance, excluding 3 earthquakes ($M = 3.6$, 5.0, 7.8) with 162 records from first stage to reduce correlation between M and R to 0.1. Also do one-stage regression although do not give coefficients.
- Use mean horizontal component because reduces uncertainty in prediction.
- Examine coefficient of variation for each earthquake using median and normalized standard deviation of recordings in inner ring of array. Find evidence for magnitude dependent uncertainty (large magnitude shocks show less uncertainty). Find that main contribution to scatter is inter-event variations again by examining coefficient of variation; although note may be because using dense array data.
- Examine mean residuals of observations from each earthquake. Find evidence for higher than predicted vertical PGA from reverse faulting earthquakes and lower than predicted vertical PGA from normal faulting earthquakes, although due to lack of information for 4 earthquakes note that difficult to draw any conclusions.
- Examine mean residuals of observations from each station in inner ring. Find mean residuals are relatively small compared with standard deviation of regression so variation between stations is less than variation between earthquakes. Find for some stations some large residuals.

2.65 *Ambraseys* et al. *(1992)*

- Ground motion model is:

$$\begin{aligned} \log(a) &= c_1 + c_2 M + c_3 r + c_4 \log r \\ r &= (d^2 + h_0^2)^{\frac{1}{2}} \end{aligned}$$

  where a is in g, $c_1 = -1.038$, $c_2 = 0.220$, $c_3 = -0.00149$, $c_4 = -0.895$, $h_0 = 5.7$ and $\sigma = 0.260$.
- Investigate equations of PML (1982) and PML (1985) using criteria:
  1. Is the chosen data set of earthquake strong-motion records suitable to represent the UK seismic environment?
  2. Are the associated seismological and geophysical parameters used in these reports reliable and consistent?
  3. Is the methodology used to derive attenuation laws and design spectra from the data set reliable?
- Investigate effect of different ground motion model, one and two-stage regression technique, record selection technique and recalculation of associated parameters. Find these choice cause large differences in predictions.
- Coefficients given above are for PML (1985) data with recalculated magnitudes and distances and addition of extra records from some earthquakes.

2.66 Kamiyama et al. (1992) & Kamiyama (1995)

- Ground motion model is (note that there is a typographical error in Kamiyama *et al.* (1992); Kamiyama (1995) because r_t has been replaced by r_c in equations):

$$\begin{aligned}
\log_{10} a_{\max} &= -1.64R_0 + b_1R_1 + b_2R_2 + c_a + \sum_{i=1}^{N-1} A_iS_i \\
R_0 &= \begin{cases} 0 & \text{for} \quad r \le r_t \\ \log_{10} r - \log_{10} r_c & \text{for} \quad r > r_t \end{cases} \\
R_1 &= \begin{cases} 0 & \text{for} \quad r \le r_t \\ 1 & \text{for} \quad r > r_t \end{cases} \\
R_2 &= \begin{cases} 0 & \text{for} \quad r \le r_t \\ M & \text{for} \quad r > r_t \end{cases}
\end{aligned}$$

  where $S_i = 1$ for i station, $S_0 = 0$ otherwise, $a_{\max}$ is in cms^{-2}, $b_1 = -1.164$, $b_2 = 0.358$, $c_a = 2.91$, $r_c = 5.3\,\mathrm{km}$ and $\sigma = 0.247$ (A_i given in publications but not reported here due to lack of space).

- Instrument correct records and filter with pass band between 0.24 and $11\,\mathrm{Hz}$.

- Model individual soil conditions at each site as amplification factors, AMP_i, as described by Kamiyama & Yanagisawa (1986).

- Most records are from hypocentral distances between 30 and $200\,\mathrm{km}$.

- Focal depths between 0 and $130\,\mathrm{km}$.

- Models peak ground accelerations independent of magnitude and distance in a fault zone, r_t, where $r_t = r_c 10^{(b_1+b_2M)/1.64}$.

- Constrain decay with distance in far field to -1.64 using results from other studies to avoid problems due to correlation between M and $\log_{10} r$.

- Use trial and error method to find r_c so that resulting values of r_t are consistent with empirical estimates of fault length from past studies.

- Also give expression using shortest distance to fault plane (rupture distance), R, by replacing the expression for $r \le r_c$ and $r > r_c$ by one expression given by replacing r, hypocentral distance, by $R + r_c$ in expression for $r > r_c$. This gives PGA independent of magnitude at distance $R = 0\,\mathrm{km}$.

- Note that use of d_h is not necessarily best choice but use it due to simplicity.

- Check residual plots; find no trends so conclude adequate from statistical point of view.

2.67 Sigbjörnsson & Baldvinsson (1992)

- Ground motion model is:

$$\begin{aligned}
\log A &= \alpha + \beta M - \log R + bR \\
\text{with: } R &= \sqrt{d^2 + h^2}
\end{aligned}$$

  where A is in g, for average horizontal PGA and $4 < M < 6$ $\alpha = -1.98$, $\beta = 0.365$, $b = -0.0039$ and $\sigma = 0.30$, for larger horizontal PGA and $4 < M < 6$ $\alpha = -1.72$, $\beta = 0.327$, $b = -0.0043$ and $\sigma = 0.30$ and for both horizontal PGAs and $2 < M < 6$ $\alpha = -2.28$, $\beta = 0.386$, $b = 0$ and $\sigma = 0.29$.

- Find that Icelandic data does not fit other published relations.

- Find equation using only records with $M \geq 4.0$, h equal to focal depth and both the horizontal components.
- Find equation using only records with $M \geq 4.0$, h equal to focal depth and larger horizontal component.
- Also repeated with all data. Anelastic coefficient constrained to zero because otherwise positive.
- Also done with h free.
- Note that large earthquakes have $h \approx 10\,\mathrm{km}$ while small events have $h \approx 5\,\mathrm{km}$.

2.68 *Taylor Castillo* et al. *(1992)*

- Ground motion model is:
$$\ln(A) = a_1 + a_2 M_s + a_3 \ln(R) + a_4 R$$
where A is in ms^{-2}, $a_1 = 0.339$, $a_2 = 0.455$, $a_3 = -0.67$, $a_4 = -0.00207$ and $\sigma = 0.61$.

2.69 *Tento* et al. *(1992)*

- Ground motion model is:
$$\begin{aligned} \ln \mathrm{PGA} &= b_1 + b_2 M + b_3 R - \ln R \\ \text{where } R &= (d^2 + h^2)^{1/2} \end{aligned}$$
where PGA is in gal, $b_1 = 4.73$, $b_2 = 0.52$, $b_3 = -0.00216$, h is mean focal depth of group into which each earthquake is classified and $\sigma = 0.67$.
- Most records from distances between $10\,\mathrm{km}$ and $40\,\mathrm{km}$.
- Correction technique based on uniform Caltech correction procedure. Most (125) were automatically digitised, rest were manually digitised. Roll-on and cutoff frequencies of Ormsby filter were selected by adopting a record dependent criteria. Cutoff frequencies range between $0.13\,\mathrm{Hz}$ and $1.18\,\mathrm{Hz}$ with a median of $0.38\,\mathrm{Hz}$.
- Records included from analysis were from free-field stations. Excluded those not complete (e.g. started during strong-motion phase). Excluded those with epicentral distances greater than that of first nontriggered station.
- Note relatively small influence of form of equation adopted although two step method seems preferable.
- Note correction procedure plays a relevant role in analysis.
- Note using d instead of R causes greater scatter in data.
- Note moderate underestimation for low magnitude in near field and for high magnitude in far field.

2.70 *Theodulidis & Papazachos (1992)*

- Ground motion model is:
$$\ln Y = C_1 + C_2 M + C_3 \ln(R + R_0) + C_4 S$$
where Y is in cms^{-2}, $C_1 = 3.88$, $C_2 = 1.12$, $C_3 = -1.65$, $R_0 = 15$, $C_4 = 0.41$ and $\sigma = 0.71$.
- Use two site categories (mean opinion of seven specialists who classified sites into three categories: soft alluvium, crystalline rock and intermediate):

S=1 Rock: 34+4 records. Japanese sites have diluvium with depth to bedrock $H < 10\,\mathrm{m}$. Alaskan sites have $\mathrm{PGV/PGA} \approx 66 \pm 7\,\mathrm{cms^{-1}g^{-1}}$.

S=0 Alluvium: 71+12 records. Japanese sites have diluvium $H > 10\,\mathrm{m}$ or alluvium $H < 10\,\mathrm{m}$, and alluvium with $H < 25\,\mathrm{m}$ as well as soft layers with thickness $< 5\,\mathrm{m}$. Alaskan sites have $\mathrm{PGV/PGA} > 66 \pm 7\,\mathrm{cms^{-1}g^{-1}}$.

- 70% of records from ground level or basement of buildings with two storeys or less. Rest from buildings with up to eight storeys.
- Some (16) Greek records manually digitized and baseline corrected, some (22) Greek records manually digitized and filtered and rest of the Greek records automatically digitized and filtered.
- Due to lack of data for $7.0 < M_s < 7.5$ include shallow subduction data from other regions with similar seismotectonic environments (Japan and Alaska) using criteria i) depth $< 35\,\mathrm{km}$, ii) M_w or M_{JMA} between 7.0 and 7.5, iii) instruments triggered before S-wave, iv) free-field recording, v) surface geology known at station. Note M_s, M_w and M_{JMA} are equivalent between 6.0 and 8.0.
- Focal depths between $0\,\mathrm{km}$ ($13\,\mathrm{km}$) and $18\,\mathrm{km}$ ($31\,\mathrm{km}$).
- Most data from $M_s < 5.5$ and from $R < 50\,\mathrm{km}$.
- Use four step regression procedure. First step use only Greek data from $M_s > 6.0$ ($9 \leq R \leq 128\,\mathrm{km}$, 14 records) for which distances are more reliable (use both hypocentral and epicentral distance find epicentral distance gives smaller standard deviation) to find geometrical coefficient C_{31} and R_0 ignoring soil conditions. Next find constant (C_{12}), magnitude (C_{22}) and soil (C_{42}) coefficients using all data. Next recalculate geometrical (C_{33}) coefficient using only Greek data with $M_s > 6.0$. Finally find constant (C_{14}), magnitude (C_{24}) and soil (C_{44}) coefficients using all the data; final coefficients are C_{14}, C_{24}, C_{33} and C_{44}.
- Plot residuals against M_s and R and find no apparent trends. Find residuals (binned into 0.2 intervals) fit normal distribution.

2.71 *Boore* et al. *(1993) & Boore* et al. *(1997)*

- Ground motion model is:

$$\begin{aligned} \log Y &= b_1 + b_2(\mathbf{M} - 6) + b_3(\mathbf{M} - 6)^2 + b_4 r + b_5 \log r + b_6 G_B + b_7 G_C \\ \text{where } r &= (d^2 + h^2)^{1/2} \end{aligned}$$

where Y is in g, for randomly-oriented horizontal component (or geometrical mean) $b_1 = -0.105$, $b_2 = 0.229$, $b_3 = 0$, $b_4 = 0$, $b_5 = -0.778$, $b_6 = 0.162$, $b_7 = 0.251$, $h = 5.57$ and $\sigma = 0.230$ (for geometrical mean $\sigma = 0.208$) and for larger horizontal component $b_1 = -0.038$, $b_2 = 0.216$, $b_3 = 0$, $b_4 = 0$, $b_5 = -0.777$, $b_6 = 0.158$, $b_7 = 0.254$, $h = 5.48$ and $\sigma = 0.205$.

- Use three site categories:

Class A $V_{s,30} > 750\,\mathrm{ms^{-1}}$, some categorised using measured shear-wave velocity, most estimated $\Rightarrow G_B = 0, G_C = 0$, 48 records

Class B $360 < V_{s,30} \leq 750\,\mathrm{ms^{-1}}$, some categorised using measured shear-wave velocity, most estimated $\Rightarrow G_B = 1, G_C = 0$, 118 records.

Class C $180 < V_{s,30} \leq 360\,\mathrm{ms^{-1}}$,some categorised using measured shear-wave velocity, most estimated $\Rightarrow G_B = 0, G_C = 1$, 105 records.

where $V_{s,30}$ is average shear-wave velocity to $30\,\mathrm{m}$.

- Define shallow earthquakes as those for which fault rupture lies mainly above a depth of $20\,\mathrm{km}$.
- Peak acceleration scaled directly from accelerograms, in order to avoid bias from sparsely sampled older data.
- Do not use data from structures three storeys or higher, from dam abutments or from base of bridge columns. Do not use data from more than one station with the same site condition within a circle of radius $1\,\mathrm{km}$ (note that this is a somewhat arbitrary choice).
- Exclude records triggered by S wave.
- Do not use data beyond cutoff distance which is defined as equal to lesser of distance to the first record triggered by S wave and closest distance to an operational nontriggered instrument.
- Note that little data beyond $80\,\mathrm{km}$.
- Due to positive values of b_4 when $b_5 = -1$, set b_4 to zero and let b_5 vary.

2.72 Campbell (1993)

- Ground motion model is:

$$\begin{aligned}\ln(Y) &= \beta_0 + a_1 M + \beta_1 \tanh[a_2(M-4.7)] - \ln(R^2 + [a_3 \exp(a_1 M)]^2)^{1/2} \\ &\quad - (\beta_4 + \beta_5 M)R + a_4 F + [\beta_2 + a_5 \ln(R)]S + \beta_3 \tanh(a_6 D)\end{aligned}$$

  where Y is in g, $\beta_0 = -3.15$, $\beta_1 = 0$, $\beta_2 = 0$, $\beta_3 = 0$, $\beta_4 = 0.0150$, $\beta_5 = -0.000995$, $a_1 = 0.683$, $a_2 = 0.647$, $a_3 = 0.0586$, $a_4 = 0.27$, $a_5 = -0.105$, $a_6 = 0.620$ and $\sigma = 0.50$.

- Uses two site categories:

  S=0 Quaternary deposits (soil).

  S=1 Tertiary or older sedimentary, metamorphic, and igneous deposits (rock).

  Also includes depth to basement rock (km), D.

- Uses two fault mechanisms:

  F=0 Strike-slip.

  F=1 Reverse, reverse-oblique, thrust, and thrust-oblique.

  Recommends use $F = 0.5$ for normal or unknown mechanisms.

- Gives estimates of average minimum depths to top of seismogenic rupture zone.
- Uses stochastic simulation model to find anelastic coefficients β_4 and β_5 because uses only near-source records.
- Uses weighted nonlinear regression method based on Campbell (1981) to control dominance of well-recorded earthquakes.

2.73 *Sadigh* et al. *(1993) & Sadigh* et al. *(1997)*

- Ground motion model is:

$$\ln \mathrm{PGA} = C_1 + C_2 M + C_3 \ln\left(r_{\mathrm{rup}} + C_4 \mathrm{e}^{C_5 M}\right) + C_6 Z_T$$

where PGA is in g, for horizontal PGA, rock sites and strike-slip faulting $C_3 = 0$ and $C_4 = -2.100$, for $M \leq 6.5$ $C_1 = -0.624$, $C_2 = 1.0$, $C_5 = 1.29649$ and $C_6 = 0.250$ and for $M > 6.5$, $C_1 = -1.274$, $C_2 = 1.1$, $C_5 = -0.48451$ and $C_6 = 0.524$. For reverse and thrust earthquakes multiply strike-slip prediction by 1.2. $\sigma = 1.39 - 0.14M$ for $M < 7.21$ and $\sigma = 0.38$ for $M \geq 7.21$. For horizontal PGA and deep soil $C_2 = 1.0$, $C_3 = 1.70$ and $C_6 = 0$, for strike-slip faulting $C_1 = -2.17$ and for reverse or thrust faulting $C_1 = -1.92$, for $M \leq 6.5$ $C_4 = 2.1863$ and $C_5 = 0.32$ and for $M > 6.5$ $C_4 = 0.3825$ and $C_5 = 0.5882$. $\sigma = 1.52 - 0.16M$ for $M \leq 7$ and $\sigma = 0.40$ for $M = 7$.

For vertical PGA, rock sites and strike-slip faulting $C_3 = 0$ and $C_4 = -2.300$, for $M \leq 6.5$ $C_1 = -0.430$, $C_2 = 1.0$, $C_5 = 1.2726$ and $C_6 = 0.228$ and for $M > 6.5$, $C_1 = -1.080$, $C_2 = 1.1$, $C_5 = -0.3524$ and $C_6 = 0.478$. For reverse and thrust earthquakes multiply strike-slip prediction by 1.1 and for oblique faulting multiply by 1.048. $\sigma = 0.48$ for $M \geq 6.5$, $\sigma = 3.08 - 0.40M$ for $6 < M < 6.5$ and $\sigma = 0.68$ for $M \leq 6$.

- Use two site categories (for horizontal motion):

  1. Rock: bedrock within about a metre of surface. Note that many such sites are soft rock with $V_s \leq 750\,\mathrm{ms}^{-1}$ and a strong velocity gradient because of near-surface weathering and fracturing, 274 records.
  2. Deep soil: greater than $20\,\mathrm{m}$ of soil over bedrock. Exclude data from very soft soil sites such as those from San Francisco bay mud, 690 records.

  Vertical equations only for rock sites.

- Crustal earthquakes defined as those that occur on faults within upper 20 to $25\,\mathrm{km}$ of continental crust.

- Use source mechanism: RV=reverse (26+2) $\Rightarrow Z_T = 1$ and SS=strike-slip (and some normal) (89+0) $\Rightarrow Z_T = 0$. Classified as RV if rake$> 45°$ and SS if rake$< 45°$. Find peak motions from small number of normal faulting earthquakes not to be significantly different than peak motions from strike-slip events so were including in SS category.

- Records from instruments in instrument shelters near ground surface or in ground floor of small, light structures.

- 4 foreign records (1 from Gazli and 3 from Tabas) supplement Californian records.

- Separate equations for $M_w < 6.5$ and $M_w \geq 6.5$ to account for near-field saturation effects and for rock and deep soil sites.

2.74 *Singh* et al. *(1993)*

- Ground motion model is:

$$\begin{aligned} \log(A) &= a_1 + a_2 M + a_3 \log[G(R_0)] + a_4 R_0 \\ \text{where } R_0^2 &= R^2 + (\mathrm{e}^{a_5 M})^2 \\ G(R_0) &= R_0 \text{ for: } R_0 \leq 100\,\mathrm{km} \\ \text{and: } G(R_0) &= \sqrt{(100 R_0)} \text{ for: } R_0 > 100\,\mathrm{km} \end{aligned}$$

where A is in cms^{-2}, $a_1 = 2.74$, $a_2 = 0.212$, $a_3 = -0.99$, $a_4 = -0.000943$, $a_5 = 0.47$ and $\sigma = 0.26$.

- Use same data as Taylor Castillo *et al.* (1992).
- Employ several different regression techniques.
- Select equation found by Bayesian method (given above) for hazard study.

2.75 *Sun & Peng (1993)*

- Ground motion model is:
$$\ln A = a + bM - c\ln(R + h) + dT_s$$
where A is in cms^{-2}, $a = 7.7$, $b = 0.49$, $c = 1.45$, $d = 0.19$, $h = 25.0$ and $\sigma = 0.46$.
- Model soil using its fundamental period of the overburden soil, T_s. Thickness of deposit defined as depth to rock base, defined either as $V_s > 800\,\mathrm{ms}^{-1}$ or when ratio of shear-wave velocity in ith layer to shear-wave velocity in $i-1$th layer is greater than 2 (only calculate period to $100\,\mathrm{m}$ because only have important effect on structure). For outcropping rock, $T_s = 0.05\,\mathrm{s}$.
- Eight distance intervals used for weighting, five $10\,\mathrm{km}$ wide up to $50\,\mathrm{km}$, 50–$69.9\,\mathrm{km}$, 70–$99.9\,\mathrm{km}$ and 100–$200\,\mathrm{km}$. Within each interval each earthquake received equal weight, inversely proportional to number of records from that earthquake in interval.
- Use resolve accelerations in direction, θ, which gives largest value. Find scatter is lower than for larger horizontal component.
- Many (27) earthquakes only have one record associated with them and 60 records are from San Fernando.

2.76 *Boore* et al. *(1994a) & Boore* et al. *(1997)*

- Based on Boore *et al.* (1993) see Section 2.71
- Ground motion model is:
$$\begin{aligned} \log Y &= b_1 + b_2(\mathbf{M} - 6) + b_3(\mathbf{M} - 6)^2 + b_4 r + b_5 \log r + b_V(\log V_S - \log V_A) \\ \text{where } r &= (d^2 + h^2)^{1/2} \end{aligned}$$
where Y is in g, b_1 to b_5, h and σ are same as for Boore *et al.* (1993) (see Section 2.71) and for randomly oriented component $b_V = -0.371$ and $V_A = 1400$ and for larger horizontal component $b_V = -0.364$ and $V_A = 1390$.
- Model site effect as a continuous function of average shear-wave velocity to $30\,\mathrm{m}$ deep, V_S.
- Coefficients b_1, b_2, b_3,b_4 and b_5 from Boore *et al.* (1993).
- Find no basis for different magnitude scaling at different distances.
- Find evidence for magnitude dependent uncertainty.
- Find evidence for amplitude dependent uncertainty.
- Find marginal statistical significance for a difference between strike-slip (defined as those with a rake angle within 30° of horizontal) and reverse-slip motions but do not model it. Modelled in Boore *et al.* (1994b) (by replacing b_1 by $b_{SS}G_{SS} + b_{RS}G_{RS}$ where $G_{SS} = 1$ for strike-slip shocks and 0 otherwise and $G_{RS} = 1$ for reverse-slip shocks and 0 otherwise) and reported in Boore *et al.* (1997). Coefficients for randomly oriented horizontal component are: $b_{SS} = -0.136$ and $b_{RS} = -0.051$[5].

[5] These are taken from Table 8 of Boore *et al.* (1997) which uses natural logarithms so they were converted into terms of logarithms to base 10.

- Analysis done using one and two-stage maximum likelihood methods; note that results are very similar.
- Earthquakes with magnitudes below 6.0 are poorly represented.
- Note that few Class A records.
- Note that V_S does not model all the effects of site because it does not model effect of the thickness of attenuating material on motion.
- Note that ideally would like to model site in terms of average shear-wave velocity to one-quarter wavelength.
- Note lack measurements from distances greater than $100\,\mathrm{km}$ so that weak-motion data from seismographic stations maybe should be used.
- Note that use of cutoff distances independent of geology or azimuth may be over strict but it is simple and objective. Note that methods based on data from nontriggered stations or using seismogram data may be better.

2.77 *Fukushima* et al. *(1994) & Fukushima* et al. *(1995)*

- Ground motion model is:
$$\log Y = aM + bX - \log X + \sum \delta_i c_i$$
where Y is in cms^{-2}, $\delta_i = 1$ at ith receiver and 0 otherwise, for horizontal PGA $a = 0.918$ and $b = -0.00846$ (σ not given) and for vertical PGA $a = 0.865$ and $b = -0.00741$ (σ not given). c_i given in paper but are not reported here due to lack of space.
- Data from three vertical arrays in Japan so predictions at surface and at different depths down to $950\,\mathrm{m}$.
- Different definition of M_{JMA} for focal depths $> 60\,\mathrm{km}$ so exclude such data. Focal depths between 2 and $60\,\mathrm{km}$.
- Exclude data from earthquakes $M < 5.0$ because errors are larger for smaller events.
- Exclude data for which predicted, using a previous attenuation relation, $\mathrm{PGV} < 0.1\,\mathrm{cms}^{-1}$ in order to find precise attenuation rate.
- Most data from earthquakes with $M \leq 6.0$ and most from $X \leq 100\,\mathrm{km}$.
- Records low-pass filtered with cutoff frequency $25\,\mathrm{Hz}$ for records from 2 sites and $30\,\mathrm{Hz}$ for records from 1 site.
- Use two-stage method because positive correlation between M and X. Also apply one step; find it is biased and two-stage method is most effective method to correct bias.
- Check residuals (not shown) against M and X find no remarkable bias.

2.78 *Lawson & Krawinkler (1994)*

- Ground motion model is:
$$\log Y = a + b(M-6) + c(M-6)^2 + d\sqrt{R^2+h^2} + e\log\sqrt{R^2+h^2} + fS_B + gS_C$$
- Use three site categories:

A Firm to hard rock: granite, igneous rocks, sandstones and shales with close to widely spaced fractures, $750 \leq V_{s,30} \leq 1400\,\mathrm{ms}^{-1} \Rightarrow S_B = 0$, $S_C = 0$.

B Gravelly soils and soft to firm rocks: soft igneous rocks, sandstones and shales, gravels and soils with $> 20\%$ gravel, $360 \leq V_{s,30} \leq 750\,\mathrm{ms}^{-1} \Rightarrow S_B = 1$, $S_C = 0$.

C Stiff clays and sandy soils: loose to very dense sands, silt loams and sandy clays, and medium stiff to hard clay and silty clays ($N > 5\,\mathrm{blows/ft}$), $180 \leq V_{s,30} \leq 360\,\mathrm{ms}^{-1} \Rightarrow S_B = 0$, $S_C = 1$.

- For shallow (fault rupture within $20\,\mathrm{km}$ of earth surface) crustal earthquakes.
- Use free-field records. Records not significantly contaminated by structural feedback, excludes records from structures with $>$2 stories.
- Chooses ground motion model because of simplicity. Note that other possible forms of equation may have significant effect on results, but including more terms complicates relationships without reducing variability.
- Does not give coefficients only predictions.

2.79 *Lungu* et al. *(1994)*

- Ground motion model is:

$$\ln \mathrm{PGA} = c_1 + c_2 M_w + c_3 \ln R + c_4 h$$

where PGA is in g, $c_1 = -2.122$, $c_2 = 1.885$, $c_3 = -1.011$, $c_4 = -0.012$ and $\sigma = 0.502$.

- Focal depth, h, between 79 and $131\,\mathrm{km}$.
- Consider to separate areas of 90° to investigate variation with respect to azimuth; find azimuthal dependence.
- Find individual attenuation equations for three earthquakes. Note faster attenuation for smaller magnitude and faster attenuation for deeper events.

2.80 *Musson* et al. *(1994)*

- Ground motion model is (model 1):

$$\ln A = a + bM - \ln(R) + dR$$

where A is in cms^{-2}, $a = 2.11$, $b = 1.23$ and $d = -0.014$.

Ground motion model is (model 2):

$$\begin{aligned} \ln A &= c_1 + c_2 M + c_4 R + \ln G(R, R_0) \\ \text{where } G(R, R_0) &= R^{-1} \quad \text{for} \quad R \leq R_0 \\ \text{and: } G(R, R_0) &= R_0^{-1} \frac{R_0}{R}^{5/6} \quad \text{for} \quad R > R_0 \end{aligned}$$

where A is in ms^{-2}, c_1 and c_2 are from Dahle *et al.* (1990b), $c_4 = -0.0148$ and σ is recommended as 0.65 (although this is from an earlier study and is not calculated in regression).

- Use data from Canada (Saguenay earthquake and Nahanni sequence) and Belgium (Roermond earthquake).
- Focal depths, h, between 1 and $30\,\mathrm{km}$ with average $14.4\,\mathrm{km}$.

- Assume peak ground acceleration equals pseudo-acceleration at $30\,\mathrm{Hz}$ due to few unclipped horizontal UK records and because instrument response of UK instruments means records unreliable above $30\,\mathrm{Hz}$. Use only digital VME records for $30\,\mathrm{Hz}$ model.
- Note poorness of data due to UK data and other data being widely separated thus preventing a comparison between the two sets. Also means straightforward regression methods would be inadequate as there would be little control on shape of curves derived.
- Note earlier models over predict UK data.
- Use two-stage least squares method to give model 1. First stage fit only UK/Belgian data to find b, in second stage use this value of b and use all data to find a and d.
- Do not recommend model 1 for general use because too influenced by limitations of data to be considered reliable. Canadian data probably insufficient to anchor curves at small R/large M and extremely high Saguenay earthquake records carry undue weight.
- Use model of Dahle *et al.* (1990b) to get model 2. Fix c_1 and c_2 to those of Dahle *et al.* (1990b) and find c_4. Prefer this model.

2.81 *Radu* et al. *(1994), Lungu* et al. *(1995a) & Lungu* et al. *(1996)*

- Ground motion model is:
$$\ln \mathrm{PGA} = c_1 + c_2 M + c_3 \ln R + c_4 h$$
where PGA is in cms^{-2}, $c_1 = 5.432$, $c_2 = 1.035$, $c_3 = -1.358$, $c_4 = -0.0072$ and $\sigma = 0.397$.
- Sites have different soil conditions, some medium and stiff sites and some very soft soil sites.
- Use some records from Moldova and Bulgaria.
- Focal depths, h, between 91 and $133\,\mathrm{km}$.
- Records from free-field or from basements of buildings.
- Originally include data from a shallower (focal depth $79\,\mathrm{km}$), smaller magnitude ($M_L = 6.1$, $M_w = 6.3$) earthquake with shorter return period than other three earthquakes, but exclude in final analysis.
- Originally do attenuation analysis for two orthogonal directions N45E (which is in direction of fault plane) and N35E (which is normal to fault plane). From this define 3 90° circular sectors based roughly on tectonic regions, and calculate attenuation relations for each of these sectors as well as for all data. Find azimuthal dependence.
- Remove 1 to 3 anomalous records per sector.
- Remove the only record from the 4/3/1977 earthquake, because it has a strong influence on results, and repeat analysis using model $\ln \mathrm{PGA} = b_1 + b_2 M + b_3 \ln R$, find lower predicted PGA.
- Find slower attenuation in direction of fault plane compared with normal to fault plane.
- Find faster attenuation and larger standard deviation (by finding attenuation equations for two different earthquakes) for deeper focus and larger magnitude shocks.

2.82 Ramazi & Schenk (1994)

- Ground motion model is:

$$\begin{aligned} a_h &= a_1(a_2 + d + H)^{a_5} \exp(a_6 M_s) \\ H &= |d - a_3|^{a_4} \end{aligned}$$

where for horizontal peak acceleration a_h is in cms^{-2}, $a_1 = 4000$, $a_2 = 20$, $a_3 = 16$ and $a_4 = 0.63$ for soil sites $a_5 = -2.02$ and $a_6 = 0.8$ and for rock sites $a_5 = -2.11$ and $a_6 = 0.79$ (σ not given). For vertical peak acceleration on soil sites a_v is in cms^{-2} a_1 to a_3 are same as horizontal and $a_4 = 0.48$, $a_5 = -1.75$ and $a_6 = 0.53$ (σ not given).

- Use two site categories (from original of four) for which derive two separate equations:
  1. Rock: mainly category (2) a) loose igneous rocks (tuffs), friable sedimentary rocks, foliated metamorphic rock and rocks which have been loosened by weathering, b) conglomerate beds, compacted sand and gravel and stiff clay (argillite) beds where soil thickness $> 60\,\mathrm{m}$ from bed rock. 29 records.
  2. Soil: mainly category (4) a) soft and wet deposits resulting from high level of water table, b) gravel and sand beds with weak cementation and/or uncementated unindurated clay (clay stone) where soil thickness $> 10\,\mathrm{m}$ from bed rock. 54 records.
- Focal depths between 10 and $69\,\mathrm{km}$.
- Find equations using hypocentral distance but find that poor fit for Rudbar (Manjil) earthquake ($M_s = 7.7$) which conclude due to use of hypocentral rather than rupture distance.
- Find equations using rupture distance[6] for Rudbar (Manjil) earthquake and hypocentral distances for other earthquakes. Coefficients given above. They conclude that it is important that equations are derived using rupture distance rather than hypocentral distance because most destructive earthquakes rupture surface in Iran.
- Do not know physical meaning of H term but find that it causes curves to fit data better.

2.83 Xiang & Gao (1994)

- Ground motion model is:

$$A_p = a\mathrm{e}^{bM_s}(R + \Delta)^c$$

where A_p is in cms^{-2} and for combined Yunnan and W. N. American data $a = 1291.07$, $b = 0.5275$, $c = -1.5785$, $\Delta = 15$ and $\sigma = 0.5203$ (in terms of natural logarithm).

- All records from basement rock.
- Most Yunnan data from main and aftershocks of Luquan and Luncang-Gengma earthquakes.
- Records from Lancang-Gengma sequence corrected.
- Most Yunnan records with $3 \leq M_s \leq 5$ and $10 \leq R \leq 40\,\mathrm{km}$.
- To overcome difficulty due to shortage of large magnitude records and sample heterogeneous distribution in near and far fields use W. N. America data, because intensity attenuation is similar.
- Fit curves to Yunnan and Yunnan with W. N. American data. Find curve for combined data has lower variance and fit to observation data for large magnitudes is better (by plotting predicted and observed PGA).

[6] They state it is ' ...closest distance from the exposure of ruptured part of the fault ...' so may not be rupture distance.

2.84 Ambraseys (1995)

- Ground motion model is:

$$\begin{aligned} \log a &= A + BM_s + Cr + D\log r \\ \text{where } r^2 &= d^2 + h_0^2 \end{aligned}$$

  where a is in g, for $4.0 \leq M \leq 7.4$: for horizontal PGA not including focal depth $A = -1.09$, $B = 0.238$, $C = -0.00050$, $D = -1$, $h_0 = 6.0$ and $\sigma = 0.28$, for vertical PGA not including focal depth $A = -1.34$, $B = 0.230$, $C = 0$, $D = -1$, $h_0 = 6.0$ and $\sigma = 0.27$, for horizontal PGA including focal depth $A = -0.87$, $B = 0.217$, $C = -0.00117$, $D = -1$, $h_0 = h$ and $\sigma = 0.26$ and for vertical PGA including focal depth $A = -1.10$, $B = 0.200$, $C = -0.00015$, $D = -1$, $h_0 = h$ and $\sigma = 0.26$.

- Reviews and re-evaluates distances, focal depths, magnitudes and PGAs because data from variety of sources with different accuracy and reliability. For $M_s > 6.0$ distances have acceptable accuracy but for $M_s < 6.0$ distance, depths and magnitudes are poorly known. Errors in locations for $M_s < 6.0$ still large with no foreseeable means of improving them. Use of d_e for $M_s < 6.0$ justified because difference between d_f and d_e for small earthquakes is not larger than uncertainty in epicentre. Check and redetermine station locations; find large differences in excess of $15\,\mathrm{km}$ for some stations.

- Focal depths poorly determined. Revises 180 depths using S-start times (time between P and S-wave arrival).

- Focal depths $h < 26\,\mathrm{km}$; most (60%+) between 4 and $14\,\mathrm{km}$.

- Does not use M_L because no M_L values for Algeria, Iran, Pakistan, Turkey and former USSR and unreliable for other regions. Does not use magnitude calculated from strong-motion records because magnitude calculation requires point source approximation to be valid. Conversion from M_L to M_s should not be done because of uncertainty in conversion which should be retained.

- Notes that M_s results in nonlinear scaling on PGA with M_w due to nonlinear relationship between $\log M_0$ and M_s.

- Uses PGAs in four forms: maximum values from accelerograms read by others (34%), from corrected records (30%), scaled directly from accelerograms (13%) and from digitised plots (23%). Notes potential bias in using both corrected and uncorrected PGAs but neglects it because small difference ($\lesssim 4\%$ for those checked). Excludes PGAs near trigger level because processing errors can be large. Some unfiltered digital records which require additional processing to simulate SMA-1 could be associated with larger differences ($\lesssim 10\%$).

- Excludes records from basements and ground floors of structures with more than 3 levels. Retains the few records from dam abutments and tunnel portals.

- Excludes records generated by close small magnitude earthquakes triggered by S-wave.

- Does not exclude records obtained at distances greater than shortest distance to an operational but not triggered instrument because of non-constant or unknown trigger levels and possible malfunctions of instruments.

- Uses weighted regression of Joyner & Boore (1988) for second stage.

- Splits data into five magnitude dependent subsets: $2.0 \leq M_s \leq 7.3$ (1260 records from 619 shocks), $3.0 \leq M_s \leq 7.3$ (1189 records from 561 shocks), $4.0 \leq M_s \leq 7.3$ (830 records from 334 shocks), , $5.0 \leq M_s \leq 7.3$ (434 records from 107 shocks), and $3.0 \leq M_s \leq 6.0$ (976 records from 524 shocks). Calculates coefficients for each subset. Finds only small differences $\pm 15\%$ over distance range 1–$200\,\mathrm{km}$ between predictions and uncertainties. Concludes results stable. Prefers results from subset with $4.0 \leq M_s \leq 7.3$.

- Finds it difficult to obtain some vertical accelerations due to low ground motion so ignores data from $> 100\,\mathrm{km}$ with PGA $< 1\%\mathrm{g}$ ($0.1\,\mathrm{ms}^{-2}$).
- Repeats regression using $r^2 = d^2 + h^2$. Finds depth important.
- Calculates using one-stage method; finds very similar results for $10 < d < 100\,\mathrm{km}$.
- Considers magnitude dependent function: $\log a = b_1 + b_2 M_s + b_3 r + b_4[r + b_5 \exp(b_6 M_s)]$. Finds b_5 is zero so drops b_3 and repeats. Finds b_5 close to zero so magnitude dependent function not valid for this dataset.
- Local shear-wave velocity, V_s, profiles known for 44 stations (268 records from 132 earthquakes between 2.5 and 7.2) although only 14 from $> 40\,\mathrm{km}$ so barely sufficient to derive equation. Use 145 records from 50 earthquakes with $M_s > 4.0$ to fit $\log a = A + BM_s + Cr + D\log r + E\log V_{s30}$, where V_{s30} is average shear-wave velocity to reference depth of $30\,\mathrm{m}$. Finds C positive so constrain to zero. Find no reduction in standard deviation.
- Uses residuals from main equation to find E. Notes that should not be used because of small number of records. Considers different choices of reference depth; finds using between 5 and $10\,\mathrm{m}$ leads to higher predicted amplifications. Notes better to use V_{s30} because no need for subjective selection of categories.

2.85 *Dahle* et al. *(1995)*

- Ground motion model is:

$$\begin{aligned} \ln A &= c_1 + c_2 M_w + c_3 \ln R + c_4 R + c_5 S \\ \text{with: } R &= \sqrt{r^2 + r_h^2} \end{aligned}$$

  where A is in ms^{-2}, $c_1 = -1.579$, $c_2 = 0.554$, $c_3 = -0.560$, $c_4 = -0.0032$, $c_5 = 0.326$, $r_h = 6$ and $\sigma = 0.3535$
- Use records from Costa Rica, Mexico, Nicaragua and El Salvador. Only Mexican earthquakes with $M_w \geq 6.5$ were used.
- Use two site categories:

$S = 0$ Rock: 92 records

$S = 1$ Soil: 88 records

- Use a Bayesian one-stage regression method to yield physically possible coefficients.
- Consider tectonic type: subduction or shallow crustal but do not model.
- Find no significant difference between Guerrero (Mexico) and other data.
- Find no significant difference between subduction and shallow crustal data.

2.86 *Garcia-Fernandez & Canas (1995)*

- Ground motion model is:

$$\ln \mathrm{PGA} = \ln C_0 + C_1 M - \frac{1}{2}\ln r - \gamma r$$

  where PGA is in cms^{-2}, for Iberian Peninsula $\ln C_0 = -5.13$, $C_1 = 2.12$ and $\gamma = 0.0039$, for NE region $\ln C_0 = -4.74$, $C_1 = 2.07$ and $\gamma = 0.0110$ and for SSE region $\ln C_0 = -5.30$, $C_1 = 2.21$ and $\gamma = 0.0175$.

- Seismogram records, from short period analogue instruments, of Lg-waves from regional earthquakes in Iberian Peninsula were used.
- Estimate peak ground acceleration from Fourier amplitude spectra at $5\,\mathrm{Hz}$
- Iberia split into two regions: NE (north east part) and SSE (south and southeastern part).
- Use $5\,\mathrm{Hz}$ γ values from Garcia-Fernandez & Canas (1992) and Vives & Canas (1992).

2.87 *Lungu* et al. *(1995b)*

- Study almost identical to Radu *et al.* (1994), see Section 2.81, but different coefficients given: $c_1 = 3.672$, $c_2 = 1.318$, $c_3 = -1.349$, $c_4 = -0.0093$ and $\sigma = 0.395$.

2.88 *Molas & Yamazaki (1995)*

- Ground motion model is:

$$\log y = b_0 + b_1 M + b_2 r + b_3 \log r + b_4 h + c_i$$

where y is in cms^{-2}, $b_0 = 0.206$, $b_1 = 0.477$, $b_2 = -0.00144$, $b_3 = -1$, $b_4 = 0.00311$, $\sigma = 0.276$ and c_i is site coefficient for site i (use 76 sites), given in paper but are not reported here due to lack of space.

- Records from accelerometers on small foundations detached from structures; thus consider as free-field.
- Exclude records with one horizontal component with $\mathrm{PGA} < 1\,\mathrm{cms}^{-2}[0.01\,\mathrm{ms}^{-2}]$ because weaker records not reliable due to resolution ($\pm 0.03\,\mathrm{cms}^{-2}[0.0003\,\mathrm{ms}^{-2}]$) of instruments.
- Exclude earthquakes with focal depths equal to $0\,\mathrm{km}$ or greater than $200\,\mathrm{km}$, due to lack of such data. Depths (depth of point on fault plane closest to site), h, between about $1\,\mathrm{km}$ to $200\,\mathrm{km}$.
- Apply a low-cut filter with cosine-shaped transition from 0.01 to $0.05\,\mathrm{Hz}$.
- Positive correlation between magnitude and distance so use two-stage method.
- Note different definition for M_{JMA} for focal depths $> 60\,\mathrm{km}$.
- Firstly do preliminary analysis with $b_4 = 0$ and no site coefficients; find b_2 is positive so constrain to 0 but find $b_3 < -1.0$ so constrain b_3 to -1.0 and unconstrain b_2. Find linear dependence in residuals on h especially for $h < 100\,\mathrm{km}$. Find significant improvement in coefficient of determination, R^2, using terms $b_4 h$ and c.
- Find singularity in matrices if apply two-stage method, due to number of coefficients, so propose a iterative partial regression method.
- Also separate data into five depth ranges (A: $h = 0.1$ to $30\,\mathrm{km}$, 553 records from 111 earthquakes; B: $h = 30$ to $60\,\mathrm{km}$, 778 records from 136 earthquakes; C: $h = 60$ to $90\,\mathrm{km}$, 526 records from 94 earthquakes; D: $h = 90$ to $120\,\mathrm{km}$, 229 records from 31 earthquakes; E: $h = 120$ to $200\,\mathrm{km}$, 112 records from 19 earthquakes) and find attenuation equations for each range. Note results from D & E may not be reliable due to small number of records. Find similar results from each group and all data together.
- Find weak correlation in station coefficients with soil categories, as defined in Iwasaki *et al.* (1980), but note large scatter.

2.89 *Ambraseys* et al. *(1996) & Simpson (1996)*

- Ground motion model is:

$$\begin{aligned} \log y &= C_1' + C_2 M + C_4 \log r + C_A S_A + C_S S_S \\ \text{where } r &= \sqrt{d^2 + h_0^2} \end{aligned}$$

  where y is in g, $C_1' = -1.48$, $C_2 = 0.266$, $C_4 = -0.922$, $C_A = 0.117$, $C_S = 0.124$, $h_0 = 3.5$ and $\sigma = 0.25$.

- Use four site conditions but retain three (because only three records from very soft (L) soil which combine with soft (S) soil category):

  R Rock: $V_s > 750\,\mathrm{ms}^{-1}$, $\Rightarrow S_A = 0, S_S = 0$, 106 records.
  A Stiff soil: $360 < V_s \leq 750\,\mathrm{ms}^{-1}$, $\Rightarrow S_A = 1, S_S = 0$, 226 records.
  S Soft soil: $180 < V_s \leq 360\,\mathrm{ms}^{-1}$, $\Rightarrow S_A = 0, S_S = 1$, 81 records.
  L Very soft soil: $V_s \leq 180\,\mathrm{ms}^{-1}$, $\Rightarrow S_A = 0, S_S = 1$, 3 records.

- Lower limit of $M_s = 4.0$ because smaller earthquakes are generally not of engineering significance.
- Focal depths less than $30\,\mathrm{km}$, 81% between 5 and $15\,\mathrm{km}$.
- Note for some records distances have uncertainty of about $10\,\mathrm{km}$.
- Most records from distances less than about $40\,\mathrm{km}$.
- For some small events need to estimate M_s from other magnitude scales.
- Most records from free-field stations although some from basements or ground floors of relatively small structures, and tunnel portals. Do not exclude records from instruments beyond cutoff distance because of limited knowledge about triggered level.
- All uncorrected records plotted, checked and corrected for spurious points and baseline shifts.
- Uniform correction procedure was applied for all records. For short records ($< 5\,\mathrm{s}$) a parabolic adjustment was made, for long records ($> 10\,\mathrm{s}$) filtering was performed with pass band 0.20 to $25\,\mathrm{Hz}$ and for intermediate records both parabolic and filtering performed and the most realistic record was chosen. Instrument correction not applied due to limited knowledge of instrument characteristics.
- Also analyze using one-stage method, note results comparable.

2.90 *Ambraseys & Simpson (1996) & Simpson (1996)*

- Based on Ambraseys *et al.* (1996), see Section 2.89.
- Coefficients are: $C_1' = -1.74$, $C_2 = 0.273$, $C_4 = -0.954$, $C_A = 0.076$, $C_S = 0.058$, $h_0 = 4.7$ and $\sigma = 0.26$.

2.91 *Bommer* et al. *(1996)*

- Ground motion model is:

$$\ln(A) = a + bM + d\ln(R) + qh$$

  where h is focal depth, A is in g, $a = -1.47$, $b = 0.608$, $d = -1.181$, $q = 0.0089$ and $\sigma = 0.54$.

- Only use subduction earthquakes.
- Do not recommend equation used for hazard analysis, since derive it only for investigating equations of Climent *et al.* (1994).

2.92 Crouse & McGuire (1996)

- Ground motion model is:

$$\ln Y = a + bM + d\ln(R + c_1 \exp\{c_2 M\}) + eF$$

where Y is in g, for site category B: $a = -2.342699$, $b = 1.091713$, $c_1 = 0.413033$, $c_2 = 0.623255$, $d = -1.751631$, $e = 0.087940$ and $\sigma = 0.427787$ and for site category C: $a = -2.353903$, $b = 0.838847$, $c_1 = 0.305134$, $c_2 = 0.640249$, $d = -1.310188$, $e = -0.051707$ and $\sigma = 0.416739$.

- Use four site categories, $\bar{V}_s$ is shear-wave velocity in upper 100 ft (30 m):

  A Rock: $\bar{V}_s \geq 2500$ fps ($\bar{V}_s \geq 750\,\mathrm{ms}^{-1}$), 33 records

  B Soft rock or stiff soil: $1200 \leq \bar{V}_s \leq 2500$ fps ($360 \leq \bar{V}_s < 750\,\mathrm{ms}^{-1}$), 88 records

  C Medium stiff soil: $600 \leq \bar{V}_s < 1200$ fps ($180 \leq \bar{V}_s < 360\,\mathrm{ms}^{-1}$), 101 records

  D Soft clay: $\bar{V}_s < 600$ fps ($\bar{V}_s < 180\,\mathrm{ms}^{-1}$), 16 records

- Use two source mechanisms: reverse (R): $\Rightarrow F = 1$, 81 records and strike-slip (S) $\Rightarrow F = 0$, 157 records. Most (77) reverse records from $M_s \leq 6.7$.

- Most (231) records from small building (up to 3 storeys in height) or from instrument shelters to reduce effect of soil-structure interaction. 6 records from 6 storey buildings and 1 record from a 4 storey building, included because lack of data in site or distance range of these records. Structures thought not to appreciably affect intermediate or long period and at large distances short period ground motion more greatly diminished than long period so less effect on predictions.

- Exclude records from Eureka-Ferndale area in N. California because may be associated with subduction source, which is a different tectonic regime than rest of data. Also excluded Mammoth Lake records because active volcanic region, atypical of rest of California.

- Include one record from Tarzana Cedar Hills although exclude a different record from this station due to possible topographic effects.

- Most records between $6 \leq Ms \leq 7.25$ and $10 \leq R \leq 80$ km.

- Apply weighted regression separately for site category B and C. Data space split into 4 magnitude (6.0–6.25, 6.25–6.75, 6.75–7.25, 7.25+) and 5 distance intervals ($\leq$ 10 km, 10–20 km, 20–40 km, 40–80 km, 80 km+). Each recording within bin given same total weight.

- So that Y is increasing function of M and decreasing function of R for all positive M and R apply constraints. Define $g = b/d$ and $h = -(g + c_2)$, then rewrite equation $\ln Y = a + d\{gM + \ln[R + c_1 \exp(c_2 M)]\} + eF$ and apply constraints $g \leq 0$, $d \leq 0$, $c \geq 0$, $c_2 \geq 0$ and $h \geq 0$.

- Check plots of residuals (not shown in paper), find uniform distribution.

- Find e not significantly different than 0 and inconsistency in results between different soil classes make it difficult to attach any significance to fault type.

- Lack of records for A and D site categories. Find scale factors $k_1 = 0.998638$ and $k_2 = 1.200678$ so that $Y_A = k_1 Y_B$ and $Y_D = k_2 Y_C$, where Y_S is predicted ground motion for site class S. Find no obvious dependence of k_1 or k_2 on acceleration from examining residuals. Find k_1 and k_2 not significantly different than 1.

- Note limited data for $R < 10$ km, advise caution for this range.

- Note equation developed to estimate site-amplification factors not for seismic hazard analysis.

2.93 *Free (1996) & Free* et al. *(1998)*

- Ground motion model is:

$$\begin{aligned} \log(Y) &= C_1 + C_2\mathbf{M} + C_3\mathbf{M}^2 + C_4\log(R) + C_5(R) + C_6(S) \\ R &= \sqrt{d^2 + h_0^2} \end{aligned}$$

  where Y is in g, for $\mathbf{M} > 1.5$ using acceleration and velocity records, for horizontal PGA $C_1 = -4.2318$, $C_2 = 1.1962$, $C_3 = -0.0651$, $C_4 = -1$, $C_5 = -0.0019$, $C_6 = 0.261$, $h_0 = 2.9$ and $\sigma = 0.432$ and for vertical PGA $C_1 = -4.1800$, $C_2 = 1.0189$, $C_3 = -0.0404$, $C_4 = -1$, $C_5 = -0.0019$, $C_6 = 0.163$, $h_0 = 2.7$ and $\sigma = 0.415$.

- Use two site categories:

$S = 0$ Rock, H: 470 records, V: 395 records.

$S = 1$ Soil, H: 88 records, V: 83 records.

  Note that not most accurate approach but due to lack of site information consider this technique makes most consistent use of available information.

- Select data using these criteria:

  1. Epicentre and recording station must be within the stable continental region boundaries defined by Johnson *et al.* (1994) because a) such regions form end of spectrum of regions described by 'intraplate' and hence allows differences with interplate regions to be seen, b) they are clearly delineated regions and c) intraplate oceanic crust is excluded.
  2. Minimum magnitude level $\mathbf{M} = 1.5$.
  3. Use records from dam abutments and downstream free-field sites but excludes records from crests, slopes, toes, galleries, or basements.
  4. Use records from acceleration and velocity instruments.
  5. Specify no minimum PGA.
  6. Specify no maximum source distance. Do not exclude records from distances greater than shortest distance to a non-triggered station.

- Data from Australia, N.W. Europe, Peninsular India and E. N. America.

- Focal depths, $2 \leq h \leq 28\,\mathrm{km}$.

- Most records from $\mathbf{M} < 4.0$.

- Visually inspect all records including integrated velocities and displacements, identify and remove traces dominated by noise, identify and correct transient errors (spikes, ramps, linear sections, back time steps and clipped peaks), identify scaling errors, identify and remove multiple event records. Linear baseline correct and elliptically filter with cut-off 0.25 to $0.5\,\mathrm{Hz}$ (determine frequency by visual inspection of adjusted record) and 33 to $100\,\mathrm{Hz}$ (generally pre-determined by Nyquist frequency).

- Large proportion of records from velocity time histories which differentiate to acceleration. Test time domain method (central difference technique) and frequency domain method; find very similar results. Use time domain method.

- Distribution with respect to magnitude did not allow two-stage regression technique.

- In many analyses distribution of data with respect to distance did not allow simultaneous determination of coefficients C_4 and C_5, for these cases constrain C_4 to -1.

- Test effect of minimum magnitude cut-off for two cut-offs $\mathbf{M} = 1.5$ and $\mathbf{M} = 3.5$. Find if include data from $\mathbf{M} < 3.5$ then there is substantial over prediction of amplitudes for $d < 10\,\mathrm{km}$ for large magnitudes unless include C_3 term. C_3 effectively accounts for large number of records from small magnitudes and so predictions using the different magnitude cut-offs are very similar over broad range of $\mathbf{M}$ and d.

- Try including focal depth, h, explicitly by replacing h_0 with h because h_0 determined for whole set (which is dominated by small shocks at shallow depths) may not be appropriate for large earthquakes. Find improved fit at small distances but it does not result in overall improvement in fit (σ increases); this increase thought due to large errors in focal depth determination.

- Find larger standard deviations than those found in previous studies which note may be due to intrinsic differences between regional subsets within whole set. Repeat analysis separately for Australia (for horizontal and vertical), N. America (for horizontal and vertical) and N.W. Europe (horizontal); find reduced standard deviations (although still large), C_5 varies significantly between 3 regions.

- Repeat analysis excluding velocity records.

- Also repeat analysis using only rock records.

2.94 *Ohno* et al. *(1996)*

- Ground motion model is:

$$\log S(T) = a(T)M - \log X_{eq} - b(T)X_{eq} + c(T) + q\Delta s(T)$$

  where $S(0.02)$ is in gal, $a(0.02) = 0.318$, $b(0.02) = 0.00164$ and $c(0.02) = 1.597$ ($\Delta s(0.02)$ and σ only given in graphs).

- Use two site conditions:

  $q = 0$ Pre-Quaternary: Rock (sandstone, siltstone, shale, granite, mudstone, etc.); thickness of surface soil overlying rock is less than $10\,\mathrm{m}$; shallow soil or thin alluvium, 160 records. S-wave velocities $> 600\,\mathrm{ms}^{-1}$.

  $q = 1$ Quaternary: Soil (alluvium, clay, sand, silt, loam, gravel, etc.), 336 records. S-wave velocities $\leq 600\,\mathrm{ms}^{-1}$.

  Exclude records from very soft soil such as bay mud or artificial fill because few such records and ground motions may be strongly affected by soil nonlinearity.

- Use equivalent hypocentral distance, X_{eq}, because strong motion in near-source region affected from points other than nearest point on fault plane.

- Use portion of record after initial S-wave arrival.

- Approximates PGA by spectral acceleration for period of $0.02\,\mathrm{s}$ and 5% damping.

- Plot the amplitude factors from first stage against M_w; find well represented by linear function.

2.95 *Sarma & Srbulov (1996)*

- Ground motion model is:

$$\begin{aligned} \log(A_p/\mathrm{g}) &= b_1 + b_2 M_s + b_3 \log r + b_4 r \\ \text{where } r &= (d^2 + h_0^2)^{0.5} \end{aligned}$$

where A_p is in g, using both horizontal components $b_1 = -1.617$, $b_2 = 0.248$, $b_3 = -0.5402$, $b_4 = -0.00392$, $h_0 = 3.2$ and $\sigma = 0.26$ and for larger horizontal component $b_1 = -1.507$, $b_2 = 0.240$, $b_3 = -0.542$, $b_4 = -0.00397$, $h_0 = 3.0$ and $\sigma = 0.26$.

- Consider two soil categories but do not model:

  1. Rock
  2. Soil

  Classify sites without regard to depth and shear-wave velocity of deposits.

- Most records from W. USA but many from Europe and Middle East.
- Focal depths between 2 and 29 km.
- Records from instruments on ground floor or in basements of buildings and structures up to 3 storeys and at free-field sites, regardless of topography.
- Records baseline corrected and low-pass filtered using elliptic filter.

2.96 *Spudich* et al. *(1996) & Spudich* et al. *(1997)*

- Ground motion model is:

$$\begin{aligned} \log_{10} Y &= b_1 + b_2(M-6) + b_3(M-6)^2 + b_4 R + b_5 \log_{10} R + b_6 \Gamma \\ \text{where } R &= \sqrt{r_{jb}^2 + h^2} \end{aligned}$$

where Y is in g, $b_1 = 0.156$, $b_2 = 0.229$, $b_3 = 0$, $b_4 = 0$, $b_5 = -0.945$, $b_6 = 0.077$, $h = 5.57$, $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2}$ where $\sigma_1 = 0.216$, $\sigma_2 = 0$, for randomly orientated component $\sigma_3 = 0.094$ and for geometric mean $\sigma_3 = 0$.

- Use two site categories (following classification of Joyner & Boore (1981)):

$\Gamma = 0$ Rock: 35 records

$\Gamma = 1$ Soil: 93 records

- Applicable for extensional regimes, i.e. those regions where lithosphere is expanding areally.
- Reject records from structures of more than two storeys or from deeply embedded basements or those which triggered on S wave.
- Include records from those instruments beyond cutoff distance, i.e. beyond first instrument which did not trigger.
- Correction technique based on uniform correction and processing. Determine passband for filtering based on visual inspection of Fourier amplitude spectra and doubly-integrated displacements. Apply instrument correction.
- Not enough data to be able to find all coefficients so use b_2 and b_3 from Boore *et al.* (1994a)
- Note that should only be used in distance range 0 to 70 km because further away ground motions tend to be over predicted.

2.97 Campbell (1997) & Campbell & Bozorgnia (1994)

- Ground motion model (horizontal component) is:

$$\begin{aligned}\ln A_H &= a_1 + a_2 M + a_3 \ln \sqrt{R_{\rm SEIS}^2 + [a_4 \exp(a_5 M)]^2} \\ &\quad + [a_6 + a_7 \ln R_{\rm SEIS} + a_8 M] F + [a_9 + a_{10} \ln R_{\rm SEIS}] S_{\rm SR} \\ &\quad + [a_{11} + a_{12} \ln R_{\rm SEIS}] S_{\rm HR}\end{aligned}$$

  where A_H is in g, $a_1 = -3.512$, $a_2 = 0.904$, $a_3 = -1.328$, $a_4 = 0.149$, $a_5 = 0.647$, $a_6 = 1.125$, $a_7 = -0.112$, $a_8 = -0.0957$, $a_9 = 0.440$, $a_{10} = -0.171$, $a_{11} = 0.405$, $a_{12} = -0.222$, $\sigma = 0.55$ for $A_H < 0.068\,\mathrm{g}$, $\sigma = 1.73 - 0.140 \ln(A_H)$ for $0.068\,\mathrm{g} \le A_H \le 0.21\,\mathrm{g}$ and $\sigma = 0.39$ for $A_H > 0.21\,\mathrm{g}$.

  Ground motion model (vertical component) is:

$$\begin{aligned}\ln A_V &= \ln A_H + b_1 + b_2 M + b_3 \ln[R_{\rm SEIS} + b_4 \exp(b_5 M)] \\ &\quad + b_6 \ln[R_{\rm SEIS} + b_7 \exp(b_8 M)] + b_9 F\end{aligned}$$

  where A_V is in g, $b_1 = -1.58$, $b_2 = -0.10$, $b_3 = -1.5$, $b_4 = 0.079$, $b_5 = 0.661$, $b_6 = 1.89$, $b_7 = 0.361$, $b_8 = 0.576$, $b_9 = -0.11$ and $\sigma_V = \sqrt{\sigma^2 + 0.36^2}$ (where σ is standard deviation for horizontal PGA prediction).

- Uses three site categories:

$S_{\rm SR} = 0, S_{\rm HR} = 1$ Hard rock: primarily Cretaceous and older sedimentary deposits, metamorphic rock, crystalline rock and hard volcanic deposits (e.g. basalt).

$S_{\rm SR} = 1, S_{\rm HR} = 0$ Soft rock: primarily Tertiary sedimentary deposits and soft volcanic deposits (e.g. ash deposits).

$S_{\rm SR} = 0, S_{\rm HR} = 0$ Alluvium or firm soil: firm or stiff Quaternary deposits with depths greater than $10\,\mathrm{m}$.

- Restricts to near-source distances to minimize influence of regional differences in crustal attenuation and to avoid complex propagation effects that have been observed at longer distances.

- Excludes recordings from basement of buildings greater than two storeys on soil and soft rock, greater than five storeys on hard rock, toe and base of dams and base of bridge columns. Excludes recordings from shallow and soft soil because previous analyses showed such sites have accelerations significantly higher than those on deep, firm alluvium. Include records from dam abutments because comprise a significant number of rock recordings and due to stiff foundations are expected to be only minimally affected by dam. Some of these could be strongly affected by local topography.

- Includes earthquakes only if they had seismogenic rupture within shallow crust (depths less than about $25\,\mathrm{km}$). Includes several large, shallow subduction interface earthquakes because previous studies found similar near-source ground motions to shallow crustal earthquakes.

- Includes only earthquakes with M about 5 or larger to emphasize those ground motions of greatest engineering interest and limit analysis to more reliable, well-studied earthquakes.

- Notes that distance to seismogenic rupture is a better measure than distance to rupture or distance to surface projection because top layer of crust is non-seismogenic and will not contribute to ground motion. Give estimates for average depth to top of seismogenic rupture for hypothetical earthquakes.

- Considers different focal mechanisms: reverse (H:6, V:5), thrust (H:9, V:6), reverse-oblique (H:4, V:2) and thrust-oblique (0), total (H:19, V:13) $\Rightarrow F = 1$ (H:278 records, V:116 records) (reverse have a dip angle greater than or equal to $45°$), strike-slip (H:27, V:13) $\Rightarrow F = 0$ (H:367 records, V:109 records) (strike-slip have an absolute value of rake less than or equal to $22.5°$ from the horizontal as measured along fault plane). There is only one normal faulting earthquakes in set of records (contributing four horizontal records) so difference is not modelled although $F = 0.5$ given as first approximation.

- Mostly W. USA with 20 records from Nicaragua(1) Mexico (5), Iran (8), Uzbekistan (1), Chile (3), Armenia (1) and Turkey (1).
- Does regression firstly with all data. Selects distance threshold for each value of magnitude, style of faulting and local site condition such that the 16th percentile estimate of A_H was equal to $0.02\,\mathrm{g}$ (which corresponds to a vertical trigger of about $0.01\,\mathrm{g}$). Repeats regression repeated only with those records within these distance thresholds. Avoids bias due to non-triggering instruments.
- Finds dispersion (uncertainty) to be dependent on magnitude and PGA, models as linear functions. Finds better fit for PGA dependency.

2.98 *Munson & Thurber (1997)*

- Ground motion model is:

$$\begin{aligned} \log_{10}\mathrm{PGA} &= b_0 + b_1(M-6) + b_2 r - log_{10} r + b_4 S \\ \text{where } r &= \sqrt{d^2+h^2} \end{aligned}$$

  PGA is in g, $b_0 = 0.518$, $b_1 = 0.387$, $b_2 = -0.00256$, $b_4 = 0.335$, $h = 11.29$ and $\sigma = 0.237$.

- Use two site categories:

$S = 0$ Lava: 38 records

$S = 1$ Ash: $60 \lesssim V_s \lesssim 200\,\mathrm{ms}^{-1}$, 13 records

- Depths between 4 and $14\,\mathrm{km}$ with average $9.6\,\mathrm{km}$ (standard deviation $2.3\,\mathrm{km}$). Limit of $15\,\mathrm{km}$ chosen to differentiate between large tectonic earthquakes and deeper mantle events.
- Attenuation greater than for western USA due to highly fractured volcanic pile.
- Peak acceleration measured directly from accelerograms. Check against one from corrected records, small difference.
- Excludes records triggered on S-wave and those beyond cutoff distance (the distance to first nontriggered instrument).
- Does weighted and unweighted least squares analysis; find some differences.

2.99 *Youngs* et al. *(1997)*

- Ground motion model for soil is:

$$\begin{aligned} \ln\mathrm{PGA} &= C_1^* + C_2\mathbf{M} + C_3^* \ln\left[r_{\mathrm{rup}} + \mathrm{e}^{C_4^* - \frac{C_2}{C_3^*}\mathbf{M}}\right] + C_5 Z_t + C_9 H + C_{10} Z_{ss} \\ \text{with: } C_1^* &= C_1 + C_6 Z_r \\ C_3^* &= C_3 + C_7 Z_r \\ C_4^* &= C_4 + C_8 Z_r \end{aligned}$$

  where PGA is in g, $C_1 = -0.6687$, $C_2 = 1.438$, $C_3 = -2.329$, $C_4 = \ln(1.097)$, $C_5 = 0.3643$, $C_9 = 0.00648$ and $\sigma = 1.45 - 0.1\mathbf{M}$ (other coefficients in equation not needed for prediction on deep soil and are not given in paper).

Ground motion model for rock is:

$$\begin{aligned} \ln \mathrm{PGA} &= C_1^* + C_2\mathbf{M} + C_3^* \ln\left[r_{\mathrm{rup}} + \mathrm{e}^{C_4^* - \frac{C_2}{C_3^*}\mathbf{M}}\right] + C_5 Z_{ss} + C_8 Z_t + C_9 H \\ \text{with: } C_1^* &= C_1 + C_3 C_4 - C_3^* C_4^* \\ C_3^* &= C_3 + C_6 Z_{ss} \\ C_4^* &= C_4 + C_7 Z_{ss} \end{aligned}$$

where PGA is in g, $C_1 = 0.2418$, $C_2 = 1.414$, $C_3 = -2.552$, $C_4 = \ln(1.7818)$, $C_8 = 0.3846$, $C_9 = 0.00607$ and $\sigma = 1.45 - 0.1\mathbf{M}$ (other coefficients in equation not needed for prediction on rock and are not given in paper).

Use different models to force rock and soil accelerations to same level in near field.

- Use three site categories to do regression but only report results for rock and deep soil:

$Z_r = 1, Z_{ds} = 0, Z_{ss} = 0$ Rock: Consists of at most about a metre of soil over weathered rock, 96 records.

$Z_{ds} = 1, Z_r = 0, Z_{ss} = 0$ Deep soil: Depth to bedrock is greater than $20\,\mathrm{m}$, 284 records.

$Z_{ss} = 1, Z_{ds} = 0, Z_r = 0$ Shallow soil: Depth to bedrock is less than $20\,\mathrm{m}$ and a significant velocity contrast may exist within $30\,\mathrm{m}$ of surface, 96 records.

- Use free-field recordings, i.e. instruments in basement or ground-floor of buildings less than four storeys in height. Data excluded if quality of time history poor or if portion of main shaking not recorded.
- Consider tectonic type: interface (assumed to be thrust) (98 records) $\Rightarrow Z_t = 0$, intraslab (assumed to be normal) (66 records) $\Rightarrow Z_t = 1$
- Focal depths, H, between 10 and $229\,\mathrm{km}$
- Not enough data to perform individual regression on each subset so do joint regression analysis.
- Both effect of depth and tectonic type significant.
- Large differences between rock and deep soil.
- Note differences between shallow crustal and interface earthquake primarily for very large earthquakes.
- Assume uncertainty to be linear function of magnitude.

2.100 *Zhao* et al. *(1997)*

- Ground motion model (Model 1) is:

$$\log_{10} \mathrm{PGA} = A_1 M_w + A_2 \log_{10} \sqrt{r^2 + d^2} + A_3 h_c + A_4 + A_5 \delta_R + A_6 \delta_A + A_7 \delta_I$$

where PGA is in ms^{-2}, $\delta_R = 1$ for crustal reverse 0 otherwise, $\delta_A = 1$ for rock 0 otherwise, $\delta_I = 1$ for interface 0 otherwise, $A_1 = 0.298$, $A_2 = -1.56$, $A_3 = 0.00619$, $A_4 = -0.365$, $A_5 = 0.107$, $A_6 = -0.186$, $A_7 = -0.124$, $d = 19$ and $\sigma = 0.230$.

- Models also given for soil sites only (Model 2), unspecified site (Model 3), focal mechanism and tectonic type unknown (Model 4) and only magnitude, depth and distance known (Model 5)
- Records from ground or base of buildings. 33 from buildings with more than 3 storeys; find no significant differences.
- Retain two site categories:

1. Rock: Topographic effects expected, very thin soil layer ($\leq 3\,\mathrm{m}$) overlying rock or rock outcrop.
2. Soil: everything else

- Use depth to centroid of rupture, h_c, $4 \leq h_c \leq 149$. Only nine are deeper than $50\,\mathrm{km}$. Exclude records from deep events which travelled through mantle.
- Consider tectonic type: C=crustal (24+17 records), I=interface (7+0 records) and S=slab (20+0 records)
- Consider source mechanism: N=normal (15+1 records), R=reverse (22+5 records) and S=strike-slip (12+11 records). Classify mixed mechanisms by ratio of components ≥ 1.0.
- For only five records difference between the distance to rupture surface and the distance to centroid could be more than 10%.
- 66 foreign near-source records ($d_r \leq 10\,\mathrm{km}$) from 17 crustal earthquakes supplement NZ data. Mainly from western North America including 17 from Imperial Valley and 12 from Northridge.
- Exclude one station's records (Atene A) due to possible topographical effects.
- Exclude records which could have been affected by different attenuation properties in the volcanic region.
- Note regional difference between Fiordland and volcanic region and rest of country but do model.
- Retain coefficients if significant at $\alpha = 0.05$.
- Anelastic term not significant.

2.101 *Bouhadad* et al. *(1998)*

- Ground motion model is:

$$A = c\exp(\alpha M)[R^k + a]^{-\beta-\gamma R}$$

- Coefficients not given, only predictions.

2.102 *Manic (1998)*

- Ground motion model is:

$$\begin{aligned} \log(A) &= c_1 + c_2 M + c_3 \log(D) + c_4 D + c_5 S \\ D &= (R^2 + d_0^2)^{1/2} \end{aligned}$$

  where A is in g, $c_1 = -1.664$, $c_2 = 0.333$, $c_3 = -1.093$, $c_4 = 0$, $c_5 = 0.236$, $d_0 = 6.6$ and $\sigma = 0.254$.

- Uses four site categories (following Ambraseys *et al.* (1996)) but only two have data within them:

$S = 0$ Rock (R): $v_s > 750\,\mathrm{ms}^{-1}$, 92 records.

$S = 1$ Stiff soil (A): $360 < v_s \leq 750\,\mathrm{ms}^{-1}$, 184 records.

  where v_s is average shear-wave velocity in upper $30\,\mathrm{m}$.

- Uses both horizontal components to get a more reliable set of data.
- Tries using M_L rather than M_s, epicentral distance rather than hypocentral distance and constraining anelastic decay coefficient, c_4, to zero. Chooses combination which gives minimum σ.

2.103 *Rinaldis* et al. *(1998)*

- Ground motion model is:

$$\ln Y = C_{14} + C_{22}M + C_{31}\ln(R + 15) + C_{43}S + C_{54}F$$

where Y is in cms^{-2}, $C_{14} = 5.57$, $C_{22} = 0.82$, $C_{31} = -1.59$, $C_{43} = -0.14$, $C_{54} = -0.18$ and $\sigma = 0.68$. Assume $15\,\mathrm{km}$ inside $\ln(R + \ldots)$ from Theodulidis & Papazachos (1992).

- Use two site categories:

$S = 0$ Rock: includes stiff sites.

$S = 1$ Alluvium: includes both shallow and deep soil sites.

- Use two source mechanism categories:

$F = 0$ Thrust and strike-slip earthquakes.

$F = 1$ Normal earthquakes.

- Use epicentral distance because in Italy and Greece the surface geology does not show any evident faulting, consequently it is impossible to use a fault distance definition.

- Good distribution and coverage of data with respect to site category and source mechanism.

- Consider six strong-motion records (three Italian and three Greek) with different associated distances, magnitudes and record length and apply the different processing techniques of ENEA-ENEL and ITSAK to check if data from two databanks can be merged. Digitise six records using same equipment. ITSAK technique: subtract the reference trace (either fixed trace or trace from clock) from uncorrected accelerogram and select band-pass filter based on either Fourier amplitude spectra of acceleration components or selected using a different technique. ENEA-ENEL technique: subtract the reference trace from uncorrected accelerogram and select band-pass filter by comparing Fourier amplitude spectra of acceleration components with that of fixed trace. Find small differences in PGA, PGV, PGD so can merge Italian and Greek data into one databank.

- Use four step regression procedure, similar to that Theodulidis & Papazachos (1992) use. First step use only data with $M \geq 6.0$ ($7 \leq R \leq 138\,\mathrm{km}$) for which distances are more accurate to find geometrical coefficient C_{31}. Next find constant (C_{12}) and magnitude (C_{22}) coefficients using all data. Next find constant (C_{13}) and soil (C_{43}) coefficients using all data. Finally find constant (C_{14}) and source mechanism (C_{54}) coefficients using data with $M \geq 6.0$ for which focal mechanism is better constrained; final coefficients are C_{14}, C_{22}, C_{31}, C_{43} and C_{54}. Investigate influence of distance on C_{54} by subdividing data in final step into three categories with respect to distance ($7 \leq R \leq 140\,\mathrm{km}$, $7 \leq R \leq 100\,\mathrm{km}$ and $7 \leq R \leq 70\,\mathrm{km}$).

- Equation intended as first attempt to obtain attenuation relations from combined databanks and site characteristics and fault rupture properties could and should be taken into account.

2.104 *Sadigh & Egan (1998)*

- Based on Sadigh *et al.* (1997), see Section 2.73.

- Ground motion model is:

$$\ln \mathrm{PGA} = C_1 + C_2M + C_3\ln[r_{\mathrm{rup}} + \exp(C_4 + C_5M)]$$

where PGA is in g, for $M < 6.5$ $C_4 = 1.29649$ and $C_5 = 0.25$ and for $M \geq 6.5$ $C_4 = -0.48451$ and $C_5 = 0.524$. For rock sites: $C_3 = -2.100$, for strike-slip mechanism and $M < 6.5$ $C_1 = -0.949$

and $C_2 = 1.05$, for strike-slip mechanism and $M \geq 6.5$ $C_1 = -1.274$ and $C_2 = 1.10$, for reverse-slip and $M < 6.5$ $C_1 = 0.276$ and $C_2 = 0.90$ and for reverse-slip and $M \geq 6.5$ $C_1 = -1.024$ and $C_2 = 1.10$. For soil sites: $C_3 = -1.75$, for strike-slip mechanism and $M < 6.5$ $C_1 = -1.1100$ and $C_2 = 0.875$, for strike-slip mechanism and $M \geq 6.5$ $C_1 = -1.3830$ and $C_2 = 0.917$, for reverse-slip mechanism and $M < 6.5$ $C_1 = -0.0895$ and $C_2 = 0.750$ and for reverse-slip mechanism and $M \geq 6.5$ $C_1 = -1.175$ and $C_2 = 0.917$ (σ not given).

- Use two site categories:

  1. Rock: bedrock within about a metre of surface. Note that many such sites are soft rock with $V_s \leq 750\,\mathrm{ms}^{-1}$ and a strong velocity gradient because of near-surface weathering and fracturing, 274 records.
  2. Deep soil: greater than $20\,\mathrm{m}$ of soil over bedrock. Exclude data from very soft soil sites such as those from San Francisco bay mud, 690 records.

- Define crustal earthquakes as those that occur on faults within upper 20 to $25\,\mathrm{km}$ of continental crust.

- Consider souce mechanism: RV=reverse (26+2) and SS=strike-slip (and some normal) (89+0). Classified as RV if rake$> 45°$ and SS if rake$< 45°$. Find peak motions from small number of normal faulting earthquakes not to be significantly different than peak motions from strike-slip events so include in SS category.

- Separate equations for $M_w < 6.5$ and $M_w \geq 6.5$ to account for near-field saturation effects, for rock and deep soil sites and reverse and strike-slip earthquakes.

- Records from instruments in instrument shelters near ground surface or in ground floor of small, light structures.

- 4 foreign records (1 from Gazli and 3 from Tabas) supplement Californian records.

2.105 Sarma & Srbulov (1998)

- Ground motion model is:

$$\log(a_p/g) = C_1 + C_2 M_s + C_3 d + C_4 \log d$$

  where a_p is in g, for soil sites $C_1 = -1.86$, $C_2 = 0.23$, $C_3 = -0.0062$, $C_4 = -0.230$ and $\sigma = 0.28$ and for rock sites $C_1 = -1.874$, $C_2 = 0.299$, $C_3 = -0.0029$, $C_4 = -0.648$ and $\sigma = 0.33$.

- Use two site categories because of limited available information (based on nature of top layer of site regardless of thickness) for which derive separate equations:

  1. Soil
  2. Rock

- Use record from free-field or in basements of buildings ≤ 3 storeys high.

- Use M_s because better represents size of shallow earthquakes and is determined from teleseismic readings with much smaller standard errors than other magnitude scales and also saturates at higher magnitudes than all other magnitude scales except M_w which is only available for relatively small portion of earthquakes. For some small earthquakes convert to M_s from other magnitude scales.

- For very short records, $\leq 5\,\mathrm{s}$ long, correct using parabolic baseline, for records $> 10\,\mathrm{s}$ long correct using elliptical filter and for records between 5 and $10\,\mathrm{s}$ long both parabolic correction and filtering applied and select best one from appearance of adjusted time histories.

- Equations not any more precise than other attenuation relations but are simply included for completeness and for a comparison of effects of dataset used with other dataset. Data did not allow distinction between different source mechanisms.

2.106 Sharma (1998)

- Ground motion model is:
$$\log A = c_1 + c_2 M - b \log(X + \mathrm{e}^{c_3 M})$$
where A is in g, $c_1 = -1.072$, $c_2 = 0.3903$, $b = -1.21$, $c_3 = 0.5873$ and $\sigma = 0.14$.

- Considers two site categories but does not model:

  R Rock: generally granite/quartzite/sandstone, 41 records.

  S Soil: exposed soil covers on basement, 25 records.

- Focal depths between 7.0 and $50.0\,\mathrm{km}$.

- Most records from distances $> 50\,\mathrm{km}$. Correlation coefficient between M and X is 0.63.

- Does not include source mechanism as parameter because not well defined and including many terms may lead to errors. Also neglects tectonic type because set is small and small differences are expected.

- Fit $\log A = -b \log X + c$ to data from each earthquake separately and find average b equal to 1.292. Then fit $\log A = aM - b \log X + c$ to data from all earthquakes and find $b = 0.6884$. Fit $\log A = -b \log X + \sum d_i l_i$ to all data, where $l_i = 1$ for ith earthquake and 0 otherwise and find $b = 1.21$, use this for rest of analysis.

- Use weighted regression, due to nonuniform sampling over all M and X. Divide data into distance bins $2.5\,\mathrm{km}$ wide up to $10\,\mathrm{km}$ and logarithmically dependent for larger distances. Within each bin each earthquake is given equal weight by assigning a relative weight of $1/n_{j,l}$, where $n_{j,l}$ is the number of recordings for jth earthquake in lth distance bin, then normalise so that sum to total number of recordings.

- Original data included two earthquakes with focal depths $91.0\,\mathrm{km}$ and $119.0\,\mathrm{km}$ and $M = 6.8$ and 6.1 which caused large errors in regression parameters due to large depths so excluded them.

- Check capability of data to compute coefficients by deleting, in turn, c_1, c_2 and c_3, find higher standard deviation.

- Makes one coefficient at a time equal to values given in Abrahamson & Litehiser (1989), finds sum of squares increases.

- Notes lack of data could make relationship unreliable.

2.107 Smit (1998)

- Ground motion model is:
$$\log Y = a + bM - \log R + dR$$
where Y is in $\mathrm{nm/s^2}$, $b = 0.868$, $d = -0.001059$, $\sigma = 0.35$, for horizontal PGA $a = 5.230$ and for vertical PGA $a = 5.054$.

- Most records from rock sites.

- Focal depths between 0 and about $27\,\mathrm{km}$ (most less than $10\,\mathrm{km}$).

- Most records from $M_L < 3.5$.
- Most earthquakes have strike-slip mechanism.
- Uses records from high gain short period seismographs and from strong-motion instruments.
- Records are instrument corrected.
- Eliminates some far-field data from small magnitude earthquakes using signal to noise ratio criterion.
- Records cover entire azimuthal range.
- Notes that need more data in near field.
- Notes that care must be taken when using equations for prediction of ground motion in strong earthquakes ($M \approx 6$) because of lack of data.

2.108 Chapman (1999)

- Ground motion model is:

$$\log_{10} Y = a + b(M-6) + c(M-6)^2 + d\log(r^2+h^2)^{1/2} + eG_1 + fG_2$$

where Y is in cms^{-2}, $a = 3.098$, $b = 0.3065$, $c = -0.07570$, $d = -0.8795$, $h = 6.910$, $e = 0.1452$, $f = 0.1893$ and $\sigma = 0.2124$.

- Use three site categories:

A & B $V_{s,30} > 760\,\mathrm{ms}^{-1}$, 24 records $\Rightarrow G_1 = 0, G_2 = 0$.

C $360 < V_{s,30} \leq 760\,\mathrm{ms}^{-1}$, 116 records $\Rightarrow G_1 = 1, G_2 = 0$.

D $180 < V_{s,30} \leq 360\,\mathrm{ms}^{-1}$, 164 records $\Rightarrow G_1 = 0, G_2 = 1$.

- Uses records from ground level or in basements of structures of two stories or less, and excludes records from dam or bridge abutments.
- Selects records which include major motion portion of strong-motion episode, represented by S wavetrain. Excludes records triggered late on S wave or those of short duration terminating early in coda.
- Most records already corrected. Some records instrument corrected and 4-pole causal Butterworth filtered (corner frequencies 0.1 and $25\,\mathrm{Hz}$). Other records instrument corrected and 4-pole or 6-pole causal Butterworth bandpass filtered (corner frequencies 0.2 and $25\,\mathrm{Hz}$). All data filtered using 6-pole causal high-pass Butterworth filter with corner frequency $0.2\,\mathrm{Hz}$ and velocity and displacement curves examined.
- Uses method of Campbell (1997) to reduce bias due to non-triggered instruments, for some recent shocks. Firstly uses all data to determine minimum distances (which are functions of magnitude and site condition) at which 16th percentile values of PGA are $< 0.02\,\mathrm{g}[0.2\,\mathrm{ms}^{-1}]$ (corresponding to $0.01\,\mathrm{g}[0.1\,\mathrm{ms}^{-1}]$ vertical component trigger threshold). Next delete records from larger distances and repeat regression.
- Check residuals against distance and magnitude for each site class; find no obvious non-normal magnitude or distance dependent trends.

2.109 *Cousins* et al. *(1999)*

- Based on Zhao *et al.* (1997) see Section 2.100
- Ground motion model is:

$$\log_{10} \mathrm{PGA} = A_1 M_w + A_2 \log_{10} R + A_3 h_c + A_4 + A_5 + A_6 + A_7 R + A_8 M_w + A_9 + A_{10} R_v$$

  where PGA is in ms^{-2}, $R = \sqrt{r^2 + d^2}$ and R_v is distance travelled by direct seismic wave through volcanic region. A_5 only for crustal reverse, A_6 only for interface, A_7 only for strong and weak rock, A_8 only for strong rock, A_9 only for strong rock, $A_1 = 0.2955$, $A_2 = -1.603$, $A_3 = 0.00737$, $A_4 = -0.3004$, $A_5 = 0.1074$, $A_6 = -0.1468$, $A_7 = -0.00150$, $A_8 = 0.3815$, $A_9 = -2.660$, $A_{10} = -0.0135$, $d = 19.0$ and $\sigma = 0.24$.

- Originally considers five site categories but retain three:
  1. Strong rock: $V_s > 700\,\mathrm{ms}^{-1}$
  2. Weak rock: $375 \leq V_s \leq 700\,\mathrm{ms}^{-1}$ and category AV those sites with a very thin layer ($\leq 3\,\mathrm{m}$) overlying rock
  3. Soil: everything else
- Depth to centroid of rupture, h_c, used, $4 \leq h_c \leq 94\,\mathrm{km}$.
- 60% on soil, 40% on rock
- Consider tectonic type: C=Crustal (12+17), I=Interface (5+0) and S=Slab(8+0)
- Consider source mechanism: N=normal (6+1), R=reverse (12+5) and S=strike-slip (7+11). Mixed classified by ratio of components ≥ 1.0.
- Mixture of analogue and digital accelerograms (72%) and seismograms (28%)
- Accelerograms sampled at 100–250 samples/sec. Bandpass frequencies chosen by analysis of Fourier amplitude spectrum compared with noise spectrum. $f_{\min}$ between 0.15 and $0.5\,\mathrm{Hz}$ and $f_{\max}$ equal to $25\,\mathrm{Hz}$. Instrument correction applied to analogue records.
- Seismograms sampled at 50–100 samples/sec. Differentiated once. Instrument corrected and high pass filtered with $f_{\min} = 0.5\,\mathrm{Hz}$. No low pass filter needed.
- Clipped seismograms usually retained.
- Directional effect noticed but not modelled.
- Most records from more than $100\,\mathrm{km}$ away. Note lack of near-source data.
- Records from accelerograms further away than first operational non-triggering digital accelerograph, which had a similar triggering level, were excluded.
- Models difference between high attenuating volcanic and normal regions.

2.110 *Spudich* et al. *(1999)*

- Update of Spudich *et al.* (1997) see Section 2.96.
- Ground motion model is:

$$\begin{aligned} \log_{10} Z &= b_1 + b_2(M-6) + b_3(M-6)^2 + b_5 \log_{10} D + b_6 \Gamma \\ \text{with: } D &= \sqrt{r_{jb}^2 + h^2} \end{aligned}$$

  where Z is in g, $b_1 = 0.299$, $b_2 = 0.229$, $b_3 = 0$, $b_5 = -1.052$, $b_6 = 0.112$, $h = 7.27$ and $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2}$ where $\sigma_1 = 0.172$, $\sigma_2 = 0.108$ and for randomly oriented horizontal component $\sigma_3 = 0.094$ and for larger horizontal component $\sigma_3 = 0$.

- Use two site categories (could not use more or $V_{s,30}$ because not enough data):

$\Gamma = 0$ Rock: includes hard rock (12 records) (plutonic igneous rocks, lava flows, welded tuffs and metamorphic rocks unless severely weathered when they are soft rock), soft rock (16 records) (all sedimentary rocks unless there was some special characteristic noted in description, such as crystalline limestone or massive cliff-forming sandstone when they are hard rock) and unknown rock (8 records). 36 records in total.

$\Gamma = 1$ Soil (alluvium, sand, gravel, clay, silt, mud, fill or glacial outwash of more than $5\,\mathrm{m}$ deep): included shallow soil (8 records) (5 to $20\,\mathrm{m}$ deep), deep soil (77 records) ($> 20\,\mathrm{m}$ deep) and unknown soil (21 records). 106 records in total.

- Applicable for extensional regimes, i.e. those regions where lithosphere is expanding areally. Significantly different ground motion than non-extensional areas.
- Criteria for selection of records is: $M_w \geq 5.0$, $d_f \leq 105\,\mathrm{km}$. Reject records from structures of more than two storeys or from deeply embedded basements or those which triggered on S wave. Also reject those close to dams which may be affected by dam. Also only use records already digitised.
- Include records from those instrument beyond cutoff distance, i.e. beyond first instrument which did not trigger, because of limited records and lack of data on non-triggering.
- Not enough data to be able to find all coefficients so use b_2 and b_3 from Boore *et al.* (1993) and b_6 from Boore *et al.* (1994a).
- One-stage maximum likelihood method used because many events used which only have one record associated with them and the two-stage method underestimates the earthquake-to-earthquake component of variation in that case.
- Correction technique based on uniform correction and processing using upper, f_h, and lower, f_l, frequencies for passband based on a visual inspection of Fourier amplitude spectrum and baseline fitting with a polynomial of degree 5.
- Check to see whether normal and strike-slip earthquakes give significantly different ground motions. No significant difference.

2.111 *Wang* et al. *(1999)*

- Ground motion model is:

$$\log A = a + bM_s + c \log R + dR$$

  where A is in cms^{-2}, using just soil records $a = 0.430$, $b = 0.428$, $c = -0.764$, $d = -0.00480$ and $\sigma = 0.271$.

- Use records from aftershocks of Tangshan earthquake.
- Focal depths between 5.7 and $12.9\,\mathrm{km}$.
- Note M_s values used may have some systematic deviation from other regions and errors, which decrease with increasing magnitude, can reach ± 0.5.
- Errors in epicentral locations not less than $2\,\mathrm{km}$. Reject 3 records because have $R < 2\,\mathrm{km}$, if include then find standard deviation increases and c obtained is unreasonable.
- Fit equation to all data (both rock and soil) but note that only for reference. Also fit equation to soil data only ($2.1 \leq R \leq 41.3\,\mathrm{km}$, $3.7 \leq M_s \leq 4.9$, 33 records from 6 earthquakes).
- Remove all four earthquakes with $M_s < 4.0$, for which error in magnitude determination is large, and fit equation to soil data only ($2.8 \leq R \leq 41.1\,\mathrm{km}$, $4.5 \leq M_s \leq 4.9$, 13 records from 2 earthquakes). Find smaller uncertainties.
- Also fit data to $\log A = a + bM_s - c\log(R + R_0)$; find similar results.
- Also use resultant of both horizontal components; find similar results to using larger component.
- Also fit eastern North America data ($3.9 \leq R \leq 61.6\,\mathrm{km}$, $2.3 \leq M_s \leq 3.8$, 7 records from 3 earthquakes); find similar attenuation characteristics.
- All equations pass F-tests.

2.112 *Ambraseys & Douglas (2000)*

- Ground motion model is:

$$\log y = b_1 + b_2 M_s + b_3 d + b_A S_A + b_s S_S$$

where y is in ms^{-2}, for horizontal PGA $b_1 = -0.659$, $b_2 = 0.202$, $b_3 = -0.0238$, $b_A = 0.020$, $b_S = 0.029$ and $\sigma = 0.214$ and for vertical PGA $b_1 = -0.959$, $b_2 = 0.226$, $b_3 = -0.0312$, $b_A = 0.024$, $b_S = 0.075$ and $\sigma = 0.270$.

Assume decay associated with anelastic effects due to large strains and cannot use both $\log d$ and d because highly correlated in near field.

- Use four site categories (often use shear-wave velocity profiles):

  L Very soft soil: approximately $V_{s,30} < 180\,\mathrm{ms}^{-1}$, (combine with category S) $\Rightarrow S_A = 0, S_S = 1$, 4 records.

  S Soft soil: approximately $180 \leq V_{s,30} < 360\,\mathrm{ms}^{-1} \Rightarrow S_A = 0, S_S = 1$, 87 records.

  A Stiff soil: approximately $360 \leq V_{s,30} < 750\,\mathrm{ms}^{-1} \Rightarrow S_A = 1, S_S = 0$, 68 records.

  R Rock: approximately $V_{s,30} > 750\,\mathrm{ms}^{-1} \Rightarrow S_A = 0, S_S = 0$, 23 records.

  where $V_{s,30}$ is average shear-wave velocity to $30\,\mathrm{m}$. Know no site category for 14 records.

- Use only records from 'near field' where importance of vertical acceleration is greatest. Select records with $M_s \geq 5.8$, $d \leq 15\,\mathrm{km}$ and focal depth $h \leq 20\,\mathrm{km}$. Do not use magnitude dependent definition to avoid correlation between magnitude and distance for the records.
- Focal depths, $1 \leq h \leq 19\,\mathrm{km}$.
- Majority (133 records, 72%) of records from W. N. America, 40 records (22%) from Europe and rest from Canada, Nicaragua, Japan and Taiwan.

- Consider three source mechanisms but do not model:
  1. Normal, 8 earthquakes, 16 records.
  2. Strike-slip, 18 earthquakes, 72 records.
  3. Thrust, 16 earthquakes, 98 records.
- Use only free-field records using definition of Joyner & Boore (1981), include a few records from structures which violate this criterion but feel that structure did not affect record in period range of interest.
- Records well distributed in magnitude and distance so equations are well constrained and representative of entire dataspace. Note lack of records from normal earthquakes. Correlation coefficient between magnitude and distance is -0.10.
- Use same correction procedure (elliptical filter with pass band 0.2 to $25\,\mathrm{Hz}$, roll-off frequency $1.001\,\mathrm{Hz}$, sampling interval $0.02\,\mathrm{s}$, ripple in pass-band 0.005 and ripple in stop-band 0.015 with instrument correction) for almost all records. Use 19 records available only in corrected form as well because in large magnitude range. Think different correction procedures will not affect results.
- Try both one-stage and two-stage regression method for horizontal PGA; find large differences in b_2 but very similar b_3. Find that (by examining cumulative frequency distribution graphs for magnitude scaling of one-stage and two-stage methods) that two-stage better represents large magnitude range than one-stage method. Examine plot of amplitude factors from first stage of two-stage method against M_s; find that amplitude factor of the two Kocaeli ($M_s = 7.8$) records is far below least squares line through the amplitude factors. Remove the two Kocaeli records and repeat analysis; find b_2 from two-stage method is changed by a lot but b_2 from one-stage method is not. Conclude two-stage method is too greatly influenced by the two records from Kocaeli and hence use one-stage method.
- Find b_2 and b_3 significantly different than 0 at 5% level but b_A and b_S not significant.

2.113 *Bozorgnia* et al. *(2000)*

- Ground motion model is:

$$\begin{aligned}\ln Y \;=\;& c_1 + c_2 M_w + c_3(8.5 - M_w)^2 \\ & + c_4 \ln(\{R_s^2 + [(c_5 S_{HS} + c_6\{S_{PS} + S_{SR}\} + c_7 S_{HR}) \\ & \exp(c_8 M_w + c_9\{8.5 - M_w\}^2)]^2\}^{1/2}) + c_{10} F_{SS} + c_{11} F_{RV} + c_{12} F_{TH} \\ & + c_{13} S_{HS} + c_{14} S_{PS} + c_{15} S_{SR} + c_{16} S_{HR}\end{aligned}$$

- Use four site categories:

  HS Holocene soil: recent alluvium $\Rightarrow S_{HS} = 1, S_{PS} = 0, S_{SR} = 0, S_{HR} = 0$.
  PS Pleistocene soil: older alluvium $\Rightarrow S_{PS} = 1, S_{HS} = 0, S_{SR} = 0, S_{HR} = 0$.
  SR Soft rock $\Rightarrow S_{SR} = 1, S_{HS} = 0, S_{PS} = 0, S_{HR} = 0$.
  HR Hard rock $\Rightarrow S_{HR} = 1, S_{HS} = 0, S_{PS} = 0, S_{SR} = 0$.

- Consider all records to be free-field.
- All earthquakes occurred in shallow crustal tectonic environment.
- Consider three source mechanisms: strike-slip ($F_{SS} = 1, F_{RV} = 0, F_{TH} = 0$) 20+ earthquakes (including 1+ normal faulting shock), reverse ($F_{RV} = 1, F_{SS} = 0, F_{TH} = 0$) 7+ earthquakes and thrust ($F_{TH} = 1, F_{SS} = 0, F_{RV} = 0$) 6+ earthquakes.
- Coefficients not given, only predictions.

2.114 Campbell & Bozorgnia (2000)

- Ground motion model is:

$$\begin{aligned}\ln Y &= c_1 + c_2 M_w + c_3(8.5 - M_w)^2 + c_4 \ln(\{R_s^2 + [(c_5 + c_6\{S_{PS} + S_{SR}\} + c_7 S_{HR}) \\ &\quad \exp(c_8 M_w + c_9\{8.5 - M_w\}^2)]^2\}^{1/2}) + c_{10} F_{SS} + c_{11} F_{RV} + c_{12} F_{TH} \\ &\quad + c_{13} S_{HS} + c_{14} S_{PS} + c_{15} S_{SR} + c_{16} S_{HR}\end{aligned}$$

  where Y is in g, for horizontal uncorrected PGA $c_1 = -2.896$, $c_2 = 0.812$, $c_3 = 0$, $c_4 = -1.318$, $c_5 = 0.187$, $c_6 = -0.029$, $c_7 = -0.064$, $c_8 = 0.616$, $c_9 = 0$, $c_{10} = 0$, $c_{11} = 0.179$, $c_{12} = 0.307$, $c_{13} = 0$, $c_{14} = -0.062$, $c_{15} = -0.195$, $c_{16} = -0.320$ and $\sigma = 0.509$, for horizontal corrected PGA $c_1 = -4.033$, $c_2 = 0.812$, $c_3 = 0.036$, $c_4 = -1.061$, $c_5 = 0.041$, $c_6 = -0.005$, $c_7 = -0.018$, $c_8 = 0.766$, $c_9 = 0.034$, $c_{10} = 0$, $c_{11} = 0.343$, $c_{12} = 0.351$, $c_{13} = 0$, $c_{14} = -0.123$, $c_{15} = -0.138$, $c_{16} = -0.289$ and $\sigma = 0.465$, for vertical uncorrected PGA $c_1 = -2.807$, $c_2 = 0.756$, $c_3 = 0$, $c_4 = -1.391$, $c_5 = 0.191$, $c_6 = 0.044$, $c_7 = -0.014$, $c_8 = 0.544$, $c_9 = 0$, $c_{10} = 0$, $c_{11} = 0.091$, $c_{12} = 0.223$, $c_{13} = 0$, $c_{14} = -0.096$, $c_{15} = -0.212$, $c_{16} = -0.199$ and $\sigma = 0.548$ and for vertical corrected PGA $c_1 = -3.108$, $c_2 = 0.756$, $c_3 = 0$, $c_4 = -1.287$, $c_5 = 0.142$, $c_6 = 0.046$, $c_7 = -0.040$, $c_8 = 0.587$, $c_9 = 0$, $c_{10} = 0$, $c_{11} = 0.253$, $c_{12} = 0.173$, $c_{13} = 0$, $c_{14} = -0.135$, $c_{15} = -0.138$, $c_{16} = -0.256$ and $\sigma = 0.520$.

- Use four site categories:

  HS Holocene soil: soil deposits of Holocene age (11,000 years or less), generally described on geological maps as recent alluvium, approximate average shear-wave velocity in top $30\,\mathrm{m}$ is $290\,\mathrm{ms}^{-1}$ $\Rightarrow S_{HS} = 1, S_{PS} = 0, S_{SR} = 0, S_{HR} = 0$.

  PS Pleistocene soil: soil deposits of Pleistocene age (11,000 to 1.5 million years) , generally described on geological maps as older alluvium or terrace deposits, approximate average shear-wave velocity in top $30\,\mathrm{m}$ is $370\,\mathrm{ms}^{-1} \Rightarrow S_{PS} = 1, S_{HS} = 0, S_{SR} = 0, S_{HR} = 0$.

  SR Soft rock: primarily includes sedimentary rock deposits of Tertiary age (1.5 to 100 million years), approximate average shear-wave velocity in top $30\,\mathrm{m}$ is $420\,\mathrm{ms}^{-1} \Rightarrow S_{SR} = 1, S_{HS} = 0, S_{PS} = 0, S_{HR} = 0$.

  HR Hard rock: primarily includes older sedimentary rock deposits, metamorphic rock and crystalline rock, approximate average shear-wave velocity in top $30\,\mathrm{m}$ is $800\,\mathrm{ms}^{-1} \Rightarrow S_{HR} = 1, S_{HS} = 0, S_{PS} = 0, S_{SR} = 0$.

- Earthquakes from shallow crustal active tectonic regions.

- Most earthquakes with $6 \leq M_w \leq 7$.

- Use three source mechanism categories:

  SS Strike-slip: primarily vertical or near-vertical faults with predominantly lateral slip (includes only normal faulting earthquake in set), $\Rightarrow F_{SS} = 1, F_{RV} = 0, F_{TH} = 0$.

  RV Reverse: steeply dipping faults with either reverse or reverse-oblique slip, $\Rightarrow F_{RV} = 1, F_{SS} = 0, F_{TH} = 0$.

  TH Thrust: shallow dipping faults with predominantly thrust slip including blind-thrust shocks, $\Rightarrow F_{TH} = 1, F_{SS} = 0, F_{RV} = 0$.

- Consider all records to be free-field. Records from ground level in instrument shelter or a building <3 storeys high (<7 if located on hard rock). Include records from dam abutments to increase number of rock records. Exclude data from basements of buildings of any size or at toe or base of dams.

- Exclude data from $R_s > 60\,\mathrm{km}$ to avoid complicating problems related to arrival of multiple reflections from lower crust. Distance range is believed to include most ground shaking amplitudes of engineering interest, except for possibly long period spectral accelerations on extremely poor soil.
- Equations for uncorrected (Phase 1 standard level of processing) and corrected (Phase 2 standard level of processing).
- Find sediment depth (depth to basement rock) has significant effect on amplitude of ground motion and should be taken into account; it will be included once its mathematical form is better understood.

2.115 *Jain* et al. *(2000)*

- Ground motion model is:
$$\ln(\mathrm{PGA}) = b_1 + b_2 M + b_3 R + b_4 \ln(R)$$
where PGA is in g, for central Himalayan earthquakes $b_1 = -4.135$, $b_2 = 0.647$, $b_3 = -0.00142$, $b_4 = -0.753$ and $\sigma = 0.59$ and for non-subduction earthquakes in N.E. India $b_1 = -3.443$, $b_2 = 0.706$, $b_3 = 0$, $b_4 = -0.828$ and $\sigma = 0.44$ (coefficients of other equations not given here because they are for a particular earthquake).
- Data from strong-motion accelerographs (SMA) and converted from structural response recorders (SRR), which consist of six seismoscopes with natural periods 0.40, 0.75 and $1.25\,\mathrm{s}$ and damping levels 5 and 10%. Conversion achieved by deriving spectral amplification factors (ratio of response ordinate and PGA) using SMA recordings close to SRR, checking that these factors were independent of distance. The mean of the six estimates of PGA (from the six spectral ordinates) from each SRR are then used as PGA values. Check quality of such PGA values through statistical comparisons and discard those few which appear inconsistent.
- Data split into four categories for which derive separate equations:
  a Central Himalayan earthquakes (thrust): (32 SMA records, 117 SRR records), 3 earthquakes with $5.5 \leq M \leq 7.0$, focal depths $10 \leq h \leq 33\,\mathrm{km}$ and $2 \leq R \leq 322\,\mathrm{km}$.
  b Non-subduction earthquakes in NE India (thrust): (43 SMA records, 0 SRR records), 3 earthquakes with $5.2 \leq M \leq 5.9$, focal depths $33 \leq h \leq 49\,\mathrm{km}$ and $6 \leq R \leq 243\,\mathrm{km}$.
  c Subduction earthquakes in NE India: (33 SMA records, 104 SRR records), 1 earthquake with $M = 7.3$, focal depth $h = 90\,\mathrm{km}$ and $39 \leq R \leq 772\,\mathrm{km}$.
  d Bihar-Nepal earthquake in Indo-Gangetic plains (strike-slip): (0 SMA records, 38 SRR records), 1 earthquake with $M = 6.8$, focal depth $h = 57\,\mathrm{km}$ and $42 \leq R \leq 337\,\mathrm{km}$.
- Limited details of fault ruptures so use epicentral distance.
- Use epicentral locations which give best correlation between distance and PGA.
- Find PGA not well predicted by earlier equations.
- Simple model and regression method because of limited data.
- Remove one PGA value from category b equation because significantly affecting equation and because epicentral location only approximate.
- Constrain b_3 for category b equation to zero because otherwise positive.
- Category c originally contained another earthquake (14 SMA records, $M = 6.1$, $200 \leq d \leq 320\,\mathrm{km}$) but gave very small b_2 so exclude it.
- Equations for category c and category d have b_2 equal to zero because only one earthquake.

- Find considerable differences between predicted PGA in different regions.
- Note lack of data hence use equations only as first approximation.

2.116 *Kobayashi* et al. *(2000)*

- Ground motion model is:

$$\log_{10} y = aM - bx - \log(x + c10^{dM}) + eh + S_k$$

where h is focal depth, y is in cms^{-2}, $a = 0.578$, $b = 0.00355$, $e = 0.00661$, $S = -0.069$, $S_R = -0.210$, $S_H = -0.114$, $S_M = 0.023$, $S_S = 0.237$ and $\sigma_T = \sqrt{\sigma^2 + \tau^2}$ where $\sigma = 0.213$ and $\tau = 0.162$.

- Use four site categories (most data from medium and hard soils):

$S_k = S_R$ Rock
$S_k = S_H$ Hard soil
$S_k = S_M$ Medium soil
$S_k = S_S$ Soft soil

S is the mean site coefficient, i.e. when do not consider site category.

- Records interpolated in frequency domain from 0.02 to $0.005\,\mathrm{s}$ interval and displacement time history calculated using a fast Fourier transform (FFT) method having perpended to beginning and appended to end at least $5\,\mathrm{s}$ of zeros to record. Number of samples in FFT is large enough that duration used in FFT is at least twice that of selected duration for processing window so that numerical errors are small. Bandpass Ormsby filter used, with limits 0.2 and $24.5\,\mathrm{Hz}$, and displacement time history plotted. If displacement in pre- and appended portions is large then increase lower frequency limit in filter until displacements are small, using smoothed Fourier spectral amplitudes from 0.05 to $25\,\mathrm{Hz}$ to make choice.
- Most earthquakes are intra-slab.
- Note lack of near-field data for all magnitudes, most data from $> 100\,\mathrm{km}$, therefore use coefficients, c and d, from an early study.
- Excludes data from distances greater than the distance at which an earlier study predicts $\mathrm{PGA} < 0.02\,\mathrm{ms}^{-2}$.
- Consider residuals of earthquakes in western Japan (a small subset of data) and find small difference in anelastic coefficient and focal depth coefficient but note may be due to small number of records or because type of source not modelled.
- Note model predicts intraslab motions well but significantly over predicts interface motions.
- Plots site correction factors (difference between individual site factor and mean factor for that category) and find rock sites have largest variation, which suggest due to hard and soft rock included.
- Examine residual plots. Find no significant bias.

2.117 Sharma (2000)

- Based on Sharma (1998), see 2.106.
- A is in g and coefficients are: $c_1 = -2.87$, $c_2 = 0.634$, $c_3 = 0.62$, $b = 1.16$ and $\sigma = 0.142$.
- Fit $\log A = -b \log X + c$ to data from each earthquake separately and find average b equal to 1.18. Then fit $\log A = aM - b \log X + c$ to data from all earthquakes and find $b = 0.405$. Fit $\log A = -b \log X + \sum d_i l_i$ to all data, where $l_i = 1$ for ith earthquake and 0 otherwise and find $b = 1.16$, use this for rest of analysis.

2.118 Si & Midorikawa (2000)

- Ground motion model for rupture distance is:

$$\log A = aM_w + hD + \sum d_i S_i + e - \log(X + c_1 10^{c_2 M_w}) - kX$$

  where A is in cms^{-2}, $a = 0.50$, $h = 0.0036$, $d_1 = 0$, $d_2 = 0.09$, $d_3 = 0.28$, $e = 0.60$, $k = 0.003$ and $\sigma = 0.27$ (c_1 and c_2 are not given).

  Ground motion model for equivalent hypocentral distance (EHD) is:

$$\log A = aM_w + hD + \sum d_i S_i + e - \log X_{eq} - kX_{eq}$$

  where A is in cms^{-2}, $a = 0.50$, $h = 0.0043$, $d_1 = 0$, $d_2 = 0.01$, $d_3 = 0.22$, $e = 0.61$, $k = 0.003$ and $\sigma = 0.28$.

- Use two site categories for most records following Joyner & Boore (1981):
  1. Rock
  2. Soil
- Records from free-field or small buildings where soil-structure interaction effects are negligible.
- Records from three different type of instrument so instrument correct. Filter with corner frequencies, chosen according to noise level, a) 0.08 & $0.15\,\mathrm{Hz}$, b) 0.10 & $0.20\,\mathrm{Hz}$ or c) 0.15 to $0.33\,\mathrm{Hz}$.
- Exclude records obviously affected by soil liquefaction.
- Focal depth (defined as average depth of fault plane), D, between 6 and $120\,\mathrm{km}$; most less than $40\,\mathrm{km}$.
- Select records satisfying: distances $< 300\,\mathrm{km}$ for $M_w > 7$, distances $< 200\,\mathrm{km}$ for $6.6 \leq M_w \leq 7$, distances $< 150\,\mathrm{km}$ for $6.3 \leq M_w \leq 6.5$ and distances $< 100\,\mathrm{km}$ for $M_w < 6.3$.
- Fix $k = 0.003$.
- Multiply rock PGAs by 1.4 to get soil PGA based on previous studies.
- Use three fault types: crustal (<719 records from 9 earthquakes) $\Rightarrow S_1 = 1, S_2 = 0, S_3 = 0$, inter-plate (<291 records from 7 earthquakes) $\Rightarrow S_2 = 1, S_1 = 0, S_3 = 0$ and intra-plate (<127 records from 5 earthquakes) $\Rightarrow S_3 = 1, S_1 = 0, S_2 = 0$.
- Use weighted regression giving more weight to near-source records (weight factor of 8 for records $< 25\,\mathrm{km}$, 4 for records between 20 and $50\,\mathrm{km}$, 2 for records between 50 and $100\,\mathrm{km}$ and 1 for records $> 100\,\mathrm{km}$). Use only three earthquakes with sufficient near-source data to find c_1 and c_2 then use all earthquakes to find a, h, d_i, e in second stage using weighted regression dependent on number of recordings for each earthquake (weight factor of 3 for >83 records, 2 for between 19 and 83 records, 1 for <19 records.

- Note that M_w and D are positively correlated so a and h may not be correctly determined when using rupture distance. Constrain a for rupture distance model to that obtained for EHD and constrain PGA to be independent of magnitude at $0\,\mathrm{km}$ and repeat regression. Coefficients given above.

2.119 *Smit* et al. *(2000)*

- Ground motion model is:

$$\begin{aligned} \log Y &= a + bM - \log R + dR \\ \text{where } T &= \sqrt{D^2 + h^2} \end{aligned}$$

  where Y is in cms^{-2}, $a = 0.72$, $b = 0.44$, $d = -0.00231$, $h = 4.5$ and $\sigma = 0.28$.

- Records from soil or alluvium sites.

- All records corrected.

- Note that scatter can be reduced by increasing number of records used (especially in near field), improving all seismological and local site parameters and increasing number of variables (especially in near field and those modelling local site behaviour) but that this requires much more information than is available.

2.120 *Takahashi* et al. *(2000)*

- Ground motion model is:

$$\log_{10}[y] = aM - bx - \log_{10}(x + c10^{dM}) + e(h - h_c)\delta_h + S_k$$

  where y is in cms^{-2}, $a = 0.446$, $b = 0.00350$, $c = 0.012$, $d = 0.446$, $e = 0.00665$, $S = 0.941$, $S_R = 0.751$, $S_H = 0.901$, $S_M = 1.003$, $S_S = 0.995$, $\sigma_T = \sqrt{\sigma^2 + \tau^2}$ where $\sigma = 0.135$ (intra-event) and $\tau = 0.203$ (inter-event), h_c is chosen as $20\,\mathrm{km}$ because gave positive depth term.

- Use four site categories:

$S_k = S_R$ Rock
$S_k = S_H$ Hard soil
$S_k = S_M$ Medium soil
$S_k = S_S$ Soft soil

  Note site conditions for many stations are uncertain. S is the mean site term for all data.

- Note ISC focal depths, h, significant reduce prediction errors compared with JMA depths. $\delta_h = 1$ for $h \geq h_c$ and $\delta_h = 0$ otherwise.

- Most Japanese data from $x > 50\,\mathrm{km}$.

- Use 166 Californian and Chilean (from 2 earthquakes) records to control model in near source.

- Due to lack of multiple records from many sites and because c and d require near-source records use a maximum likelihood regression method of two steps. Firstly, find all coefficients using all data except those from sites with only one record associated with them and unknown site class. Next, use individual site terms for all sites so as to reduce influence of uncertainty because of approximate site classifications and find a, b, e and site terms using c and d from first step.

- Intra-event and inter-event residuals decrease with increasing magnitude.

- Conclude variation in residuals against distance is due to small number of records at short and large distances.
- Individual site factors means prediction error propagates into site terms when number of records per station is very small.
- Note model may not be suitable for seismic hazard studies because model prediction errors are partitioned into σ_T and mean site terms for a given site class. Suitable model can be derived when accurate site classifications are available.

2.121 Wang & Tao (2000)

- Ground motion model is:
$$\log Y = C + (\alpha + \beta M) \log(R + R_0)$$
where Y is in cms^{-2}, $C = 4.053$, $\alpha = -2.797$, $\beta = 0.251$, $R_0 = 8.84$ and $\sigma = 0.257$.
- Use same data as Joyner & Boore (1981), see Section 2.23.
- Use a two-stage method based on Joyner & Boore (1981). Firstly fit data to $\log Y = C + \sum_{i=1}^{n} (a_i E_i) \log(R_i + R_0)$, where $E_i = 1$ for records from ith earthquake and $E_i = 0$ otherwise, to find C and a_i for each earthquake. Next fit $a = \alpha + \beta M$ to find α and β using a_i from first stage.

3. GENERAL CHARACTERISTICS OF ATTENUATION RELATIONS FOR PEAK GROUND ACCELERATION

Table 3.1 gives the general characterisitics of published attenuation relations for peak ground acceleration. The columns are:

H Number of horizontal records (if both horizontal components are used then multiply by two to get total number)

V Number of vertical components

E Number of earthquakes

$M_{\min}$ Magnitude of smallest earthquake

$M_{\max}$ Magnitude of largest earthquake

M scale Magnitude scale (scales in brackets refer to those scales which the main M values were sometimes converted from, or used without conversion, when no data existed), where:

- m_b Body-wave magnitude
- M_C Chinese surface wave magnitude
- M_{CL} Coda length magnitude
- M_D Duration magnitude
- M_{JMA} Japanese Meteorological Agency magnitude
- M_L Local magnitude
- M_{bLg} Magnitude calculated using Lg amplitudes on short-period, vertical seismographs
- M_s Surface-wave magnitude
- M_w Moment magnitude

$d_{\min}$ Shortest source-to-site distance

$d_{\max}$ Longest source-to-site distance

d scale Distance measure, where:

- d_c Distance to rupture centroid
- d_e Epicentral distance
- d_E Distance to energy centre
- d_f Distance to projection of rupture plane on surface (Joyner & Boore, 1981)
- d_h Hypocentral (or focal) distance
- d_q Equivalent hypocentral distance (EHD) (Ohno *et al.*, 1993)
- d_r Distance to rupture plane
- d_s Distance to seismogenic rupture plane (assumes near-surface rupture in sediments is non-seismogenic) (Campbell, 1997)

S Number of different site conditions modelled, where:

- C Continuous classification
- I Individual classification for each site

C Use of the two horizontal components of each accelerogram, where:

- B Both components
- C Randomly chosen component
- G Geometric mean
- L Larger component
- M Mean (not stated what type)
- O Randomly oriented component
- R Resolved component
- U Unknown
- V Vectorially resolved component, i.e. square root of sum of squares of the two components

R Regression method used, where:

- 1 Ordinary one-stage
- 1B Bayesian one-stage
- 1M Maximum likelihood one-stage (Joyner & Boore, 1993)
- 1W Weighted one-stage
- 2 Two-stage (Joyner & Boore, 1981)
- 2M Maximum likelihood two-stage (Joyner & Boore, 1993)
- 2W Two-stage with second staged weighted as described in Joyner & Boore (1988)
- O Other (see section refering to study)
- U Unknown

M Source mechanisms (and tectonic type) of earthquakes (letters in brackets refer to those mechanism which are separately modelled), where:

- A All (this is assumed if no information is given in the reference)
- B Interslab
- F Interface
- I Intraplate
- N Normal
- O Oblique
- R Reverse
- S Strike-slip
- T Thrust

'+' refers to extra records from outside region used to supplement data. (...) refer either to magnitudes of supplementing records or to those used for part of analysis. * means information is approximate because either read from graph or found in another way.

Tab. 3.1: Characteristics of published peak ground acceleration relations

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Milne & Davenport (1969)	W. USA	U	-	U	U	U	U	U	U	d_e	1	U	U	A
Esteva (1970)	W. USA	U	-	U	U	U	U	15*	500*	d_h	1	U	U	A
Denham & Small (1971)	Yonki, New Guinea	8	-	8	U	U	M_L[1]	U	U	d_h	1	U	U	A
Donovan (1973)	Mostly W. USA but 100+ foreign	678	-	U	<5	>8	U	3*	450*	d_h	1	U	U	A
Esteva & Villaverde (1973)	W. USA	U	-	U	U	U	U	15*	150*	d_h	1	B	U	A
Orphal & Lahoud (1974)	California	140	-	31	4.1	7.0	M_L	15	350	d_h	1	U	O	A
Ambraseys (1975)	Europe	58	-	U[2]	3.5	5.0	M_L	5	35	d_h	1	U[3]	U	A
Trifunac & Brady (1975),Trifunac (1976) & Trifunac & Brady (1976)	W. USA	181	181	57	3.8	7.7	Mostly M_L	6[4]*	400[5]*	d_e	3	B	O	A
Blume (1977)	California & W. Nevada	795[6]	-	U	U	U	M_L	U	U	d_h	2 (1)	B	U	A
McGuire (1977)	W. USA	34	-	22	5.3	7.6	M_L	14	125	d_h	1	B	U	A
Milne (1977)	W. USA	200*	-	U	3.5	7.7	U	1	380	d_h	1	U	U	A
Donovan & Bornstein (1978)	W. USA	59	-	10	5.0	7.7	U[7]	0.1	321	d_E, d_r and d_h	1	B	O	A
Faccioli (1978)	Mostly W. USA & Japan, some foreign	47[8]	-	23	4.9	7.8	U[9]	15	342	d_h	1	B	U	A
McGuire (1978)	W. USA	70	-	17+*	4.5*	7.7	U[10]	11*	210*	d_h	2	B	U	A

continued on next page

[1] State that it is Richter magnitude which assume to be M_L

[2] Ambraseys & Bommer (1995) state that uses 38 earthquakes.

[3] Ambraseys & Bommer (1995) state that uses larger component.

[4] Note only valid for $R \geq 20\,\mathrm{km}$

[5] Note only valid for $R \leq 200\,\mathrm{km}$

[6] Total earthquake components (does not need to be multiplied by two) for magnitude and distance dependence. Uses 2713 underground nuclear explosion records for site dependence.

[7] Idriss (1978) finds magnitudes to be mixture of M_L and M_s.

[8] Total earthquake components (does not need to be multiplied by two)

[9] Idriss (1978) believes majority are M_s.

[10] Idriss (1978) finds magnitudes to be mixture of M_L, m_b and M_s.

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
A. Patwardhan *et al.* (1978)[11]	Worldwide	63 (32)	-	25 (23)	4 (5.3)	7.7 (7.8)	M_s	U	U	d_r	2	B	U	A
Cornell *et al.* (1979)	W. USA	70	-	U	U	U	M_L	U	U	d_h	1	C	U	A
Aptikaev & Kopnichev (1980)	Worldwide	Many 100s	- (70*)	U (59)	U	U	U	U	U	d_h	1	U	U	A (T, TS, S, SN, N)[12]
Blume (1980)	W. USA	816	-	U	2.1	7.6	U	0	449	d_h	1	B	1, O	A
Iwasaki *et al.* (1980)	Japan	301	-	51	>5.0	<7.9	M_L[13]	<20	>200	d_e	4	U	1	A
Ohsaki *et al.* (1980a)	Japan	75	75	U	4	7.4	U	6	500	d_h	1	U	1	A
Campbell (1981)	W. USA+8 foreign	116	-	27	5.0	7.7	M_L for $M < 6.0$ and M_s otherwise	0.08	47.7	d_r	1	M	O	A
Chiaruttini & Siro (1981)	Europe & Mid. East	224	-	117	2.7	7.8	M_L (m_b)	3	480	d_h	1	L	1	A
Joyner & Boore (1981)	W. N. America	182	-	23	5.0	7.7	M_w (M_L)	0.5	370	d_f	2	L	2	A
Bolt & Abrahamson (1982)	W. N. America	182	-	23	5.0	7.7	M_w (M_L)	0.5	370	d_f	1	L	O	A
PML (1982)	Europe + USA + others	113	-	32	4.3	8	M_s	0.1	330	d_e or d_f	1	U	U	A
Schenk (1982)	Unknown	3500	-	U	2.5	6.5	M_s	2	600	d_h	1	U	O	A
Joyner & Fumal (1984)	W. N. America	182	-	23	5.0	7.7	M_w (M_L)	0.5	370	d_f	C	L	2	A
Kawashima *et al.* (1984) & Kawashima *et al.* (1986)	Japan	197	-	90	5.0	7.9	M_{JMA}	5*	550*	d_e	3	R	1	A
McCann Jr. & Echezwia (1984)	N. America + foreign	83	-	18	5.0+	U	M_w	U	U	d_r	1	U	O	A
Schenk (1984)	Unknown	3500	-	U	2.5	6.5	U	2	600	d_h	1	U	O	A
Xu *et al.* (1984)	N. China	19	-	10	4.5	7.8	M_w (M_L for $M < 6.0$, M_s for $M \geq 6.0$)	10.1	157	d_e	1	L	1	A
Kawashima *et al.* (1985)	Japan	-	119	90*	5.0*	7.5*	M_{JMA}	5*	500*	d_e	3	-	1	A

continued on next page

[11] Reported in Idriss (1978).

[12] Assume dip-slip means normal mechanism.

[13] State that it is Richter magnitude which assume to be M_L

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Peng *et al.* (1985)	N.E. China	73	-	20	3.7	7.8	M_C	2	442.5	d_e	1	U	1	A
PML (1985)	USA + Europe + others	203	-	46	3.1	6.9	M_s	0.1	40	d_r	1	U	U	A (S, T)
McCue (1986)	E. Australia	U	-	U	1.7	5.4	M_L	2.5	134	d_h	1	U	U	A
C.B. Crouse (1987)[14]	S. California	U	-	U	U	U	M_s	U	U	d_r	1	B	U	A
Sabetta & Pugliese (1987)	Italy	95	-	17	4.6	6.8	M_s for $M \geq 5.5$, M_L otherwise	1.5, 1.5	179, 180	Both d_f & d_e	2	L	1	A
K. Sadigh (1987)[15]	W. USA + others	U	-	U	U	U	M_w	U	U	d_r	2	B	U	A (S, R)
Singh *et al.* (1987)	Mexico	16	-	16	5.6	8.1	M_s	282	466	d_r	1	U	1	A
Algermissen *et al.* (1988)	Vicinity of San Salvador	82	-	U	U	U	M_s	U	U	d_h	1	M	U	A
Annaka & Nozawa (1988)	Japan	U	-	45	U	U	U	U	U	U	1	U	1	A
K.W. Campbell (1988)[16]	Worldwide	U	-	U	≥ 5	U	M_L for $M < 6.0$ and M_s otherwise	U	<50	d_s	2	M	U	A (S, R)
Fukushima *et al.* (1988) & Fukushima & Tanaka (1990)	Japan+200 W. USA	486+200	-	28+15	4.6(5.0)	8.2(7.7)	M_s (M_{JMA})	16 (0.1)	303 (48)	d_h, d_r for 2 Japanese & all US	4	G	2	A
Gaull (1988)	S.W. W. Australia	25+	-	12+	2.6	6.9	M_L	2.5	175	d_h	1	U	O	A
Joyner & Boore (1988)	W. N. America	182	-	23	5.0	7.7	M_w (M_L)	0.5	370	d_f	2	L, O	2W	A
McCue *et al.* (1988)	S.E. Australia	62	-	U	0.5*	6*	M_L	5*	833	d_e	1	U	O	A
Petrovski & Marcellini (1988)	Europe	120	-	46	3	7	U	8	200	d_h	1	L	U	A
Tong & Katayama (1988)	Kanto (Japan)	<227	-	<27	4.5*	7.9*	U	10*	750*	d_e	C	L	O	A
Yamabe & Kanai (1988)	Japan	U	-	22	5.3	7.9	U	U	U	d_h	1	U	O	A
Youngs *et al.* (1988)	Worldwide subduction zones	197+389	-	60	5	8.1 (8.2)[17]	M_w (M_s, m_b)	15* (20*)	450* (450*)	d_r, d_h for $M_w \lesssim 7.5$	1	G	1W	A (B,F)

continued on next page

[14] Reported in Joyner & Boore (1988).
[15] Reported in Joyner & Boore (1988).
[16] Reported in Joyner & Boore (1988).
[17] Consider equations valid for $M_w \leq 8$

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Abrahamson & Litehiser (1989)	75%+ California, rest foreign	585	585	76	5.0	8.1	M_s for $M_s \geq 6.0$, M_L (m_b) otherwise	0.08	400	d_r	1	L	O	A (R & RO, I)
Campbell (1989)	W. N. America + 3 from Managua	190	-	91	2.9	5.0	M_L	0.6	18.3	d_e	1	M	O	A
Alfaro *et al.* (1990)	Guatemala, Nicaragua & El Salvador	20	-	12	4.1	7.5	M_s	1	27	d_e	1	L	U	A
Ambraseys (1990)	W. N. America	182	-	23	5.03	7.7	M_w (M_L)	0.5	370	d_f	2	L	2	A
Campbell (1990)	Unknown	U	-	U	U	U	M_L for $M < 6$, M_s for $M \geq 6$	U	U	d_s	1	U	U	A
Dahle *et al.* (1990b) & Dahle *et al.* (1990a)	Worldwide intraplate regions	87	-	56	2.9	7.8	M_s (M_L, m_b, M_{CL})	6	1300	d_h	1	L	2	A
Jacob *et al.* (1990)	E. N. America	U	-	8	1.8	6.4	m_b	$\leq$ 20	820	U[18]	1	U	O	A
Sen (1990)	Whittier Narrows area	72*	-	11	2.2	3.5	M_L	12*	21*	d_h	1	U	1M	A (T)
Tsai *et al.* (1990)	Worldwide	<217	-	<51	4.9*	7.4	M_w	3*	150*	d_r	1	M	U	T (S,O)
Ambraseys & Bommer (1991) & Ambraseys & Bommer (1992)	Europe & Mid. East	529	459	H:219, V:191	4	7.34	M_s	1	H:313, V:214	d_f for $M_s \gtrsim 6.0$, d_e otherwise	1	L	1, 2	A
Crouse (1991)	Worldwide subduction zones	697[19]	-	U	4.8	8.2	M_w (M_s, M_{JMA})	>8	>866	d_E, d_h for $M < 7.5$	1	B	1	A
Huo & Hu (1991)	W. USA with 25 foreign	383+25	-	14+2	5.0	7.4 (7.3)	M_L or m_b for $M < 6.0$ and M_s otherwise	0.1	227 (265)	d_f	2	B	O	A
I.M. Idriss (1991) reported in Idriss (1993)	Unknown	572	-	30*	4.6	7.4	M_L for $M < 6$, M_s for $M \geq 6$	1	100	d_r, d_h for $M < 6$	1	U	U	A

continued on next page

[18] Free (1996) believes it is d_h.

[19] Total number of components, does not need to be multiplied by two.

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Niazi & Bozorgnia (1991)	SMART-1 array, Taiwan	236	234	12	3.6	7.8	M_L (M_D) for $M_L < 6.6$, else M_s	3.1[20]	119.7[20]	d_h	1	M	2W	A
Ambraseys *et al.* (1992)	USA + Europe + others	504	-	45	3.1	6.87	M_s	0.5	39	d_f, d_e for some	1	L	1	A
Kamiyama *et al.* (1992) & Kamiyama (1995)	Japan	357	-	82	4.1	7.9	M_{JMA}	3.4	413.3	d_h	I	B	O	A
Sigbjörnsson & Baldvinsson (1992)	Iceland	262	-	39	2.0	6.0	U	2	80	d_f	2	B,L	2	A
Taylor Castillo *et al.* (1992)	Nicaragua, El Salvador & Costa Rica	89	-	27	3.0	7.6	M_s	6	210	d_h	1	L	U	A
Tento *et al.* (1992)	Italy	137	-	40	4	6.6	M_L	3.2	170	d_f for $M_L \geq 5.7$, d_e otherwise	1	L	2	A
Theodulidis & Papazachos (1992)	Greece+16 foreign	105+16[21]	-	36+4	4.5 (7.2)	7.0 (7.5)	M_s, M_w, M_{JMA}	1 (48)	128 (236)	d_e	2	B	O	A
Boore *et al.* (1993) & Boore *et al.* (1997)	W. N. America	271	-	20	5.1[22]	7.7	M_w	0	118.2	d_f	3	L, G	2M	A
Campbell (1993)	Worldwide	U	-	U	U[23]	U	M_L for $M < 6.0$ and M_s otherwise	U	U[24]	d_s	2	M	O	A (T,S)
Sadigh *et al.* (1993) & Sadigh *et al.* (1997)	California with 4 foreign	960+4	U	119+2	3.8 (6.8)	7.4 (7.4)	M_w	0.1 (3)	305 (172)[25]	d_r for some, d_h for small ones	2	G	U	A(R,S)
Singh *et al.* (1993)	Nicaragua, El Salvador & Costa Rica	89	-	27	3.0	7.6	M_s	6	210	d_h	1	V	O	A

continued on next page

[20] Distance to centre of array
[21] Total number of components does not need to be multiplied by two
[22] Boore *et al.* (1997) revise this magnitude to 5.87. New minimum magnitude is 5.2.
[23] Considers equation valid for $M \geq 4.7$.
[24] Considers equation valid for $d \leq 300\,\text{km}$.
[25] Equations stated to be for distances up to $100\,\text{km}$

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Sun & Peng (1993)	W. USA with 1 foreign	150+1	-	42+1	4.1	7.7	M_L for $M < 6$, else M_s	2*	150*	d_e	C	R	1	A
Boore *et al.* (1994a) & Boore *et al.* (1997)	W. N. America	271 (70)	-	20 (9)	5.1[26] (5.3)	7.7 (7.4)	M_w	0	118.2 (109)	d_f	C	L, G	1M, 2M	A (R,S)[27]
Fukushima *et al.* (1994) & Fukushima *et al.* (1995)	3 vertical arrays in Japan	285	284	42	5.0	7.7	$M_{\rm JMA}$	60*	400*	d_h	I	B	1,2	A
Lawson & Krawinkler (1994)	W. USA	250+	-	11	5.8	7.4	M_w	U	100	d_f	3	U	1M	A
Lungu *et al.* (1994)	Romania	≈ 300	125	4	6.3	7.4	M_w	U	U	d_h	1	U	1	A
Musson *et al.* (1994)	UK + 30* foreign	15 + 30*	-	4+16	3 (3.7)	3.5 (6.4)	M_L	70* ($>$1.3)	$>$477.4 (200*)	d_h	1	U[28]	O	A
Radu *et al.* (1994), Lungu *et al.* (1995a) & Lungu *et al.* (1996)	Romania	106	-	3	6.7(M_L) or 7.0(M_w)	7.2(M_L) or 7.5(M_w)	U[29]	90*	320*	d_h	1	L	1	A
Ramazi & Schenk (1994)	Iran	83	83	20	5.1	7.7	M_s[30]	≤ 8	$\geq$ 180	d_h for most, d_r for 19[31]	2	U	U	A
Xiang & Gao (1994)	Yunnan, China + 114 W. N. America	131+114	-	U	2.5*	7.6*	M_s (M_L)	2*	120*	d_e	1	L	U	A
Ambraseys (1995)	Europe and Mid. East	830	620	334	4.0	7.3	M_s	0*	260*	d_f for $M_s > 6.0$, d_e otherwise	1	L	2W	A
Dahle *et al.* (1995)	Cen. America	280	-	72	3*	8*	M_w (M_s, m_b,M_D)	6*	490*	d_h	2	L	1B	A
Garcia-Fernandez & Canas (1995)	Iberian Pen.	57	367	U	3.1	5.0	M_{bLg}	U	U	d_e	1	U	U	A

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[26] Boore *et al.* (1997) revise this magnitude to 5.87. New minimum magnitude is 5.2.
[27] Coefficients given in Boore *et al.* (1994b)
[28] Free (1996) believes it is largest horizontal component.
[29] It is not clear whether use Richter magnitude (M_L) or M_w.
[30] Some may be m_b because in their Table 1 some earthquakes to not have M_s given but do have m_b. If so new minimum is 5.0.
[31] They state it is 'closest distance from the exposure of ruptured part of the fault, instead of focal distances' so may not be rupture distance.

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Lungu *et al.* (1995b)	Romania	106	-	3	6.7(M_L) or 7.0(M_w)	7.2(M_L) or 7.5(M_w)	U[32]	U	U	d_h	1	L	1	A
Molas & Yamazaki (1995)	Japan	2166	-	387	4.1*	7.8*	M_{JMA}	8*	1000*	d_r for 2 earthquakes, d_h otherwise	I	L	O	A
Ambraseys *et al.* (1996) & Simpson (1996)	Europe & Mid. East	422	-	157	4.0	7.9	M_s (unspecified)	0	260	d_f for $M_s > 6.0$, d_e otherwise	3	L	2W[33]	A
Ambraseys & Simpson (1996) & Simpson (1996)	Europe & Mid. East	-	417	157	4.0	7.9	M_s (unspecified)	0	260	d_f for $M > 6.0$, d_e otherwise	3	-	2W[34]	A
Bommer *et al.* (1996)	El Salvador & Nicaragua	36	-	20	3.7	7.0	M_s	62	260	d_h	1	L	U	A
Crouse & McGuire (1996)	Cen. & S. California	238	-	16	6.0	7.7	M_s	0.1	211	d_r	4	G	1W	R,S (R,S)
Free (1996) & Free *et al.* (1998)	Stable continental regions	558	478	H: 222, V: 189	1.5	6.8	M_w	0	820	d_f for some, d_e for most	2	L	1	A
Ohno *et al.* (1996)	California	248	-	17	5.0	7.5	M_w (M_L)	7.2	99.6	d_q for $M > 5.3$, d_h otherwise	2	B	2M	A
Sarma & Srbulov (1996)	Worldwide	350	-	114	3.9	7.7	M_s	1	213	d_f & d_e	1	B, L	U	A
Spudich *et al.* (1996)	Worldwide extensional regimes	128	-	30	5.10	6.90	M_w	0	102.1	d_f	2	G, C	2M	NS
Campbell (1997) & Campbell & Bozorgnia (1994)	Worldwide	645	225	H:47, V:26	4.7	H:8.0, V:8.1	M_w	3	60	d_s	3	G	1	A(S,R,N)

continued on next page

[32] It is not clear whether use Richter magnitude (M_L) or M_w.
[33] Ambraseys *et al.* (1996) state it is two-stage of Joyner & Boore (1981) but in fact it is two-stage method of Joyner & Boore (1988).
[34] Ambraseys *et al.* (1996) state it is two-stage of Joyner & Boore (1981) but in fact it is two-stage method of Joyner & Boore (1988).

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Munson & Thurber (1997)	Hawaii	51	-	22	4.0	7.2	M_s for $M_s \geq 6.1$, M_L otherwise	0	88	d_f	2	L	2M	A
Youngs *et al.* (1997)	Worldwide subduction zones	476	-	164	5.0	8.2	M_w (M_s,m_b)	8.5	550.9	d_r, d_h for some	2	G	1M	NT
Zhao *et al.* (1997)	NZ with 66 foreign	461[35]+66	-	49+17	5.08	7.23(7.41)	M_w	11 (0.1)	573 (10)	d_r for some, d_c for most	2	U	1	A(R)
Bouhadad *et al.* (1998)	Algeria	U	-	2	5.6	6.1	M_s	20	70	d_h	1	L, M	1	A
Manic (1998)	N.W. Balkans	276[36]	-	56	4	7	M_s	U	U	d_h	2	B	1	A
Rinaldis *et al.* (1998)	Italy & Greece	137*	-	24*	4.5	7	M_s or M_w	7	138	d_e	2	U	O	A (N,ST)
Sadigh & Egan (1998)	California with 4 foreign	960+4	-	119+2	3.8	7.4	M_w	0.1	305[37]	d_r for some, d_h for small ones	2	G	U	A(R,SN)
Sarma & Srbulov (1998)	Worldwide	690[38]	-	113	3.9	7.7	M_s (U)	0	197	d_f, d_e	2	B	1	A
Sharma (1998)	Indian Himalayas	66	-	5	5.5	6.6	U	8	248	d_h	1	L	1W	A
Smit (1998)	Switzerland + some from S. Germany	$\ll$ 1546	<1546	H: <120, V: 120	2.0	5.1	M_L	1	290	d_h	1	U	2	A
Chapman (1999)	W. N. America	304	-	23	5.0	7.7	M_w	0.1	189.4	d_f	3	G	2M	A
Cousins *et al.* (1999)	NZ with 66 foreign	610+66	-	25+17	5.17	7.09(7.41)	M_w	0.1	400	d_r for some, d_c for most	3	U	U	A(R)
Spudich *et al.* (1999)	Worldwide extensional regimes	142	-	39	5.1	7.2	M_w	0	99.4	d_f	2	G, O	1M	NS
Wang *et al.* (1999)	Tangshan, N. China	44	-	6	3.7	4.9	M_s (M_L)	2.1	41.3	d_e	1	L	1	A
Ambraseys & Douglas (2000)	Worldwide	186	183	44	5.83	7.8	M_s	0	15	d_f	3	L	1	A
Bozorgnia *et al.* (2000)	Worldwide	2823	2823	48	4.7	7.7	M_w	U	$\leq$ 60	d_s	4	G	U	A (R,S,T)

continued on next page

[35] Includes some not used for regression
[36] Total number of components do not need to be multiplied by two.
[37] Equations stated to be for distances up to $100\,\mathrm{km}$
[38] Total number of components do not need to be multiplied by two.

Tab. 3.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	C	R	M
Campbell & Bozorgnia (2000)	Worldwide	960[39]	941[40]	49[41]	4.7	7.7	M_w	1*	60*	d_s	4	G	1	A (S,R,T)
Jain *et al.* (2000)	Central Himalayas	32 (117)	-	3	5.5	7.0	U	2 (4)	152 (322)	d_e	1	U	1	T
Kobayashi *et al.* (2000)	Japan	U	-	U	5.0	7.8	M_w	0.9*	400*	U	4	B	1M	A
Sharma (2000)	Indian Himalayas	-	66	5	5.5	6.6	U	8	248	d_h	1	-	1W	A
Si & Midorikawa (2000)	Japan	856	-	21	5.8	8.3	M_w	0*	280*	Both d_q & d_r	2	L	O	A
Smit *et al.* (2000)	Caucasus	84	-	26	4.0	7.1	M_s	4	230	d_e[42]	1	L	2	A
Takahashi *et al.* (2000)	Japan+166 foreign	1332	-	U+7*	5* (5.8*)	8.3* (8*)	M_w	1* (0.1*)	300* (100*)	d_r, d_h for some	4	G	O	A
Wang & Tao (2000)	W. N. America	182	-	23	5.0	7.7	M_w (M_L)	0.5	370	d_f	2	L	O	A

[39] Equation for corrected PGA uses 443 records.
[40] Equation for corrected PGA uses 439 records.
[41] Equation for corrected PGA uses data from 36 earthquakes.
[42] Smit *et al.* (2000) give d_h but this is typographical error (Smit, 2000).

4. SUMMARY OF PUBLISHED ATTENUATION RELATIONS FOR SPECTRAL ORDINATES

4.1 Johnson (1973)

- Ground motion model is:

$$\mathrm{PSRV} = C10^{\alpha m_b} R^m$$

- Response parameter is pseudo-velocity for 5% damping.
- Most (76%) records from $R < 70\,\mathrm{km}$.
- Uses only shallow focus earthquakes of 'normal' or less depth, to minimize variables, except for one record from deeper earthquake ($m_b = 6.5$, $R = 61.1\,\mathrm{km}$) which produces no distortion in statistical calculations.

4.2 Kobayashi & Nagahashi (1977)

- Ground motion model is:

$$\log_{10} S_{V0} = a(\omega)M - b(\omega)\log_{10} x - c(\omega)$$

- Response parameter is velocity for unspecified[1] damping.
- Do regression iteratively. Assume $a(\omega)$, $b(\omega)$ and $c(\omega)$. Find amplification factors, $G_i(\omega)$, for each response spectra, $R_i(\omega)$: $G_i = R_i(\omega)/S_{V0}$. Calculate amplification factor, G, for each site: $G = \sqrt[n]{\prod_{i=1}^{n} G_i(\omega)}$. Estimate bedrock spectrum, $B_i(\omega)$, for each record: $B_i(\omega) = R_i(\omega)/G(\omega)$. Find $a(\omega)$, $b(\omega)$ and $c(\omega)$ by least squares. Repeat these steps until convergence. Hence find attenuation relation for bedrock and amplification function for each site.

4.3 McGuire (1977)

- See Section 2.10.
- Response parameter is pseudo-velocity for 0, 2, 5 and 10% damping.
- Residuals pass Kolmogorov-Smirnov goodness-of-fit test at 5% significance level for normal distribution, so it is concluded that pseudo-velocities are lognormally distributed.
- Feels that using 16 natural periods presents a very good picture of spectral trends throughout entire period range.
- Only gives graphs of coefficients not actual calculated values.

[1] It is probably 5%.

4.4 *Faccioli (1978)*

- See Section 2.13.
- Response parameter is pseudo-velocity for 5% damping.
- Plots all spectra. 2 records have abnormally high values in long period range, so remove and repeat. Results practically unaffected so leave them in.
- Notes that due to small size of sample, site and source correlation can introduce some error in coefficients because all data treated as statistically independent. Assume correlations are small so neglect error.

4.5 *McGuire (1978)*

- See Section 2.14.
- Response parameter is pseudo-velocity for 2% damping.

4.6 *Cornell* et al. *(1979)*

- See Section 2.16.
- Response parameter is pseudo-velocity for 0, 2 and 10% damping.
- Consider different paths, e.g. going through intensities, Fourier spectra and PGA, to predict PSV. Note that direct paths have minimum variance but that going through intermediate steps does not significantly increase prediction uncertainty provided that intermediate parameters are representative of frequency band of structural system.
- Do not give coefficients.

4.7 *Ohsaki* et al. *(1980b)*

- Ground motion model is:
$$\log S_v = a'M - b' \log x - c'$$
- Response parameter is velocity for 5% damping.
- Use two soil conditions:

Group A Hard rock: geology consists of granite, andesite and shale of Miocene or earlier geological age, having S wave velocity $\gtrsim 1500\,\mathrm{ms}^{-1}$ or P wave velocity $\gtrsim 3000\,\mathrm{ms}^{-1}$, 60 records

Group B Rather soft rock: geology consists of mudstone of Pliocene or late Miocene age, having S wave velocity of about 500–$1000\,\mathrm{ms}^{-1}$, 35 records.

- Use records where geological and geotechnical conditions investigated in detail and considered to represent free-field rock motions. Exclude records suspected to be amplified by surface soil or affected by high topographical relief.
- Most records from $\geq 30\,\mathrm{km}$.
- Do regression on both site categories separately and give graphs of coefficients not tables.

4.8 *Ohsaki* et al. *(1980a)*

- See Section 2.20.
- Response parameter is velocity for 5% damping.
- Also give smoothed results using correction factors based on derived PGV equation.

4.9 *Trifunac (1980)*

- Ground motion model is:

$$\log_{10}\mathrm{PSV}(T) = \begin{cases} M - \log_{10} A_0(R) - b(T)M_{\min} - c(T) - d(T)h - e(T)v \\ \qquad - f(T)M_{\min}^2 - g(T)R \\ \qquad \text{for} \quad M \le M_{\min} \\ M - \log_{10} A_0(R) - b(T)M - c(T) - d(T)h - e(T)v \\ \qquad - f(T)M^2 - g(T)R \\ \qquad \text{for} \quad M_{\min} < M < M_{\max} \\ M_{\max} - \log_{10} A_0(R) - b(T)M_{\max} - c(T) - d(T)h - e(T)v \\ \qquad - f(T)M_{\max}^2 - g(T)R \\ \qquad \text{for} \quad M \ge M_{\max} \end{cases}$$

  where $\log_{10} A_0(R)$ is an empirically determined attenuation function from Richter (1958) used for calculation of M_L, v is component direction ($v = 0$ for horizontal and 1 for vertical), $M_{\min} = -b(T)/(2f(T))$ and $M_{\max} = (1 - b(T))/(2f(T))$.

- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Characterises site condition by depth of sedimentary and alluvial deposits beneath station, h. Uses records with $0 \le h \le 6\,\mathrm{km}$, with most $< 4\,\mathrm{km}$.
- Performs analysis to minimize possible bias due to uneven distribution of data among magnitude, site conditions and from abundance of data for some earthquakes.
- Tries terms with higher powers of h but coefficients are undistinguishable from zero at 95% confidence level.
- Assumes probability that $\log_{10}\mathrm{PSV}(T) - \log_{10}\bar{\mathrm{PSV}}(T) \le \epsilon$, where $\log_{10}\mathrm{PSV}(T)$ is measured PSV and $\bar{\mathrm{PSV}}(T)$ is predicted PSV and ϵ is a probability, can be expressed as $p(\epsilon, T) = [1 - \exp(-\exp(\alpha(T)\epsilon(T) + \beta(T)))]^{N(T)}$. This assumption passes Kolmogorov-Smirnov and χ^2 tests at 95% level.
- Finds $a(T)$ through $g(T)$ significantly different than zero for large subregions of whole period range. $d(T)$ is only significantly different than zero for $T \gtrsim 0.3\,\mathrm{s}$.
- Gives coefficients of smoothed results for 11 periods.
- Notes only preliminary. Improvements should be based on physical nature of phenomenon using a functional form predicted by theory and experiment but due to lack of data cannot be done.

4.10 *Devillers & Mohammadioun (1981)*

- Ground motion model is:

$$V(f) = C10^{\alpha M} R^n$$

- Response parameter is pseudo-velocity for 2, 5, 10 and 20% damping.

- Most records from between 20 and $40\,\mathrm{km}$. No records from $R < 10\,\mathrm{km}$ so equation does not apply there.
- Eliminate suspect and/or redundant (San Fernando) records.
- Split data into intensity groups: VI (126 records), VII (56 records), V+VI (186 records), VI+VII (182 records) and VII+$\geq$ VIII (70 records) and calculates coefficients for each group.
- Note not adjusted for local site conditions. Try to distinguish effect but correlations do not reveal significant variations. Notes very few records on hard rock.
- Do not give coefficients only graphs of results.

4.11 *Kobayashi & Midorikawa (1982)*

- Ground motion model is:

$$\begin{aligned} \log \mathrm{Sv}_0(T) &= a(T)(\log M_0 - c) - b(T)\log X + d \\ \text{where } a(T) &= a_1 + a_2 \log T \\ \text{and: } b(T)) &= \begin{cases} b_1(\log T)^2 + b_2 \log T + b_3 & \text{for: } 0.1 \leq T \leq 0.3\,\mathrm{s} \\ b_4 - b_5 \log T & \text{for: } 0.3 \leq T \leq 5\,\mathrm{s} \end{cases} \end{aligned}$$

- Response parameter is velocity for 5% damping.
- Magnitudes converted to seismic moment, M_0, by using empirical formula.
- Observed surface spectra divided by amplification over bedrock (assumed to have shear-wave velocity of $3\,\mathrm{kms}^{-1}$), calculated for each of the 9 sites.
- Note equation not for near field because earthquake is not a point source.

4.12 *Joyner & Fumal (1984) & Joyner & Boore (1988)*

- See Section 2.27.
- Use data from Joyner & Boore (1982).
- Response parameter is pseudo-velocity for 5% damping.
- shear-wave velocity not significant, at 90%, for periods 0.1, 0.15 and $0.2\,\mathrm{s}$ but significant for longer periods.
- Regression using shear-wave velocity and depth to rock shows significant correlation (decreasing ground motion with increasing depth) for long periods but small coefficients. Short periods do not show significant correlation.
- State inappropriate to use depth to rock for present data due to limited correlation and because San Fernando data is analysed on its own does not show significant correlation.

4.13 *Kawashima* et al. *(1984)*

- See Section 2.28.
- Response parameter is acceleration for 5% damping.

4.14 *Kawashima* et al. *(1985)*

- See section 2.32.
- Response parameter is acceleration for 5% damping.
- Variation of a and b with respect to T is due to insufficient number of records.

4.15 *Kamiyama & Yanagisawa (1986)*

- Ground motion model is:

  $$\log_{10} V(T) = a(T)M_J - b(T)\log_{10}(\Delta + 30) - d(T)D - c(T) + A_1(T)S_1 + \ldots + A_{N-1}(T)S_{N-1}$$

  where $S_i = 1$ for ith site and 0 otherwise.
- Response parameters are acceleration, velocity and displacement for 0, 2, 5 and 10% damping
- Model site amplification of each of the 26 sites individually by using S_i. Choose one site as bed rock site, which has S-wave velocity of about $1000\,\mathrm{ms}^{-1}$.
- Use records with PGA$> 20\mathrm{gal}$ ($0.2\,\mathrm{ms}^{-2}$).
- Focal depths, D, between 0 and $130\,\mathrm{km}$, with most between 10 and $50\,\mathrm{km}$.
- Find no significant differences between site amplification spectra for different response parameters or different damping levels.
- Compare amplification spectra from regression for different sites with those predicted using S-wave theory and find good agreement.
- Coefficients only given for velocity for 5% damping.

4.16 *C.B. Crouse (1987) reported in Joyner & Boore (1988)*

- See Section 2.36.
- Response parameter is pseudo-velocity for 5% damping.

4.17 *K. Sadigh (1987) reported in Joyner & Boore (1988)*

- See Section 2.38.
- Response parameter is pseudo-acceleration for 5% damping.

4.18 *Annaka & Nozawa (1988)*

- See Section 2.41.
- Response parameter is acceleration for 5% damping.
- Give only graphs of coefficients.

4.19 *Crouse* et al. *(1988)*

- Ground motion model is:
$$\ln[\mathrm{PSV}(T)] = a + bM + c\ln[R] + dh$$
- Most data from shallow stiff soil and sedimentary deposits between about 5 and $25\,\mathrm{m}$ deep on Tertiary or older bedrock.
- Response parameter is pseudo-velocity for 5% damping.
- All earthquakes from Benioff-Wadati zones.
- Exclude data with magnitudes or distances well outside range of most selected records.
- Focal depths, h between 14 and $130\,\mathrm{km}$.
- No strong correlations between h, R and M.
- Try terms eM^2 and fR but find not significant (using t-test).
- Try term $R + C_1\exp(C_2M)$ instead of R; find similar standard errors.
- Find d is insignificant for 0.6 to $2\,\mathrm{s}$; find d does not significantly reduce standard errors.
- Find residuals are normally distributed (by plotting on normal probability paper and by Kolmogorov-Smirnov test).
- Split data by fault mechanism (thrust: 49 records, normal: 11 records, strike-slip: 4 records) and find attenuation equation for each subset; results are not significantly different (at 95% using F test). Also check by examining normal deviates (normalised residuals) for each subset and period; find no significant differences.
- Use 131 records from six other subduction zones (Nankai, Kuril, Alaska, Peru/N. Chile, Mexico and New Britain/Bougainville) to examine whether ground motions from all subduction zones are similar.
- Examine normal deviates for residuals between other zones' ground motion and N. Honshu equation. Find no significant differences (although obtain significant results for some periods and focal mechanisms) between N. Honshu, Kuril and Nankai motions. Find differences for Alaskan and Mexican data but could be due to site effects (because some data from soft soil sites). Find differences for Peru/N. Chile and New Britain/Bougainville which are probably source effects.
- Plot seismotectonic data (age, convergence rate, dip, contact width, maximum subduction depth, maximum historical earthquake (M_w), maximum rupture length, stress drop and seismic slip) against decreasing ground motion at stiff sites for $T > 0.8\,\mathrm{s}$. Find weak correlations for stress drop and M_w (if ignore Mexican data) but due to variability in stress drop estimates note lack of confidence in results.

4.20 *Yokota* et al. *(1988)*

- Ground motion model is:
$$\log S_v(T) = a(T)M + b(T)\log X + c(T)$$
- Response parameter is velocity for 5% damping.
- Focal depths between about 20 and $100\,\mathrm{km}$.
- Records from two stations in lowlands of Tokyo $3.7\,\mathrm{km}$ apart.

- Also analyse another region, using 26 records from 17 earthquakes with distances between 95 and 216 km. Note difference in results between regions.
- Analyses vertical spectra from three small regions separately, one with 24 records with $4.0 \leq M \leq 6.1$ and $60 \leq X \leq 100\,\mathrm{km}$, one with 22 records with $4.2 \leq M \leq 6.0$ and $68 \leq X \leq 99\,\mathrm{km}$ and one with 5 records with $4.4 \leq M \leq 6.0$ and $59 \leq X \leq 82\,\mathrm{km}$.
- Give no coefficients, only results.

4.21 *Youngs* et al. *(1988)*

- See Section 2.50.
- Ground motion model is:
$$\ln(S_v/a_{\max}) = C_6 + C_7(C_8 - M_w)^{C_9}$$
- Response parameter, S_v, is velocity[2] for 5% damping
- Develop relationships for ratio $S_v/a_{\max}$ because there is a much more data for PGA than spectral ordinates and use of ratio results in relationships that are consistent over full range of magnitudes and distances.
- Calculate median spectral shapes from all records with $7.8 \leq M_w \leq 8.1$ (choose this because abundant data) and $R < 150\,\mathrm{km}$ and one for $R > 150\,\mathrm{km}$. Find significant difference in spectral shape for two distance ranges. Since interest is in near-field ground motion use smoothed $R < 150\,\mathrm{km}$ spectral shape. Plot ratios $[S_v/a_{\max}(M_w)]/[S_v/a_{\max}(M_w = 8)]$ against magnitude. Fit equation given above, fixing $C_8 = 10$ (for complete saturation at $M_w = 10$) and $C_9 = 3$ (average value obtained for periods $> 1\,\mathrm{s}$). Fit C_7 by a linear function of $\ln T$ and then fix C_6 to yield calculated spectral amplifications for $M_w = 8$.
- Calculate standard deviation using residuals of all response spectra and conclude standard deviation is governed by equation derived for PGA.

4.22 *Trifunac & Lee (1989)*

- Ground motion model is:
$$\begin{aligned}
\log_{10}[\mathrm{PSV}(T)] &= M + \mathrm{Att}(\Delta, M, T) + b_1(T)M + b_2(T)h + b_3(T)v + b_5(T) \\
&\quad + b_6(T)M^2 \\
\text{where } \mathrm{Att}(\Delta, M, T) &= A_0(T)\log_{10}\Delta \\
A_0(T) &= \begin{cases} -0.732025 \quad \text{for:} \quad T > 1.8\,\mathrm{s} \\ -0.767093 + 0.271556\log_{10}T - 0.525641(\log_{10}T)^2 \\ \qquad\qquad\qquad\qquad \text{for:} \quad T < 1.8\,\mathrm{s} \end{cases} \\
\Delta &= S\left(\ln\frac{R^2 + H^2 + S^2}{R^2 + H^2 + S_0^2}\right)^{-1/2} \\
S &= 0.2 + 8.51(M - 5)
\end{aligned}$$

  where $v = 0$ for horizontal component and 1 for vertical, Δ is representative distance, S_0 is correlation radius of source function (or coherence size of source) (which can be approximated by $C_sT/2$, where C_s is shear wave velocity), h is depth of sedimentary deposits beneath recording station and H is focal depth.

[2] In paper conversion is made between S_v and spectral acceleration, S_a, suggesting that it is pseudo-velocity.

- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Screen data to minimize possible bias due to uneven distribution of data among different magnitude ranges and site conditions or from excessive contribution to database from several abundantly recorded earthquakes.
- Include term, $b_4(T)\Delta/100$, but insignificant for most periods so remove.
- Equation only applies for $M_{\min} \leq M \leq M_{\max}$, where $M_{\min} = -b_1(T)/(2b_6(T))$ and $M_{\max} = -(1 + b_1(T))/(2b_6(T))$. For $M \leq M_{\min}$ use $M_{\min}$ everywhere except first term. For $M \geq M_{\max}$ use $M_{\max}$ everywhere.
- Use method of Trifunac (1980) for uncertainties.
- Note estimates should only be used where signal to noise ratio (based on estimated digitisation noise) not much less than unity or slope in log-log scale is not significantly greater than -1.
- Also fit data to $\log_{10} \mathrm{PSV}(T) = M + \mathrm{Att}(\Delta, M, T) + b_1(T)M + b_2(T)s + b_3(T)v + b_5(T) + b_6(T)M^2$ (where $s = 0$ for sediment sites, 1 for intermediate sites and 2 for basement rock sites) because depth of sediment not always known.

4.23 *Atkinson (1990)*

- Ground motion model is:

$$\log y = c_1 + c_2(\mathbf{M} - 6) + c_3(\mathbf{M} - 6)^2 - \log R - c_4 R$$

- Response parameter is pseudo-velocity for 5% damping.
- All data from rock sites.
- Includes only if a reliable seismic moment estimate exists.
- Converts ECTN vertical seismograms to equivalent horizontal component by multiplying by 1.4.
- Includes Nahanni (western Canada) earthquakes because exhibit dominant characteristics of eastern North American shocks (low seismicity area, high horizontal compressive stress, thrust mechanisms dominant, no surface ruptures despite shallow focus and rocks have high seismic velocity).
- Excludes US digital strong-motion Saguenay records due to low resolution. Two effects on response spectra: i) high frequencies contaminated by a 'mathematical noise' floor, ii) significant errors in amplitudes of low to intermediate frequencies (severity dependent on resolution degree). Inclusion of such data could lead to significant misinterpretation of these earthquakes.
- Most records (66, 65%) from $R \geq 111\,\mathrm{km}$ and $\mathbf{M} \leq 5.22$.
- Examines residuals from equations. Finds no persistent trends except for Saguenay data ($\mathbf{M} = 6$) between $63 \leq R \leq 158\,\mathrm{km}$.
- Notes data very limited in large magnitude range and that one or two earthquakes are controlling predictions.
- Notes different regression technique could change predictions for large magnitudes but i) data too limited to warrant more sophisticated analysis and ii) may be other factors, in addition to number of recordings, which should be considered in weighting each earthquake.

4.24 *Campbell (1990)*

- See Section 2.55.
- Response parameter is pseudo-velocity for 5% damping.

4.25 *Dahle* et al. *(1990b) & Dahle* et al. *(1990a)*

- See Section 2.56.
- Response parameter is pseudo-velocity for 5% damping.
- Coefficients only given for 7 periods; graphs for others.

4.26 *Tamura* et al. *(1990)*

- Ground motion model is:

$$S_A(T_i, \mathrm{GC}) = a(T_i, \mathrm{GC}) 10^{b(T_i,\mathrm{GC})M} (\Delta + 30)^{C(T_i,\mathrm{GC})}$$

- Response parameter is acceleration for 2 and 5% damping.
- Use three site categories (GC) for which perform separate regression:

Group 1 Ground characteristic index $\lesssim 0.67$, 29 records.

Group 2 Ground characteristic index between about 0.67 and 1.50, 46 records.

Group 3 Ground characteristic index $\gtrsim 1.50$, 22 records.

where the ground characteristic index is calculated from statistical analysis of amplitude of records. Thought to reflect the characteristic of deep soil deposits at site (1.0 means amplification is average for Japan, < 1.0 or > 1.0 means amplification is lower or greater, respectively, than average for Japan).

- Records from JMA low-magnification mechanical seismographs (natural period $6\,\mathrm{s}$, damping ratio 0.55) which were instrument corrected (because sensitivity for periods $> 10\,\mathrm{s}$ is substantially suppressed) , filtered (cut-offs 1.3–$2\,\mathrm{s}$ and 20–$30\,\mathrm{s}$ chosen from a study of recording accuracy of instruments) and differentiated in frequency domain to find ground velocity and acceleration. Hence limit analysis to 2 to $20\,\mathrm{s}$.
- Do not use resultant of two horizontal components because two components not synchronous.
- Find difference in predicted ground motion using derived equations and those from earlier equations for short periods. Find that b for earlier equations increases almost linearly with logarithm of natural period, T, so find equation, by least squares, connecting b and $\log T$. Assume this equation holds for 2 to $20\,\mathrm{s}$ and so fix b and recalculate a and c; find predictions now agree.
- Only give graphs for original coefficients for 5% damping. Give tables of coefficients for preferred second analysis.

4.27 *Tsai* et al. *(1990)*

- See Section 2.59.
- Response parameter is acceleration for 5% damping.
- Also give equations for average acceleration for 2 period bands 0.12–$0.33\,\mathrm{s}$ and 0.07–$0.2\,\mathrm{s}$.

4.28 Crouse (1991)

- See Section 2.61.
- Response parameter is pseudo-velocity for 5% damping.
- Focal depths, h, between 10 and $238\,\mathrm{km}$.
- Notes that spectral database is biased to higher ground motions (because only higher ground motions are digitised). Suggest either using a different form of equation or impose constraints. Do not do either because (1) consider sample adequate for regression and (2) although overestimate smaller, more distant motion, it would properly estimate larger motions which are of greater concern for design applications.
- Sets p_3, p_5 and p_6 to those for PGA equation after trial regressions; does not appreciably affect standard deviation.
- Finds relatively larger standard deviation for 3.0 and $4.0\,\mathrm{s}$ which suggests form of equation may be inappropriate for longer periods.
- Plots normalised residuals (not shown) which show uniform distribution.

4.29 I.M. Idriss (1991) reported in Idriss (1993)

- See section 2.63.
- Response parameter is pseudo-acceleration for 5% damping.

4.30 Mohammadioun (1991)

- Ground motion model is:
$$\log \mathrm{PSV}(f) = k(f) + a(f)M + n(f)R$$
- Response parameter is pseudo-velocity for 5%.
- Records not baseline corrected so no equations for periods $> 2\,\mathrm{s}$.
- Does not split up data into subsets by intensity because risk of creating data populations which are not statistically significant.
- Notes that could be inconsistency with using both d_h and d_r.
- Notes that results are preliminary.
- Also analyses wide range of Californian data for 96 periods between 0.013 and $5\,\mathrm{s}$ split into two intensity dependent subsets: those records with site intensities VI-VII (326 records) and those with site intensities VII+ (156 records). Uses d_r except for Imperial Valley earthquake where uses d_E. Does not use include soil or other variables because poorly defined and lead to selection of records that are not statistically valid.

4.31 Benito et al. (1992)

- Ground motion model is:
$$\ln \begin{matrix}\mathrm{PSA} \\ \mathrm{PSV}\end{matrix} = c_1 + c_2 M + c_3 \ln(R + R_0) + c_4(R + R_0)$$

- Response parameters are pseudo-acceleration, PSA, and pseudo-velocity, PSV, for 5% damping[3].
- Use three soil conditions (revised when cross hole information was available):

$S = 0$ Hard and rock sites, 50 records.

$S = 1$ Intermediate soil, 10 records.

$S = 2$ Soft soil, 12 records.

- Use M_L because most suitable for distance range of majority of records.
- Try including $c_5 S$ term but find low significance values for c_5. Repeat regression for each soil category separately. Give results when coefficient of determination $R^2 > 0.80$, standard errors $< 25\%$ and coefficients have high significance levels.
- For PSA for $S = 0$ give coefficients for all periods, for $S = 1$ give coefficients for 0.17 to $0.2\,\mathrm{s}$ and for $S = 2$ give coefficients for 1 to $10\,\mathrm{s}$.
- Also consider Friuli records ($4.2 \leq M_L \leq 6.5$, epicentral distances between 2 and $192\,\mathrm{km}$, 14 records for $S = 0$, 23 records for $S = 1$ and 16 records for $S = 2$).
- Note need to include term in model reflecting explicitly local amplification dependent on natural period of soil as well as predominant period of incident radiation to bed rock.

4.32 *Niazi & Bozorgnia (1992)*

- See Section 2.64.
- Response parameter is pseudo-velocity for 5% damping.
- For some periods ($0.20\,\mathrm{s}$ for vertical and 0.10 and $0.111\,\mathrm{s}$ for horizontal) constrain c_2 to zero so that predicted amplitude would not decrease with increasing magnitude at zero distance. Note that does not affect uncertainty.
- Note that long period filter cutoff may be too long for records from small shocks but if a shorter period was used then information on long period spectral ordinates would be lost. Note that insufficient data for well constrained results at $M = 5$ or $M > 7$.
- Find evidence for long period noise in d and in Degree of Magnitude Saturation ($\mathrm{DMS} = -(c_2 d/b) * 100$).
- Examine median and normalized standard deviation (coefficient of variation) and find evidence for decreasing uncertainty with increasing magnitude.

4.33 *Tento* et al. *(1992)*

- See Section 2.69.
- Response parameter is pseudo-velocity for 5% damping.
- Note that correction procedure significantly affects results for $T > 2\,\mathrm{s}$. Correction procedure introduces dishomogeneity and errors due to subjectivity of choice of low frequency filter limits.

[3] Although coefficients should only differ by a constant because $\mathrm{PSA} = (2\pi/T)\mathrm{PSV}$ they do not; hence response parameters are probably not those stated.

4.34 *Boore* et al. *(1993) & Boore* et al. *(1997)*

- See Section 2.71
- Response parameter is pseudo-velocity for 2, 5, 10 and 20% damping.
- Cutoff distance is lesser of distance to first digitized record triggered by S wave, distance to closest non-digitized recording, and closest distance to an operational nontriggered instrument.
- Note that can only use response spectral values between 0.1 and $2\,\mathrm{s}$ because of low sampling rate of older data (sometimes only 50 samples/sec) and low signal to noise ratios and filter cutoffs.
- Site categories same as in Section 2.71 but due to smaller dataset number of records in each category is less. Class A: 12 records, B: 51 records, C: 49 records.
- Smoothed coefficients using a least-squares fit of a cubic polynomial.

4.35 *Caillot & Bard (1993)*

- Ground motion model is:
$$\ln y = \beta_1 + \beta_2 M + \beta_3 \ln \mathrm{HYPO} + \beta_4 S_1$$
- Response parameter is acceleration for 5% damping.
- Consider three site conditions but only retain two:
  1. Rock: ENEA/ENEL S0 classification $\Rightarrow S_1 = 0$, 49 records.
  2. Thin alluvium: depth of soil between 5 and $20\,\mathrm{m}$,ENEA/ENEL S1 classification $\Rightarrow S_1 = 1$, 34 records.
- Selected records have $d_e < 60\,\mathrm{km}$ and focal depth less than $30\,\mathrm{km}$. Data selected so that mean and standard deviation of magnitude and hypocentral distance in each site category are equal, in this case 5.1 and $20\,\mathrm{km}$ respectively.
- All records processed using common procedure. High pass filtered with $f_l = 0.5\,\mathrm{Hz}$, instrument corrected and low pass filtered with $f_h = 30\,\mathrm{Hz}$.
- Considered three things when choosing method of analysis:
  1. Attenuation equation must have some physical basis.
  2. Parameters must be available for original data set.
  3. Attenuation equation must be easy to use in a predictive manner.
- Hypocentral distance used because rupture not known for most earthquakes. Note that only important for magnitudes greater than about 6.5 and distances less than about $15\,\mathrm{km}$.
- Originally included another set of data (32 records) from thick soil with depth greater than about $20\,\mathrm{m}$ (ENEA/ENEL classification S2) but note that results for this category are much more uncertain, possibly due to diversity of geotechnical characteristics of soils. Therefore excluded.
- Regression was done using two-stage algorithm (Joyner & Boore, 1981) and a weighted one-stage method. Weight by splitting the magnitude and distance ranges into four intervals and weighting data in each interval inversely proportionally to number of points in the bin. Thus gives roughly equal weight to each part of magnitude-distance space.

- Note that results from two-stage regression for this set of data may be misleading because for some periods it does not bring any 'explanation' to the variance of initial data. The two-stage and normal one-stage and weighted one-stage yield significant changes in predictions.
- Repeat analysis using only S0 subset and using only S1 subset but no significant changes in magnitude or distance scaling between the two subsets so consider complete set and include a constant scaling between rock and shallow soil. If set is reduced to 53 records with similar spread of magnitude, distance and sites then difference between shallow soil and rock is not significant.
- Note that confidence interval should be given by formula in Weisburg (1985) not normal way of simply using standard deviation.

4.36 Campbell (1993)

- See Section 2.72.
- Response parameter is pseudo-acceleration for 5% damping.
- Notes that equation can predict smaller pseudo-acceleration than PGA for short periods, which is impossible in practice. Hence pseudo-acceleration for periods $\leq 0.2\,\mathrm{s}$ should be constrained to be $\geq$ PGA.

4.37 Lee (1993)

- Ground motion model is:

$$\begin{aligned}
\log_{10}[\widehat{\mathrm{PSV}}(T)] &= M_{<} + \mathrm{Att}(\Delta, M, T) + \hat{b}_1(T) M_{<>} + \hat{b}_2(T) h + \hat{b}_3(T) v \\
&\quad + \hat{b}_4(T) hv + \hat{b}_5(T) + \hat{b}_6(T) M_{<>}^2 + \hat{b}_7^{(1)}(T) S_L^{(1)} + \hat{b}_7^{(2)}(T) S_L^{(2)} \\
\text{where } M_{<} &= \min(M, M_{\max}) \\
M_{<>} &= \max(M_{\min}, M_{<}) \\
M_{\min} &= -\hat{b}_1/(2\hat{b}_6(T)) \\
M_{\max} &= -(1 + \hat{b}_1(T))/(2\hat{b}_6(T))
\end{aligned}$$

  where $v = 0$ for horizontal component, 1 for vertical, h is depth of sedimentary deposits beneath recording station and $\mathrm{Att}(\Delta, M, T)$ is same as Trifunac & Lee (1989) (see Section 4.22).

- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Uses three site categories:

$S_L = 0$ Rock: 1 sediment site ($h > 0$), 11 intermediate sites ($h \sim 0$) and 13 bedrock sites ($h = 0$) $\Rightarrow$ $S_L^{(1)} = 0$ & $S_L^{(2)} = 0$.

$S_L = 1$ Stiff soil ($\leq 45 - 60\,\mathrm{m}$ deep): 37 sediment sites ($h > 0$), 24 intermediate sites ($h \sim 0$) and 3 bedrock sites ($h = 0$) $\Rightarrow$ $S_L^{(1)} = 1$ & $S_L^{(2)} = 0$.

$S_L = 2$ Deep soil: 44 sediment sites ($h > 0$) and 2 intermediate sites ($h \sim 0$) $\Rightarrow$ $S_L^{(1)} = 0$ & $S_L^{(2)} = 1$.

- For $M > 6.5$ uses different (unspecified) magnitude scales because for seismic risk analysis often catalogues do not specify scale and often estimates are not homogeneous.
- Free-field records with both soil and alluvial depth information.
- Screens data to minimize possible bias due to uneven distribution of soil classification or excessive contribution from several abundantly recorded earthquakes.

- Gives smoothed coefficients for 12 periods.
- Uses method of Trifunac (1980) for uncertainties.
- Also uses method where site coefficients, $\hat{b}_7^{(1)}$ & $\hat{b}_7^{(2)}$, are found from residues from equation without site coefficients; find similar results.

4.38 *Sadigh* et al. *(1993) & Sadigh* et al. *(1997)*

- See Section 2.73
- Ground motion model for deep soil is:

$$\ln y = C_1 + C_2 M - C_3 \ln(r_{\mathrm{rup}} + C_4 \mathrm{e}^{C_5 M}) + C_6 + C_7 (8.5 - M)^{2.5}$$

  where C_6 is different for reverse and strike-slip earthquakes.

  Ground motion model for rock is:

$$\ln y = C_1 + C_2 M + C_3 (8.5 - M)^{2.5} + C_4 \ln(r_{\mathrm{rup}} + \exp(C_5 + C_6 M)) + C_7 \ln(r_{\mathrm{rup}} + 2)$$

  where C_1 is different for reverse and strike-slip earthquakes.

  Vertical equations do not include C_7.
- Response parameter is acceleration for 5% damping.
- Perform analysis on spectral amplification $\ln(\mathrm{SA}/\mathrm{PGA})$.
- Give smooth coefficients.
- Find standard errors to be dependent on magnitude and fit to a linear relation.

4.39 *Sun & Peng (1993)*

- See section 2.75.
- Response parameter is acceleration for 5% damping.
- Coefficients not given.

4.40 *Boore* et al. *(1994a) & Boore* et al. *(1997)*

- See Section 2.76
- Find no evidence for magnitude dependent uncertainty for spectral values.
- Find no evidence for amplitude dependent uncertainty for spectral values.
- Note that effect of basin-generated surface waves can have an important effect but probably not at periods between 0.1 and $2\,\mathrm{s}$.

4.41 *Climent* et al. *(1994)*

- Inspect observed and predicted values and conclude no clear difference between upper-crustal and subduction zone ground motions. Equations are for region regardless of earthquake source type.

4.42 *Fukushima* et al. *(1994) & Fukushima* et al. *(1995)*

- See Section 2.77.
- Response parameter is pseudo-velocity for 5% damping.
- Only give graphs of coefficients.
- Note possible noise contamination, for periods $< 0.1\,\mathrm{s}$, in coefficients.

4.43 *Lawson & Krawinkler (1994)*

- See Section 2.78.
- Response parameter is acceleration for 5% damping.

4.44 *Lee & Manić (1994) & Lee (1995)*

- Ground motion model is:

$$\begin{aligned}
\log_{10}\widehat{\mathrm{PSV}} &= M_{<} + \mathrm{Att} + b_1 M_{<>} + b_2^{(1)} S^{(1)} + b_2^{(2)} S^{(2)} + b_3 v + b_4 + b_5 M_{<>}^2 \\
&\quad + b_6^{(1)} S_L^{(1)} \\
M_{<} &= \min(M, M_{\max}) \\
\text{where } M_{\max} &= \frac{-(1+b_1)}{2b_5} \\
M_{<>} &= \max(M_{<}, M_{\min}) \\
\text{where } M_{\min} &= \frac{-b_1}{2b_5} \\
\mathrm{Att} &= \begin{cases} A_0 \log_{10}\Delta & \text{for} \quad R \le R_0 \\ A_0 \log_{10}\Delta_0 - \frac{(R-R_0)}{200} & \text{for} \quad R > R_0 \end{cases} \\
\text{with: } A_0 &= \begin{cases} -0.761 & \text{for} \quad T \ge 1.8\,\mathrm{s} \\ -0.831 + 0.313\log_{10} T - 0.161(\log_{10} T)^2 & \text{for} \quad T < 1.8\,\mathrm{s} \end{cases} \\
\Delta &= S\left[\ln\left(\frac{R^2+H^2+S^2}{R^2+H^2+S_0^2}\right)\right]^{-\frac{1}{2}} \\
\Delta_0 &= \Delta(R_0) \\
\text{where } R_0 &= \frac{1}{2}\left\{\frac{-200A_0(1-S_0^2/S^2)}{\ln 10} + \left[\left[\frac{200A_0(1-S_0^2/S^2)}{\ln 10}\right]^2 - 4H^2\right]\right\}
\end{aligned}$$

  where Δ is 'representative' distance, S is 'size' of fault, S_0 is coherence radius of source and v is component orientation ($v = 0$ for horizontal, $v = 1$ for vertical).

- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Consider three geological site conditions:

  $s = 0$ Sediment: $\Rightarrow S^{(1)} = 0, S^{(2)} = 0$, 151 records.

  $s = 1$ Intermediate sites: $\Rightarrow S^{(1)} = 1, S^{(2)} = 0$, 106 records.

  $s = 2$ Basement rock: $\Rightarrow S^{(1)} = 0, S^{(2)} = 1$, 54 records.

- Consider three local site categories but only retain two:

$s_L = 0$ Rock: $\Rightarrow S_L^{(1)} = 0$, 100 records.

$s_L = 1$ Stiff soil: $\Rightarrow S_L^{(1)} = 1$, 205 records.

- Cannot include those records from deep soil sites ($s_L = 2$) because only six records.
- Most earthquakes are shallow, depth $H < 25\,\mathrm{km}$.
- Most records have epicentral distances, $R < 50\,\mathrm{km}$.
- Most have magnitudes between 3 and 6.
- Only use records with high signal-to-noise ratio. Quality of records is not adequate for response spectrum calculation outside range 0.04 to $2\,\mathrm{s}$.
- Analysis performed using residue 2-step method. In first step use only records from $M \geq 4.25$ to force a concave form to magnitude scaling (if all records used then find a convex parabola), s_L parameter is not included. In second step find s_L dependence from residuals of first stage including all magnitudes.
- Give expressions to describe distribution of residuals so that can find confidence limits, unlike normal standard deviation based method, see Trifunac (1980).
- Note difference between western USA and Yugoslavian ground motions.

4.45 Mohammadioun (1994a)

- Ground motion model is:

$$\log \mathrm{SR}(f) = k(f) + \alpha(f)M + n(f)\log R$$

- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Uses records from rock sites ($V_s \geq 750\,\mathrm{ms}^{-1}$).
- Half of records from $R < 30\,\mathrm{km}$ and significant number from $R < 10\,\mathrm{km}$.
- Most (82%) records from earthquakes with $6.2 \leq M \leq 7.0$.
- Coefficients not given, only results.

4.46 Mohammadioun (1994b)

- Ground motion model is:

$$\log V(f) = k(f) + \alpha(f)M + n(f)\log R$$

- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Choose W. USA to make data as homogeneous as possible in terms of seismotectonic context and parameter quality.
- Notes recording site-intensities may only be average intensity values, thereby neglecting possible microzoning effects.
- Uses M_L because generally available and uniformly determined. Notes may not be best choice.
- Records from free-field and typical of different intensity classes.

- Does regression for records associated with three different intensities: V (184 records, $5.5 \lesssim R \lesssim 200\,\mathrm{km}$), VI (256 records, $3 \lesssim R \lesssim 250\,\mathrm{km}$, VII (274 records, $1 \lesssim R \lesssim 150\,\mathrm{km}$) and four different intensity groups: V-VI, VI-VII, VII and more (extra 25 records, $1 \lesssim R \lesssim 100\,\mathrm{km}$) and V and less (extra 30 records, $25 \lesssim R \lesssim 350\,\mathrm{km}$.
- Graph of $\alpha(f)$ given for horizontal component for the four intensity groups and graph of $n(f)$ for vertical component for intensity VI.

4.47 *Musson* et al. *(1994)*

- See section 2.80.
- Response parameter is pseudo-velocity for 5% damping.
- More data because use analogue records as well.

4.48 *Theodulidis & Papazachos (1994)*

- Use same data, equation and procedure as Theodulidis & Papazachos (1992), see Section 2.70.
- Response parameter is pseudo-velocity for 0, 2, 5, 10 and 20% damping.
- Note lack of near-field data ($R < 20\,\mathrm{km}$, $M > 6.2$) to constrain R_0.
- Only give graphs of original coefficients but give table of smoothed (using a ($\frac{1}{4}$+$\frac{1}{2}$+$\frac{1}{4}$ running average along $\log T$) coefficients for 13 periods and all 5 damping levels.
- Note large residuals for $T > 0.5\,\mathrm{s}$ due mainly to different digitising and processing procedures which significantly affect long period spectral values.
- Check histograms of residuals for 0.1, 0.3, 0.5, 1, 3 and $5\,\mathrm{s}$ and find similar to normal distribution.
- Note no data from $R < 30\,\mathrm{km}$ for $M > 6.5$ so state caution is required for use of equations in that range. Also suggest do not use equations for $M > 7.5$ or for $R > 130\,\mathrm{km}$.
- Note may not apply for very soft soils.
- Note lack of data.

4.49 *Dahle* et al. *(1995)*

- See Section 2.85.
- Derive spectral attenuation relations for almost double number of periods given. Coefficients smoothed using a third degree polynomial.

4.50 *Ambraseys* et al. *(1996) & Simpson (1996)*

- See Section 2.89.
- Response parameter is acceleration for 5% damping.
- Do no smoothing because if plotted on a normal scale then smoothing should be done on T, but if on log-log plot then smoothing should be done on $\log T$.

4.51 *Ambraseys & Simpson (1996) & Simpson (1996)*

- See Section 2.90.
- Response parameter is acceleration for 5% damping.

4.52 *Bommer* et al. *(1996)*

- See section 2.91.
- Response parameter is pseudo-velocity for unspecified damping.

4.53 *Crouse & McGuire (1996)*

- See section 2.92.
- Response parameter is pseudo-velocity for 5% damping.
- Find k_1 not significantly different than 1 for $T \leq 0.15\,\mathrm{s}$ and k_2 not significantly different than 1 for $T \leq 0.50\,\mathrm{s}$.

4.54 *Free (1996) & Free* et al. *(1998)*

- See Section 2.93.
- Response parameter is acceleration for 5% damping.
- Finds including focal depth, h, explicitly has dramatic effect on predicted spectra at short distances but insignificant effect at large distances.
- Repeats analysis using only E. N. American data. Finds significantly larger amplitudes than predictions from combined set for short and intermediate distances for periods $> 0.3\,\mathrm{s}$ but similar spectra for large distances.

4.55 *Ohno* et al. *(1996)*

- See Section 2.94.
- Response parameter is acceleration for 5% damping.
- Plot amplitude factors from first stage against M_w; find well represented by linear function.
- Do not give table of coefficients only graphs of coefficients.

4.56 *Sabetta & Pugliese (1996)*

- Ground motion model used is:

$$\log_{10} Y = a + bM - \log_{10} \sqrt{d^2 + h^2} + e_1 S_1 + e_2 S_2$$

- Response parameter, Y, is pseudo-velocity for 5% damping
- Use data from Sabetta & Pugliese (1987).

- Remove anelastic decay term because it was not significant at $\alpha = 0.1$ and sometimes it was positive. Originally geometrical decay coefficient c was allowed to vary but find it is close to -1 so constrain.
- Use three site categories:

$S_1 = 1, S_2 = 0$ Shallow: depth $H \leq 20\,\mathrm{m}$ alluvium $400 \leq V_s \leq 800\,\mathrm{ms}^{-1}$.
$S_1 = 0, S_2 = 1$ Deep: depth $H > 20\,\mathrm{m}$ alluvium $400 \leq V_s \leq 800\,\mathrm{ms}^{-1}$.
$S_1 = 0, S_2 = 0$ Stiff: $V_s > 800\,\mathrm{ms}^{-1}$.

- Accelerograms digitised at 400 samples/sec. Bandpass frequencies chosen by an analysis of signal and fixed trace Fourier spectra. $f_{\min}$ between 0.2 and $0.7\,\mathrm{Hz}$ most about $0.3\,\mathrm{Hz}$ and $f_{\max}$ between 20 and $35\,\mathrm{Hz}$ most about $25\,\mathrm{Hz}$. Instrument correction applied.
- Use one-stage method although two-stage method yields similar results.
- Also present smoothed coefficients.

4.57 *Spudich* et al. *(1996)*

- See Section 2.96
- Response parameter is pseudo-velocity for 5% damping.
- Only use spectral values within the passband of the filter used to correct records hence number of records used for each period varies, lowest number is 99 for periods between 1.7 and $2.0\,\mathrm{s}$.
- Smooth coefficients using cubics or quadratics.

4.58 *Abrahamson & Silva (1997)*

- Ground motion model is:

$$
\begin{aligned}
\ln \mathrm{Sa} &= f_1 + F f_3 + HW f_{HW}(M) f_{HW}(R_{\mathrm{rup}}) + S f_5 \\
f_1 &= \begin{cases} a_1 + a_2(M - c_1) + a_{12}(8.5 - M)^n + [a_3 + a_{13}(M - c_1)] \ln R & \\ \qquad\qquad \text{for} \quad M \leq c_1 \\ a_1 + a_4(M - c_1) + a_{12}(8.5 - M)^n + [a_3 + a_{13}(M - c_1)] \ln R & \\ \qquad\qquad \text{for} \quad M > c_1 \end{cases} \\
\text{where } R &= \sqrt{r_{\mathrm{rup}} + c_4^2} \\
f_3 &= \begin{cases} a_5 & \text{for} \quad M \leq 5.8 \\ a_5 + \frac{a_6 - a_5}{c_1 - 5.8} & \text{for} \quad 5.8 < M < c_1 \\ a_6 & \text{for} \quad M \geq c_1 \end{cases} \\
f_{HW}(M) &= \begin{cases} 0 & \text{for} \quad M \leq 5.5 \\ M - 5.5 & \text{for} \quad 5.5 < M < 6.5 \\ 1 & \text{for} \quad M \geq 6.5 \end{cases} \\
f_{HW}(r_{\mathrm{rup}}) &= \begin{cases} 0 & \text{for} \quad r_{\mathrm{rup}} < 4 \\ a_9 \frac{r_{\mathrm{rup}} - 4}{4} & \text{for} \quad 4 < r_{\mathrm{rup}} < 8 \\ a_9 & \text{for} \quad 8 < r_{\mathrm{rup}} < 18 \\ a_9 \left(1 - \frac{r_{\mathrm{rup}} - 18}{7}\right) & \text{for} \quad 18 < r_{\mathrm{rup}} < 24 \\ 0 & \text{for} \quad r_{\mathrm{rup}} > 25 \end{cases} \\
f_5 &= a_{10} + a_{11} \ln(\widehat{\mathrm{PGA}} + c_5)
\end{aligned}
$$

where $\widehat{\mathrm{PGA}}$ is expected peak acceleration on rock as predicted by the attenuation equation with $S = 0$.

- Response parameter is acceleration for unspecified[4] damping.
- Use two site categories:

$S = 0$ Rock: rock ($V_s > 600\,\mathrm{ms}^{-1}$), very thin soil ($< 5\,\mathrm{m}$) over rock or soil 5 to $20\,\mathrm{m}$ thick over rock.

$S = 1$ Deep soil: deep soil in narrow canyon (soil $> 20\,\mathrm{m}$ thick and canyon $< 2\,\mathrm{km}$ wide) or deep soil in broad canyon (soil $> 20\,\mathrm{m}$ thick and canyon $> 2\,\mathrm{km}$ wide).

- All records reprocessed using common procedure. Interpolated to 400 samples/sec, low-pass filtering with corner frequency selected for each record based on visual examination of Fourier amplitude spectrum, instrument corrected, decimated to 100 to 200 samples/sec depending on low-pass corner frequency, baseline correction using 0 to 10 degree polynomial, high-pass filtered based on integrated displacements.
- Only use response spectral data within frequency band $1.25f_h$ to $0.8f_l$ to avoid effects of filter roll-off. Hence number of records used for regression at each period varies, minimum number is less than 100 records for $0.01\,\mathrm{s}$.
- Well distributed dataset in terms of magnitude and distance.
- Supplement data with records from Gazli, Friuli, Tabas, Taiwan, Nahanni and Spitak.
- Consider source mechanism: reverse $\Rightarrow F = 1$, reverse/oblique $\Rightarrow F = 0.5$, others (strike-slip and normal) $\Rightarrow F = 0$).
- Consider hanging wall effect: if over hanging wall $HW = 1$, otherwise $HW = 0$.
- Note that interpretation of c_4 is not clear for their distance measure but yields better fit.
- Model nonlinear soil response by f_5.
- Model uncertainty as magnitude dependent.
- Fix some coefficients to be independent of period so that response spectral values vary smoothly with distance, magnitude and period.
- Smooth coefficients using piecewise continuous linear fits on log period axis. For highly correlated coefficients, smooth one coefficient and re-estimate other coefficients.

4.59 Atkinson (1997)

- Ground motion model used is:

$$\begin{aligned} \log \mathrm{PSA} &= c_0 + c_1(M_w - 6) + c_2(M_w - 6)^2 + c_3 h - c_{a_1} \log R - c_{a_2} R + c_s S \\ \text{with: } c_{a_2} &= c_{a_3} + c_{a_4} h \end{aligned}$$

- Response parameter is pseudo-acceleration for 5% damping.
- Uses two site categories (no soil profiles were available for Cascadia region):

$S = 0$ Rock: average V_s assumed to be about $2000\,\mathrm{ms}^{-1}$

$S = 1$ Soil: average V_s assumed to be about $255\,\mathrm{ms}^{-1}$ (although includes some soft soil sites with average V_s about $125\,\mathrm{ms}^{-1}$).

- Tectonic type of earthquakes used: crustal, subcrustal and subduction

[4] It is probably 5%.

- Most Cascadia data is from seismograms. Converts vertical measurements from these to one horizontal component.
- Supplements in large magnitude range ($6.7 < M_w \leq 8.2$) with data from 9 subduction earthquakes in Alaska, Mexico, Japan and Chile
- Most magnitudes below 5.3 and no data between 6.8 and 7.5.
- Focal depths between 1 and $60\,\mathrm{km}$
- Only uses events recorded at 3 or more stations. Improves ability of regression to distinguish between magnitude and distance dependencies in data.
- Most low magnitude events were recorded on rock and most high magnitude events were on soil. Thus to stabilize regression takes the coefficients c_s from Boore *et al.* (1994a) and not derived from this data.
- Magnitude partitioning, in first step, into 0.5 unit intervals gave evidence for magnitude dependent attenuation. Uses $c_{a_1} = 1$ for $4.1 \leq M_w \leq 6.7$ and $c_{a_1} = 0.5$ (largest which yielded positive c_{a_2}) for $M_w \geq 7.5$. Thought to show breakdown of point source assumption.
- Demonstrates depth dependence in anelastic decay by performing regression in four $15\,\mathrm{km}$ deep subsets for range $4.1 \leq M_w \leq 6.7$. c_{a_3} and c_{a_4} then finds by regression for each period. No depth dependence for $M_w \geq 7.5$ because of lack of different depths.
- Includes depth dependence in second step because gave better fit for short periods.
- Checks dependence on crustal, interface and intra-slab events; finds no dependence.

4.60 Campbell (1997)

- See Section 2.97
- Ground motion model (horizontal component) is:

$$\begin{aligned}
\ln \mathrm{SA}_H &= \ln A_H + c_1 + c_2 \tanh[c_3(M-4.7)] + (c_4 + c_5 M) R_{\mathrm{SEIS}} + 0.5 c_6 S_{\mathrm{SR}} \\
&\quad + c_6 S_{\mathrm{HR}} + c_7 \tanh(c_8 D)(1 - S_{\mathrm{HR}}) + f_{\mathrm{SA}} \\
f_{\mathrm{SA}} &= \begin{cases} 0 & \text{for} \quad D \geq 1\,\mathrm{km} \\ c_6(1 - S_{\mathrm{HR}})(1-D) + 0.5 c_6 (1-D) S_{\mathrm{SR}} & \text{for} \quad D < 1\,\mathrm{km} \end{cases}
\end{aligned}$$

- Ground motion model (vertical component) is:

$$\begin{aligned}
\ln \mathrm{SA}_V &= \ln \mathrm{SA}_H + c_1 + b_2 M + c_2 \tanh[d_1(M-4.7)] + c_3 \tanh[d_2(M-4.7)] \\
&\quad + b_3 \ln[R_{\mathrm{SEIS}} + b_4 \exp(b_5 M)] + b_6 \ln[R_{\mathrm{SEIS}} + b_7 \exp(b_8 M)] + b_9 F \\
&\quad + c_4 \tanh(d_3 D) + c_5 \tanh(d_4 D)
\end{aligned}$$

- Response parameter is pseudo-acceleration for 5% damping.
- Notes importance of depth to basement rock, D, for modelling long period site response. For shallow sediments defines D as depth to top of Cretaceous or older deposits, for deep sediments determine D from crustal velocity profiles where define basement as crystalline basement rock or sedimentary deposits having a P-wave velocity $\geq 5\,\mathrm{kms}^{-1}$ or shear-wave velocity $\geq 3\,\mathrm{kms}^{-1}$ (called 'seismic basement' by geophysicists).

- Uses different data than for PGA equations hence: reverse (3), thrust (H:9, V:6), reverse-oblique (2) and thrust-oblique (0), total (H:14, V:11) (H:140 records, V:85 records), strike-slip (H:124 records, V:88 records). Only two normal faulting earthquakes in horizontal set of records (contributing 2 records) so a difference in not modelled although $F = 0.5$ is given as first approximation to use as for PGA case.
- Only excludes records from toe and base of dams, included those from buildings and bridge columns which were excluded from PGA study, because of lack of data.
- Uses weighted regression analysis. Assigns recordings from a given earthquake that fell within the same distance interval (ten logarithmical spaced) same weight as those recordings from other earthquakes that fell within the same distance interval. Gives recordings from a given earthquake that occurred at the same site location the same cumulative weight as a single recording at that distance, thus reducing the bias.
- Performs analysis on spectral ratio $\ln(\mathrm{PSA}/\mathrm{PGA})$ because of unacceptably large period-to-period variability in regression coefficients when direct regression is applied and strongly correlated coefficients. Notes that are too many regression coefficients so it was necessary to perform analysis in many steps, at each step different coefficients are determined and detrended and residuals examined to find appropriate functional forms for trends present. Yields more stable results.
- No consideration of nontriggering instruments made, unlike PGA study.

4.61 *Youngs* et al. *(1997)*

- See Section 2.99
- Ground motion model used is:

$$\ln(\mathrm{SA}/\mathrm{PGA}) = B_1 + B_2(10 - \mathbf{M})^3 + B_3 \ln\left[r_{\mathrm{rup}} + \mathrm{e}^{\alpha_1 + \alpha_2 \mathbf{M}}\right]$$

  where α_1 and α_2 are set equal to C_4 and C_5 of appropriate PGA equation.
- Response parameter, SA, is acceleration for 5% damping.
- Do analysis on response spectral amplification because digitised and processed accelerograms used for spectral attenuation is only a subset of PGA database and they are often those with strongest shaking. Hence analysis directly on spectral accelerations may be biased.
- Smooth coefficients.

4.62 *Bommer* et al. *(1998)*

- Ground motion model is:

$$\begin{aligned} \log(\mathrm{SD}) &= C_1 + C_2 M + C_4 \log r + C_A S_A + C_S S_S \\ r &= \sqrt{d^2 + h_0^2} \end{aligned}$$

- Response parameter is displacement for 5, 10, 15, 20, 25 and 30% damping.
- Use three site conditions:

  R Rock: $V_s > 750\,\mathrm{ms}^{-1}$, $S_A = 0$, $S_S = 0$, 30–45 records.

  A Stiff soil: $360 < V_s \leq 750\,\mathrm{ms}^{-1}$, $S_A = 1$, $S_S = 0$, 56–92 records.

  S Soft soil: $180 < V_s \leq 360\,\mathrm{ms}^{-1}$, $S_A = 0$, $S_S = 1$, 32–43 records.

- Use subset of data of Ambraseys *et al.* (1996) (see 2.89) data with a few changes and exclusion of records from earthquakes with $M_s < 5.5$ because ground motion at long periods was of interest and to increase likelihood of acceptable single-to-noise ratio at longer periods.
- Each record individually filtered. Firstly filter record with sharp low cut-off at $0.1\,\mathrm{Hz}$ and plot velocity and displacement time-histories. Check, visually, whether contaminated by noise and if so increase cutoff frequency by small amount and repeat procedure until resulting velocity and displacement time-histories are deemed acceptable and no significant improvement is observed by further increase of cutoff frequency. Instrument correction not applied because high frequency distortion caused by transducer characteristics not important for displacement spectra. Only use each record for regression for periods up to $0.1\,\mathrm{s}$ less than filter cutoff used for that record to avoid distortion by filter, hence as period increases number of data points decreases.
- Regression procedure same as Ambraseys *et al.* (1996), see 2.89.

4.63 Perea & Sordo (1998)

- Ground motion model is:
$$\ln \mathrm{Pa} = \beta_1 + \beta_2 M + \beta_3 \ln(R + 25)$$
- Response parameter is pseudo-acceleration for 5% damping.
- All records from five medium soft soil sites.
- Use m_b for $M < 6$ and M_s otherwise, because m_b is more representative of released energy for small earthquakes and M_s better represents energy release for large earthquakes because m_b saturates starting from $M > 6$.
- Try including anelastic decay term, $\beta_4 R$ but it does not significantly affect standard deviation.
- Also repeat analysis for three other zones. Zone 1: 3 earthquakes, 3 records ($5.0 \leq M \leq 6.4$, $80 \leq R \leq 156\,\mathrm{km}$) for which conclude has too limited data for reliable equation. Zone 3[5]: 11 earthquakes, 13 records ($4.5 \leq M \leq 7.7$, $251 \leq R \leq 426\,\mathrm{km}$) for which find fits spectra of medium sized shocks better than large shocks because of lack of data for large earthquakes. Zone 4: 4 earthquakes, 7 records ($5.1 \leq M \leq 6.2$, $356 \leq R \leq 573\,\mathrm{km}$) for which find β_2 is negative and β_3 is positive for some periods (which is nonphysical) which state is due to limited number of earthquakes and their similar epicentral distances.
- Find fit spectra of medium sized earthquakes than large earthquakes because of lack of data from large earthquakes.
- Only give graphs of coefficients.

4.64 Shabestari & Yamazaki (1998)

- Ground motion model is:
$$\log y(T) = b_0(T) + b_1(T)M + b_2(T) - \log r + b_4(T)h + c_i(T)$$
where $c_i(T)$ is the station coefficient, reflecting relative site effect for each period, assuming zero mean for all stations.
- Response parameters are acceleration and velocity for 5% damping.

[5] The following values are from their Table 1 which does not match with their Figure 3.

- Include at least five earthquakes with $M_{\mathrm{JMA}} \geq 7.2$.
- Exclude earthquakes with focal depths, h, equal to $0\,\mathrm{km}$ or greater than $200\,\mathrm{km}$.
- Exclude records with vectorial composition of PGA less than $0.01\,\mathrm{ms}^{-2}$.
- Use three-stage iterative partial regression method.
- For $T \geq 6\,\mathrm{s}$ constrain horizontal anelastic coefficient to zero because get positive coefficient.
- See Yamazaki *et al.* (2000) for examination of station coefficients.

4.65 *Chapman (1999)*

- See Section 2.108.
- Response parameter is pseudo-velocity for 2, 5 and 10% damping.

4.66 *Spudich* et al. *(1999)*

- See Section 2.110.
- Response parameter is pseudo-velocity for 5% damping.
- Use only use response spectral data within frequency band $1.25f_h$ to $0.75f_l$ to avoid effects of filter roll-off. Eight records were not processed like the rest so use only response spectral values within 0.1 to $1\,\mathrm{s}$. Hence number of records used for regression at each period varies, minimum number used is 105 records for $2\,\mathrm{s}$.
- Give smoothed coefficients using cubic function.

4.67 *Ambraseys & Douglas (2000)*

- See Section 2.112.
- Response parameter is acceleration for 5% damping.
- Find b_2 and b_3 significantly different than 0 at 5% level for all periods but b_A and b_S not significant for many periods (especially for vertical component).
- Find deamplification for vertical component on soft and stiff soil compared with rock. Check by removing all 34 Northridge records (many of which were on soft soil) and repeat analysis; find little change.
- Also derive equations for horizontal response under influence of vertical acceleration using a bending SDOF model; find little change in response.

4.68 *Bozorgnia* et al. *(2000)*

- See Section 2.113.
- Response parameter is acceleration for 5% damping.
- Different set of data than for PGA hence: strike-slip: 20 earthquakes (including one normal faulting shock), reverse: 7 earthquakes and thrust: 6 earthquakes.
- Find considerable period-to-period variability in coefficients causing predicted spectra to be very jagged near limits of magnitude and distance ranges so carried out partial smoothing of coefficients.

4.69 *Campbell & Bozorgnia (2000)*

- See Section 2.114.
- Response parameter is pseudo-acceleration for 5% damping.

4.70 *Chou & Uang (2000)*

- Ground motion model is:

$$\log Y = a + b(M-6) + c(M-6)^2 + d\log(D^2+h^2)^{1/2} + eG_c + fG_d$$

- Response parameter is pseudo-velocity for 5% damping.
- Use three site categories (based on average shear-wave velocity, V_s, over top $30\,\mathrm{m}$):

Classes A+B Hard rock or rock: $V_s > 760\,\mathrm{ms}^{-1}$, $G_c = 0$, $G_d = 0$, 35 records.
Class C Very dense soil and soft rock: $360 < V_s \leq 760\,\mathrm{ms}^{-1}$, $G_c = 1$, $G_d = 0$, 97 records.
Class D Stiff soil: $180 \leq V_s \leq 360\,\mathrm{ms}^{-1}$, $G_c = 0$, $G_d = 1$, 141 records.

- Records from free-field or ground level of structures no more than two storeys in height.
- Smooth coefficients using cubic polynomial.
- Do not give coefficients for all periods.
- Find cannot use equation to predict near-field ground motions.

4.71 *Kawano* et al. *(2000)*

- Ground motion model is:

$$\log S_i(T) = a(T)M - \{b(T)X_{eq} + \log X_{eq}\} + c_i(T)$$

  where $c_i(T)$ is an individual site amplification factor for each of 12 stations.
- Response parameter is acceleration for 5% damping.
- Focal depths between 0 and $60\,\mathrm{km}$.
- Use data either recorded at ground surface where $0.5 \leq V_s \leq 2.7\,\mathrm{kms}^{-1}$ ($1.7 \leq V_p \leq 5.5\,\mathrm{kms}^{-1}$) or obtained by analytically removing effects of uppermost surface layers of ground from underground observation data (or by stripping-off analysis) using underground structure.
- Use only ground motion after arrival of first S wave because most important for aseismic design.
- Do not give table of coefficients, only graphs of coefficients.
- Define amplification factors, $d_i(T) = c_i(T) - c_0(T)$ for horizontal motion and $d_i(T) = c_{v,i}(T) - c_0(T)$ for vertical motion, where $c_0(T)$ is the regression coefficient for data observed at ground layer equivalent to seismic bedrock.
- Find $S_h(T) = S_b(T)\alpha_h(T)\beta_h(T)$ where $S_b(T)$ is $S_0(T)$. $\alpha_h(T) = (V_s/V_{s,b})^{-\delta_h(T)}$ for $T \leq T_{s,1}$ and $\alpha_h(T) = \alpha_h(T_{s,1})$ for $T > T_{s,1}$ where $T_{s,1}$ is the primary predominant period of surface layer. $\beta_h(T) = 1$ for $T \leq T_{s,1}$, $\beta_h(T) = (T/T_{s,1})^{-\log(\alpha_h(T_{s,1}))}$ for $10T_{s,1} > T > T_{s,1}$ and $\beta_h(T) = 10^{-\log(\alpha_h(T_{s,1}))}$ for $T \geq 10T_{s,1}$. $V_{s,b} = 2.2\,\mathrm{kms}^{-1}$. Similar relationships are defined for vertical motion, $S_v(T)$.
- Note that relation does not include effect of source mechanism or rupture propagation, so probably less valid in near-fault region.

4.72 *Kobayashi* et al. *(2000)*

- See Section 2.116.
- Response parameter is pseudo-velocity for 5% damping.
- Use significantly less records for $T > 1.5\,\mathrm{s}$.

4.73 *McVerry* et al. *(2000)*

- Ground motion model for crustal earthquakes is (using form from Abrahamson & Silva (1997), see Section 4.58):

$$\begin{aligned}\ln \mathrm{SA}'(T) &= C_1(T) + C_{4AS}(M-6) + C_{3AS}(T)(8.5-M)^2 + C_5(T)r \\ &\quad + (C_8(T) + C_{6AS}(M-6))\ln(r^2 + C_{10AS}^2(T))^{1/2} + C_{46}(T)r_{\mathrm{VOL}} \\ &\quad + \{C_2(T)r + C_{44}(T) + (C_9(T) + C_7(T)(M-6))(\ln(r^2 + C_{10AS}^2(T))^{1/2} \\ &\quad - \ln C_{10AS})\} \\ &\quad + \{C_{29}(T)\} \\ &\quad + \{C_{30AS}(T)\ln(\mathrm{PGA}'_{WA} + 0.03) + C_{43}(T)\} \\ &\quad + C_{32}\mathrm{CN} + C_{33AS}(T)\mathrm{CR}\end{aligned}$$

  Also add on hanging wall term, see Section 4.58. Subscript AS denotes those coefficients from Abrahamson & Silva (1997). Three parts of equation within $\{\ldots\}$ are for site conditions MA/SA, Class B and Class C respectively. PGA'_{WA} is the predicted PGA ($\mathrm{SA}'(0)$) for weak rock category. $\mathrm{CN} = -1$ for normal mechanism and 0 otherwise. $\mathrm{CR} = 0.5$ for reverse/oblique, 1.0 for reverse and 0 otherwise. Ground motion model for subduction zone earthquakes is (using form from Youngs *et al.* (1997), see Section 4.61):

$$\begin{aligned}\ln \mathrm{SA}'(T) &= C_{11}(T) + [C_{12Y} + (C_{17Y}(T) - C_{17}(T))C_{19Y}] \\ &\quad + C_{13Y}(T)(10-M)^3 + C_{17}(T)\ln(r + C_{18Y}\exp(C_{19Y}M)) + C_{20}(T)H_C \\ &\quad + C_{24}(T)\mathrm{SI} + C_{46}(T)r_{\mathrm{VOL}}(1-\mathrm{DS}) \\ &\quad + \{C_{44}(T) + C_{16}(T)(\ln(r + C_{18Y}\exp(C_{19Y}M)) \\ &\quad - \ln(C_{18Y}\exp(C_{19Y}M)))\} \\ &\quad + \{C_{29}(T)\} \\ &\quad + \{C_{30Y}(T)\ln(\mathrm{PGA}'_{WA} + 0.03) + C_{43}(T)\}\end{aligned}$$

  Subscript Y denotes those coefficients from Youngs *et al.* (1997). Three parts of equation within $\{\ldots\}$ are for site conditions MA/SA, Class B and Class C respectively. $\mathrm{SI} = 1$ for subduction interface and 0 otherwise. $\mathrm{DS} = 1$ for deep slab and 0 otherwise. r_{VOL} is length of path that lies in the volcanic zone.

- Response parameter is acceleration for 5% damping.
- Use four site conditions (mostly based on geological descriptions rather than measured shear-wave velocity):

WA	Weak rock sites, or sites with soil layer of thickness $\leq 3\,\mathrm{m}$ overlying weak rock.
MA/SA	Moderate-strength or strong rock sites, or sites with soil layer of thickness $\leq 3\,\mathrm{m}$ overlying moderate-strength or strong rock.
Class B	Intermediate soil sites or sites with soil layer of thickness $> 3\,\mathrm{m}$ overlying rock.
Class C	Flexible or deep soil sites with natural periods $> 0.6\,\mathrm{s}$.

Justify soil categories using statistical studies of residuals at early stage. Exclude response spectra from very soft soil sites ($V_s < 150\,\mathrm{ms}^{-1}$ for depths of $\gtrsim 10\,\mathrm{m}$).

- Use data for PGA equation from Zhao *et al.* (1997), see Section 2.100.
- Exclude records from bases of buildings with >4 storeys.
- Use less records for long periods because noise.
- Lack of data prevent development of robust model purely from NZ data. Plot residuals of predicted response using published attenuation relations (base models) for other areas to find relations which gave good representations of NZ data. Then modify some coefficients to improve match; imposing constraints so that the selected models control behaviour at short distances where NZ data lacking. Require crustal and subduction zone expressions for rock sites to match magnitude dependence of base models at $r = 0\,\mathrm{km}$. Constrain coefficients that occur nonlinearly and nonlinear site response coefficient for Class C to base model values.
- Find anelastic attenuation term and additive terms for shallow slab earthquakes for subduction earthquakes not statistically significant. Also differences in attenuation rates for shallow slab, deep slab and interface earthquakes not statistically significant.
- Exclude deep slab earthquakes because of high attenuation in mantle; note equation should not be used for such earthquakes.
- Different attenuation rate for site category MA/SA because of magnitude dependence apparent in residuals for simpler model.
- Eliminate nonlinear site response term for Class B because find unacceptable (positive) values of coefficient and constraining to negative values produces poorer fit.
- Predicted PGA ($\mathrm{SA}'(0)$) from response spectrum set of records considerably smaller than those, $\mathrm{SA}(0)$, from the complete PGA set of records. Thus scale $\mathrm{SA}'(T)$ by ratio $\mathrm{SA}(0)/\mathrm{SA}'(0)$.
- Standard error has a magnitude dependent intra-event component and a magnitude independent inter-event component.
- Note lack of data for large magnitude subduction zone earthquakes and large magnitude near source data for crustal earthquakes.
- Do not give coefficients, only predictions.

4.74 *Shabestari & Yamazaki (2000)*

- Ground motion model is:

$$\log y(T) = b_0(T) + b_1(T)M + b_2(T) - \log r + b_4(T)h + c_i(T)$$

where $c_i(T)$ is the station coefficient, reflecting the relative site effect for each period, assuming zero mean for all stations.

- Response parameters are acceleration and velocity for 5% damping.
- Depths between 1 (includes earthquakes with depths reported as $0\,\mathrm{km}$) and $158\,\mathrm{km}$. Exclude earthquakes with focal depths greater than $200\,\mathrm{km}$.
- Exclude records with vectorial composition of PGA less than $0.01\,\mathrm{ms}^{-2}$.

- Exclude data from stations which have recorded less than two records, because the station coefficient could not be determined adequately. Use records from 823 stations.
- Most records from distances between 50 and $300\,\mathrm{km}$.
- Use three-stage iterative partial regression method.
- For $T \geq 5\,\mathrm{s}$ constrain horizontal anelastic coefficient to zero because get positive coefficient.

4.75 *Smit* et al. *(2000)*

- See Section 2.119.
- Response parameter is acceleration for 5% damping.

4.76 *Takahashi* et al. *(2000)*

- See Section 2.120.
- Response parameter is pseudo-velocity for 5% damping.
- For periods $\geq 1\,\mathrm{s}$ long period noise in records leads to reduction in number of records.
- Set b and e to zero at long periods because estimates not statistically significant.
- Find that soft soil site correction terms may be affected by different processing procedures for data from different sources.

5. GENERAL CHARACTERISTICS OF ATTENUATION RELATIONS FOR SPECTRAL ORDINATES

Table 5.1 gives the general characterisitics of published attenuation relations for spectral ordinates. The columns are the same as in Table 3.1 with three extra columns:

Ts Number of periods for which attenuation equations are derived

$T_{\min}$ Minimum period for which attenuation equation is derived

$T_{\max}$ Maximum period for which attenuation equation is derived

Tab. 5.1: Characteristics of published spectral relations

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
Johnson (1973)	W. USA	41	-	23	5.3	7.7	m_b	6.3	149.8	d_e	1	14	0.055	2.469	M	1	A
Kobayashi & Nagahashi (1977)	Japan	U	-	U	5.4*	7.9*	U	60*	210*	d_h	I	U	0.1	5	R[1]	O	A
McGuire (1977)	W. USA	34	-	22	5.3	7.6	M_L	14	125	d_h	1	16	0.1	8	B	U	A
Faccioli (1978)	W. USA, Japan, Papua New Guinea, Mexico & Greece	26[2]	-	11	5.3	7.8	U	15	342	d_h	1	15	0.1	4	B	U	A
McGuire (1978)	W. USA	70	-	17+*	4.5*	7.7	U[3]	11*	210*	d_h	2	1	1	1	B	U	A
Cornell *et al.* (1979)	W. USA	70	-	U	U	U	M_L	U	U	d_h	1	7	0.17	5	C	U	A
Ohsaki *et al.* (1980b)	Japan	95	-	29+	3.9*	7.2*	U	3*	500*	d_h	2	86	0.02	5	U	1	A
Ohsaki *et al.* (1980a)	Japan	75	-	U	4	7.4	U	6	500	d_h	1	U	0.02	5	U	1	A
Trifunac (1980)	W. USA	U	-	U	U	U	U	U	U	d_e	C	91	0.04	7.5	U	U	A
Devillers & Mohammadioun (1981)	W. USA	186	-	U	3.3*	7.7*	U	$\geq$ 10	250*	d_h	1	46	0.04	10	U	1	A
Kobayashi & Midorikawa (1982)	Japan	45	-	U	5.1	7.5	U	50	280	d_h	1	U	0.1	5	U	O	A
Joyner & Fumal (1984) & Joyner & Boore (1988)	W. N. America	U	-	U	5.0	7.7	$M_w(M_L)$	U	U	d_f	C	12	0.1	4	L	U	A
Kawashima *et al.* (1984)	Japan	197	-	90	5.0	U	M_{JMA}	U	U	d_e	3	10	0.1	3	R	1	A
Kawashima *et al.* (1985)	Japan	-	119	90*	5.0*	7.5*	M_{JMA}	5*	500*	d_e	3	10	0.1	3	-	1	A
Kamiyama & Yanagisawa (1986)	Japan	228	-	69	4.5	7.9	M_{JMA}	3	323	d_e	I	45	0.1	10	U	1	A

continued on next page

[1] They state it is two dimensional response spectrum which assume to be resolved component.
[2] Total earthquake components (does not need to be multiplied by two)
[3] Idriss (1978) finds magnitudes to be mixture of M_L, m_b and M_s.

Tab. 5.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
C.B. Crouse (1987)[4]	S. California	U	-	U	U	U	M_s	U	U	d_r	1	10	0.05	6	B	U	A
K. Sadigh (1987)[5]	W. USA + others	U	-	U	U	U	M_w	U	U	d_r	2	7	0.1	4	B	U	A (S, R)
Annaka & Nozawa (1988)	Japan	U	-	45	U	U	U	U	U	U	1	U	0.04*	4*	U	1	A
Crouse *et al.* (1988)	N. Honshu	64	-	U	5.1	8.2	M_w, M_s & M_{JMA} for < 7.5	42	407	d_E, d_h for $M < 7.5$	1	10	0.1	4	B	1	A
Yokota *et al.* (1988)	Tokyo	154	24	75 (U)	4.0	6.1	M_{JMA}	59 (60)	206 (100)	d_h	1	U	0.1 (0.05)	10 (5)	U	U	A
Youngs *et al.* (1988)	Worldwide subduction zones	20 + 197 + 389	-	16* (60)	5.6* (5)	8.1* (8.1, 8.2)[6]	M_w (M_s, m_b)	U (15*, 20*)	U (450*, 450*)	d_r, d_h for $M_w \lesssim 7.5$	1	15	0.07	4	G	1W	A (B,F)
Trifunac & Lee (1989)	Mostly California	438	438	104	U	U	U	U	U	d_e	C	12	0.04	14	B	U	A
Atkinson (1990)	E. N. America + 10 others	92+10[7]	-	8+3	3.60 (5.16)	6.00 (6.84)	M_w	8 (8)	1215 (23)	d_h	1	4	0.1	1	B	2	A
Campbell (1990)	Unknown	U	-	U	U	U	M_L for $M < 6$, M_s for $M \geq 6$	U	U	d_s	1	15	0.04	4	U	U	A
Dahle *et al.* (1990b) & Dahle *et al.* (1990a)	Worldwide intraplate regions	87	-	56	2.9	7.8	M_s (M_L, m_b, M_{CL}	6	1300	d_h	1	89	0.025	4	L	2	A
Tamura *et al.* (1990)	Japan	97	-	7	7.1	7.9	M_{JMA}	U	U	d_e	3	13	2	20	L	1, O	A
Tsai *et al.* (1990)	Worldwide	$<$88	-	$<$51	4.9*	7.4	M_w	3*	150*	d_r	1	14	0.07	1	U	M	T (S,O)
Crouse (1991)	Worldwide subduction zones	235	-	U	5.1	8.2	M_w (M_s, M_{JMA})	$>$8	$>$469	d_E, d_h for $M < 7.5$	1	10	0.1	4	B	1	A

continued on next page

[4] Reported in Joyner & Boore (1988).
[5] Reported in Joyner & Boore (1988).
[6] Consider equations valid for $M_w \leq 8$
[7] Total earthquake components (does not need to be multiplied by two). 79+10 records for $0.1\,\mathrm{s}$ equation.

Tab. 5.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
I.M. Idriss (1991)[8]	Unknown	572	-	30*	4.6	7.4	M_L for $M < 6$, M_s for $M \geq 6$	1	100	d_r, d_h for $M < 6$	1	23	0.03	5	U	U	A
Mohammadioun (1991)	Italy	144	-	46	3.0	6.5	U	6	186	d_h, 1 eq. with d_r	1	81	0.013	1.95	B	U	A
Niazi & Bozorgnia (1992)	SMART-1 array, Taiwan	236	234	12	3.6	7.8	M_L (M_D) for $M_L < 6.6$, else M_s	3.1[9]	119.7[9]	d_h	1	23	0.03	10	M	2W	A
Benito *et al.* (1992)	Campano Lucano	84	-	U	4.7	6.5	M_L	3.4*	142*	d_h	3	15	0.04	10	L	1	A
Tento *et al.* (1992)	Italy	137	-	40	4	6.6	M_L	3.2	170	d_f for $M_L \geq 5.7$, d_e otherwise	1	12	0.04	2.75	L	2	A
Boore *et al.* (1993) & Boore *et al.* (1997)	W. N. America	112	-	14	5.30	7.70	M_w	0	109	d_f	3	46	0.1	2	L, G	2M	A
Caillot & Bard (1993)	Italy	83	-	≤ 40	3.2	6.8	M_s if M_L & $M_s \geq 6.0$ else M_L	10	63	d_h	2	25	0.05	1.98	U	2, 1W	A
Campbell (1993)	Worldwide	U	-	U	U[10]	U	M_L for $M < 6.0$ and M_s otherwise	U	U[11]	d_s	2	15	0.04	4	M	O	A (T,S)
Lee (1993)	Mostly California	494	494	106	U	U	M_L for $M \lesssim 6.5$, others for $M > 6.5$	U	U	d_e	3	91	0.04	15	B	U	A

continued on next page

[8] Reported in Idriss (1993).
[9] Distance to centre of array
[10] Considers equation valid for $M \geq 4.7$.
[11] Considers equation valid for $d \leq 300\,\mathrm{km}$.

Tab. 5.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
Sadigh *et al.* (1993) & Sadigh *et al.* (1997)	California with 4 foreign	960+4	U	119+2	3.8 (6.8)	7.4 (7.4)	M_w	0.1 (3)	305 (172)[12]	d_r for some, d_h for small ones	2	21	0.05[13]	7.5[14]	G	U	A(R,S)
Sun & Peng (1993)	W. USA with 1 foreign	150+1	-	42+1	4.1	7.7	M_L for $M < 6$, else M_s	2*	150*	d_e	C	U	0.04	10	R	1	A
Boore *et al.* (1994a) & Boore *et al.* (1997)	W. N. America	112 (70)	-	14 (9)	5.30	7.70 (7.40)	M_w	0	109	d_f	C	46	0.1	2	L, G	1M, 2M	A (R,S)[15]
Climent *et al.* (1994)	Central America & Mexico	280	U	72	U	U	U	U	U	U	U	U	0.05*	≥ 2	U	U	A
Fukushima *et al.* (1994) & Fukushima *et al.* (1995)	3 vertical arrays in Japan	285	284	42	5.0	7.7	M_{JMA}	60*	400*	d_h	I	U	0.05	2	B	1,2	A
Lawson & Krawinkler (1994)	W. USA	250+	-	11	5.8	7.4	M_w	U	100	d_f	3	38	0.1	4	U	1M	A
Lee & Manić (1994) & Lee (1995)	Former Yugoslavia	313	313	183	3.75	7.0	U	4	250	d_e	6	12	0.04	2	U	2R	A
Mohammadioun (1994a)	California	108[16]	56	23	5.3	7.7	M_L	3	136	Often d_r, d_h in far field	1	96	0.013	5	B	1	A
Mohammadioun (1994b)	W. USA	530[17]	≈ 265	U	U	U	M_L	1	250	d_r, d_E if more appropriate, d_h in far field	1	96	0.013	5	B	1	A

continued on next page

[12] Equations stated to be for distances up to $100\,\mathrm{km}$
[13] Minimum period for vertical equations is $0.04\,\mathrm{s}$.
[14] Maximum period for vertical equations is $3\,\mathrm{s}$.
[15] Coefficients given in Boore *et al.* (1994b).
[16] Total number, does not need to be multiplied by two.
[17] Total number, does not need to be multiplied by two.

Tab. 5.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
Musson *et al.* (1994)	UK + 28* foreign	88*+28*[18]	-	15+16	3 (3.7)	4.1 (6.4)	M_L	70* (>1.3)	>477.4 (200*)	d_h	1	4	0.1	1	U[19]	O	A
Theodulidis & Papazachos (1994)	Greece+16 foreign	105+16[20]	-	36+4	4.5 (7.2)	7.0 (7.5)	M_s, M_w, M_{JMA}	1 (48)	128 (236)	d_e	2	73	0.05	5	B	O	A
Dahle *et al.* (1995)	Cen. America	280	-	72	3*	8*	M_w (M_s, m_b, M_D)	6*	490*	d_h	2	8	0.025	4	L	1B	A
Ambraseys *et al.* (1996)	Europe & Mid. East	422	-	157	4.0	7.9	M_s (unspecified)	0	260	d_f for $M > 6.0$, d_e otherwise	3	46	0.1	2	L	2	A
Ambraseys & Simpson (1996)	Europe & Mid. East	-	417	157	4.0	7.9	M_s (unspecified)	0	260	d_f for $M > 6.0$, d_e otherwise	3	46	0.1	2	L	2	A
Bommer *et al.* (1996)	El Salvador & Nicaragua	36	-	20	3.7	7.0	M_s	62	260	d_h	1	10	0.1	2	L	U	A
Crouse & McGuire (1996)	Cen. & S. California	238	-	16	6.0	7.7	M_s	0.1	211	d_r	4	14	0.04	14	G	1W	R,S (R,S)
Free (1996) & Free *et al.* (1998)	Stable continental regions	399–410	347–477	H: 137–138, V: 126–132	1.5	6.8	M_w	0	820	d_f for some, d_e for most	2	52	0.04	2	L	1	A
Ohno *et al.* (1996)	California	248	-	17	5.0	7.5	M_w (M_L)	7.2	99.6	d_q for $M > 5.3$, d_h otherwise	2	U	0.02	2	B	2M	A
Sabetta & Pugliese (1996)	Italy	95	95	17	4.6	6.8	M_s if M_L & $M_s \geq 5.5$ else M_L	1.5, 1.5	179, 180[21]	Both d_f & d_e	3	14	0.04	4	L	1	A

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[18] There are 116 records in total.
[19] Free (1996) believes it is largest horizontal component.
[20] Total number of components does not need to be multiplied by two
[21] State equations should not be used for distances $> 100\,\mathrm{km}$

Tab. 5.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
Spudich *et al.* (1996)	Worldwide extensional regimes	99–118	-	27–29	5.10	6.90	M_w	0	102.1	d_f	2	46	0.1	2	G, C	2M	NS
Abrahamson & Silva (1997)	California with some others	≤ 655*	≤ 650*	≤ 58	4.4	7.4	U	0.1	220*	d_r	2	28	0.01	5	G	1M	A (S,O,T)
Atkinson (1997)	Cascadia with some foreign	U	-	11+9	4.1	6.7(8.2)	M_w	20*	580*	d_c for some, d_h for small ones	2	12	0.1	2	B	2	A
Campbell (1997)	Worldwide	266 [22]	173	H:30, V:22	4.7	8.1	M_s for $M_s \geq 6$, M_L for $M_s < 6$	3	50	d_s	3	13	0.05	4	G	IW	A (S,R,N)
Youngs *et al.* (1997)	Worldwide subduction zones	≤ 476	-	≤ 164	5.0	8.2	M_w (M_s,m_b)	8.5	550.9	d_r, d_h for some	2	11	0.075	3	G	1M	NT (N,T)
Bommer *et al.* (1998)	Europe & Mid. East	121–183	-	34–43	5.5	7.9	M_s	3	260	d_f for most, d_e otherwise	3	66	0.04	3	L	2	A
Perea & Sordo (1998)	Urban area of Puebla, Mexico	10[23]	-	8	5.8	8.1	m_b for $M < 6$, M_s otherwise	274	663	d_e	1	195	0.01	3.5	L	1	A
Shabestari & Yamazaki (1998)	Japan	3990	-	1020	U	8.1	M_{JMA}	U	U	d_r	U	35	0.04	10	L	O	A
Chapman (1999)	W. N. America	304	-	23	5.0	7.7	M_w	0.1	189.4	d_f	3	24	0.1	2	G	2M	A
Spudich *et al.* (1999)	Worldwide extensional regimes	105–132	-	≤ 38	5.1	7.2	M_w	0	99.4	d_f	2	46	0.1	2	G	1M	NS
Ambraseys & Douglas (2000)	Worldwide	186	183	44	5.83	7.8	M_s	0	15	d_f	3	46	0.1	2	L	1	A
Bozorgnia *et al.* (2000)	Worldwide	1308	1308	33	U	U	M_w	U	≤ 60	d_s	4	U	0.05	4	G	U	A (R,S,T)

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[22] Typographic error in Table 3 of Campbell (1997) does not match number of recordings in Table 4

[23] Typographical error in Figure 3b) of Perea & Sordo (1998) because it does not match their Table 1.

Tab. 5.1: *continued*

Reference	Area	H	V	E	$M_{\min}$	$M_{\max}$	M scale	$d_{\min}$	$d_{\max}$	d scale	S	Ts	$T_{\min}$	$T_{\max}$	C	R	M
Campbell & Bozorgnia (2000)	Worldwide	275–435	274–434	≤ 36	≥ 4.7	≤ 7.7	M_w	$\geq$ 1*	$\leq$ 60*	d_s	4	14	0.05	4	G	1	A (S,R,T)
Chou & Uang (2000)	California	273	-	15	5.6	7.4	M_w	0*	120	d_f	3	25	0.1	3	G	2M	A
Kawano *et al.* (2000)	Japan	107	107	44	5.5	7.0	M_{JMA}	27	202	d_q	I, C	U	0.02	5	U	O	A
Kobayashi *et al.* (2000)	Japan	U	-	U	5.0	7.8	M_w	0.9*	400*	U	4	17	0.1	5	B	1M	A
McVerry *et al.* (2000)	NZ with 66 foreign	$\leq$ 224 (461+66)	-	(51+17)	(5.08)	(7.23(7.41))	M_w	(0.1)	(573)	(d_r for some, d_c for most)	4	U	0.01*	4*	U	O	A (N, R, RO)
Shabestari & Yamazaki (2000)	Japan	6017	-	94	5.0	6.6	M_{JMA}	7*	950*	d_r	I	35	0.04	10	L	O	A
Smit *et al.* (2000)	Caucasus	84	-	26	4.0	7.1	M_s	4	230	d_h	1	22	0.05	1	L	2	A
Takahashi *et al.* (2000)	Japan+166 foreign	$\leq$1332	-	U+7*	5* (5.8*)	8.3* (8*)	M_w	1* (0.1*)	300* (100*)	d_r, d_h for some	4	20	0.05	5	G	O	A

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ESEE RESEARCH PUBLICATIONS

The Engineering Seismology and Earthquake Engineering Section at Imperial College publishes reports on research work undertaken by members of the Section and the Department; a list of currently available reports is given below. Copies may be obtained from the ESEE Section Secretary, Department of Civil and Environmental Engineering, Imperial College, London SW7 2BU (Prices on application, Reports marked N are not for sale and some are out-of-print. The reports are sent free to selected libraries in Universities and Research Centres).

Shake Table Facility Time History Matching and Exploratory Analysis, A.C. Hargreaves, A.H. Al- Sheikh and S.H. Perry, ESEE-86/1, April 1986.

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The Spitak (Armenia, USSR) Earthquake of 7 December 1988; A Preliminary Engineering Seismology Report, J.J. Bommer and N.N. Ambraseys, ESEE-89/1, March 1989.

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